

Humidity-responsive controlled-release 1-MCP packaging

A literature, technology, commercial and patent-landscape assessment

Whether a humidity-triggered 1-MCP packaging material — optionally carrying an essential-oil antimicrobial — is novel, buildable, and worth pursuing for climacteric fruit.

Every value in the evidence tables is traceable to a quote in a document that was actually retrieved. Values that could not be verified are marked, never guessed. This report expresses **no freedom-to-operate or patentability opinion**; those require claim-by-claim review by qualified counsel. Proposed formulations are labelled as experimental hypotheses, not literature-established facts.

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Executive conclusion

The short answer

The broad idea is not novel, and the closest prior art is closer than the brief assumed. A 2024 preprint (Cao et al., Nanjing Forestry University) already combines carvacrol/ β -cyclodextrin **and** 1-MCP/ α -cyclodextrin on **coated paper** made by a **water-free method**, validated on **peach**. Separately, the coated- insert *format* is commercial — Verdant Technologies' HarvestHold Fresh is a 1-MCP sheet applied in the box — and delayed, gradual in-package 1-MCP release is commercially claimed by Fresh Inset (Vidre+). And **Verdant Technologies — which sells HarvestHold Fresh and now holds the Cellresin estate — has an application claiming a coated substrate whose coating consists essentially of a 1-MCP clathrate of α -cyclodextrin**, which is the recommended architecture in claim language.

The proposed starting formulation is worse than the simpler alternative. A chitosan/CNC film carrying a 1-MCP/cyclodextrin complex *and* a CNC-stabilised essential-oil Pickering emulsion scores **last of six architectures** (43/100), because its central feature — an aqueous, hygroscopic matrix — attacks the one thing the product must protect: a payload that water discharges.

What remains defensible is narrow, quantitative, and worth testing. Not a new format, not a new material set, and not "controlled release" in the abstract, but a **specified humidity activation threshold and coat weight delivering a reproducible headspace 1-MCP concentration across varying package volume and fruit mass**, demonstrated by GC quantification and a fitted model. No retrieved source — academic or commercial — reports a coat weight, a release model, an activation threshold, or a dose-reproducibility figure for a 1-MCP paper.

The principal risks are not technical, and one may already be decisive. The patent estate is systematically enclosed by three parties — Verdant, AgroFresh and Hazel — covering the complex, the film, the coating, the coated substrate and the dual active. Of the ten most important open questions, three are regulatory, legal or commercial, and two can end the project regardless of any experimental result. 1-MCP is a registered pesticide active (EPA Reg. 71297-2, since 2002) and registrations are crop- and format-specific.

Direct answers to the brief's fifteen questions

#	Question	Answer
1	Is the broad idea novel?	No. Anticipated in the literature, in patents, and in commercial products.
2	Which parts are already known?	All of them: 1-MCP/CD complexes, humidity activation, coated paper and sheet inserts, labels and sachets, dual 1-MCP + essential oil on paper, and delayed release.
3	Where is the most credible white space?	Dose specification and measurement — a defined RH threshold and coat weight giving reproducible headspace concentration across variable packages. Measured across 82 papers: only 4 report a kinetic model, 4 report package volume and mass, 9 report loading. But this is a <i>method</i> claim in an article space that is already enclosed, so it is weak-to-moderate at best.
4	Film, coated paper, label, patch, insert, pad or sachet?	Coated paper insert — highest score on every project-critical criterion. Chosen as the right engineering answer, not as a novel format.
5	Same layer or separate layers?	Separate — and omit the oil in v1. Cavity competition, oil-phase partitioning, plasticisation, and divergent release requirements all argue against co-location.
6	Is CNC necessary?	No. With no oil phase to stabilise it has no function; it adds cost and process steps.

#	Question	Answer
7	Is chitosan necessary?	No. Hygroscopic (wrong for a moisture-triggered payload), needs acidic aqueous processing, and its contact-antimicrobial mechanism cannot operate in a non-contact insert.
8	Is the Pickering emulsion justified?	No. It requires the aqueous phase the design exists to avoid. Cao et al. achieved a dual-active coated paper without one.
9	Which antimicrobial is best?	Carvacrol , if any — pure compound, documented against <i>Monilinia</i> , and directly comparable to the closest prior art. Deferred to v2.
10	Which commodity first?	Tomato (fast, cheap, unambiguous response), then peach (commercial relevance, decay pressure, direct comparability).
11	Simplest proof-of-concept?	1-MCP/ α -CD + one binder on paper, Meyer-rod coated, protective overwrap. Three materials.
12	Most scalable route?	Meyer-rod in the lab → slot-die at scale , both specified in $\text{g}\cdot\text{m}^{-2}$. Reject extrusion coating (thermal loss) and solvent casting (no scale-up path, unscalable units).
13	Principal patent risk?	US20210331990A1 (Verdant, priority 2020-04-27) — claims a coated substrate whose coating consists essentially of a 1-MCP clathrate of α -cyclodextrin. That is the recommended architecture, and it needs no second substrate, so the single-layer design-around does not clear it.
14	What justifies continuing?	A sharp, reproducible RH activation threshold plus a low delivered-dose CV across package volumes — evidence nobody currently publishes.
15	What should cause abandonment?	Release versus RH smooth and shallow across all binders; or dose CV that stays high after optimisation; or a blocking claim with no design-around; or a registration requirement beyond project resources.

What we recommend

Primary: a single-layer 1-MCP/ α -CD coated paper insert with a protective overwrap. Three materials, one coating pass, mild drying, no fruit contact.

Backup: complex adsorbed on absorbent paper with no binder at all — two materials, built to answer one question fast.

High-risk/high-reward: a condensate-managing coated pad that uses the moisture it manages as its trigger.

Do first, before any formulation work: assay the complex's actual 1-MCP content, get a supplier quotation, and ask a FIFRA specialist whether this product can reach market without a registrant partner.

How much confidence this deserves

The evidence base is real and verifiable — every value in the literature and patent tables is traceable to a quote in a document that was actually retrieved, and values that could not be verified are marked rather than guessed. But three limitations should temper any conclusion about *absence* of prior work:

- **Four indexes named in the brief were inaccessible** (Web of Science, Scopus, CAB Abstracts, AGRICOLA). CAB Abstracts and AGRICOLA index precisely the postharvest and extension literature where a 1-MCP paper insert might be described.

- **Patent coverage rests on a single provider**, with no independent legal-status cross-check, and unexamined applications were not searchable.
- **Open-access coverage of this literature is thin** — 34 of 82 extracted papers yielded full text, so of 2 050 table cells, 265 are full-text-verified, 248 abstract-level, 454 confirmed *not stated*, and 837 unverifiable. Process-detail cells are often legitimately empty rather than genuinely absent from the source.

Nothing here constitutes a freedom-to-operate or patentability opinion, and none should be inferred.

Correction of the original novelty assumption

The brief already anticipates that the broad concept is not new and asks that this be tested rather than assumed. The evidence retrieved here confirms it, and more sharply than the brief's own framing suggests.

The broad concept is not novel — and the closest prior art is very close

The proposed combination is: **1-MCP held in a cyclodextrin complex, coated on a paper substrate, releasing on exposure to humidity, optionally alongside an essential-oil antimicrobial, for climacteric fruit.**

Each element is disclosed, and — critically — so is the *combination*.

Cao et al. (2024), Research Square preprint, DOI 10.21203/rs.3.rs-4251760/v1 — “Preparation of carvacrol β -cyclodextrin inclusions and 1-methylcyclopropene- α -CD coated paper by water-free method for postharvest preservation of peach” (Nanjing Forestry University). Full text retrieved and verified.

This single document combines:

Proposed element	Status in Cao et al.
1-MCP in α -cyclodextrin	present
Essential oil (carvacrol) in β -cyclodextrin	present
Both actives on a coated paper substrate	present
Water-free / low-moisture preparation to limit premature 1-MCP loss	present — it is the paper's headline method
Climacteric fruit validation (peach)	present
Antimicrobial efficacy vs. <i>Monilinia fructicola</i> (brown rot)	present

It even articulates the exact problem this project proposes to solve, in its own words:

“1-MCP- α -CD is sensitive to high moisture, which may result in the complete release of 1-MCP gas in a short period of time. The respiration of the fruit as well as the storage environment usually results in high relative humidity (RH). So, the 1-MCP- α -CD was not stable enough for long term release or in a controlled release manner of 1-MCP to reach the desired effect.”

— and offers a solution (a polyethylene-glycol matrix moderating release).

Any claim that “1-MCP + essential oil + cyclodextrin + coated paper” is novel would not survive contact with this document. It should be treated as the closest known prior art and cited in any invention disclosure.

The recommended architecture is claimed almost verbatim

Forty patent families were retrieved and their claims read. One document dominates the analysis:

US20210331990A1 — Verdant Technologies LLC, priority **2020-04-27** Claim 1: “A coated substrate comprising a coating disposed on a substrate surface, the coating consisting essentially of a **1-methylcyclopropene clathrate of α -cyclodextrin...**”

That is a coated substrate bearing a 1-MCP/ α -cyclodextrin coating — the architecture this report otherwise recommends on engineering grounds, described in the claim language of a granted-family application. It requires **no second substrate**, so the “single layer, not a laminate” design-around that would distinguish the Kimberly-Clark laminate does **not** clear it.

The wider estate, by priority date:

Priority	Publication	Assignee	Claim substance
1998-08-20	US6017849A	Biotechnologies for Horticulture / AgroFresh	A complex formed from a molecular encapsulation agent and a cyclopropene derivative — the foundational complex claim, cited on the SmartFresh InBox label
2000-09-29	US6426319B1	AgroFresh (current) / Rohm and Haas	Cyclopropene delivery systems requiring less water
2002-12-10	US8129450B2	Verdant (current) / Cellresin	Cyclodextrin-grafted polymer articles
2007-01-17	US8603524B2	AgroFresh	A film comprising a cyclopropene/cyclodextrin complex intermingled with a water-soluble polymer
2009-12-14	US9642356B2	Verdant	Controlled-release composition: cyclodextrin-modified thermoplastic + 1-MCP
2011-03-27	US8481127B2	Verdant	Cyclodextrin composition with radiation-polymerizable monomers + a 1-MCP inclusion complex
2012-11-30	US10212931B2	Kimberly-Clark / Cellresin	A treated laminate: substrate, cyclodextrin coating, second substrate over it
2013-07-11	PH12016500070A1	AgroFresh	Delivery systems in the form of coatings or films giving controlled release of 1-MCP in vapour form
2016-02-19	US20190037839A1	Hazel Technologies	Porous carbon/silicate carriers releasing 1-MCP, optionally with essential oils
2020-02-03	US20210235692A1	Fresh Inset SA	Stabilised 1-MCP compositions
2020-04-27	US20210331990A1	Verdant	Coated substrate; coating consisting essentially of a 1-MCP clathrate of α-cyclodextrin

Three structural facts matter more than any single document:

1. **The estate has consolidated.** Patents originally prosecuted by Cellresin Technologies now list **Verdant Technologies LLC** as current assignee. The same company sells HarvestHold Fresh — the commercial 1-MCP sheet. One party therefore holds the product, the foundational cyclodextrin estate, and the coated-substrate claim.
2. **Coatings and films were claimed long before this project.** AgroFresh's US8603524B2 (2007) claims a film containing a cyclopropene/cyclodextrin complex; PH12016500070A1 (2013) claims coatings and films for vapour-phase 1-MCP release.
3. **Even the dual-active idea is claimed.** Hazel's US20190037839A1 covers 1-MCP release from porous carriers *optionally including essential oils*.

No freedom-to-operate or patentability conclusion is drawn here — that requires claim-by-claim legal review. What can be said technically is that this is not a crowded field with gaps; it is a **systematically enclosed** one, held by three well-resourced parties (Verdant, AgroFresh, Hazel) who between them cover the complex, the film, the coating, the coated substrate, and the dual active.

2.2b What the corpus measures about the literature

The claim in §2.3 — that quantitative release and dose parameters are missing — is not an impression from one paper. Across **82 papers** whose 25 fields were extracted and grounded:

Parameter	Papers reporting it
A fitted kinetic model	4 / 82

Parameter	Papers reporting it
Package volume and produce mass	4 / 82
1-MCP loading	9 / 82
Manufacturing / drying temperature	11 / 82
Relative-humidity conditions	14 / 82
Release-measurement method	8 / 82

This literature reports *fruit outcomes*, not *delivery-system characterisation*. That is the measured shape of the gap, and it is consistent across the corpus rather than peculiar to the closest prior art.

(Counts are papers with a grounded value; cells we could not verify are counted as absent, so these are lower bounds — see the coverage table in Section 4.)

Where the evidence suggests white space may actually remain

The value of the Cao et al. paper is not only that it anticipates the concept — it is that its *gaps are measurable*, and they are the same gaps across much of this literature. In that full text:

- **No release-kinetics modelling** — zero occurrences of Higuchi, Korsmeyer-Peppas, or any kinetic model.
- **No gas-chromatographic quantification of 1-MCP** — release was followed gravimetrically. There is no measured headspace 1-MCP concentration in $\mu\text{L}\cdot\text{L}^{-1}$.
- **No defined humidity activation threshold** — humidity is discussed as an influence on rate, not engineered as a trigger with a specified onset.
- **No coat weight or coating-thickness specification**, and no dose-per-package logic tying active loading to package volume or fruit mass.

So the credible white space is **not** “1-MCP plus essential oil on paper”, which is taken. It is narrower and more quantitative:

*a specified humidity-activation threshold and release window, demonstrated by **headspace GC quantification of 1-MCP in $\mu\text{L}\cdot\text{L}^{-1}$** , with a defined coat weight that delivers a reproducible dose across varying package volume and fruit mass.*

That is a claim about measurement rigour and dose control rather than about materials — and it is consistent with the brief’s own suspicion that novelty, if any, lies in a defined threshold, a delayed-onset profile, or a dose-control method rather than in the ingredient list.

This remains a hypothesis to be tested, not an established finding. It rests on gaps observed in the retrieved corpus, and four indexes named in the brief (Web of Science, Scopus, CAB Abstracts, AGRICOLA) were not accessible. Absence of evidence here is weaker than usual.

What this means for the project

- Do not pursue “a chitosan/CNC film containing 1-MCP-CD and an essential-oil Pickering emulsion” as a novel concept. It is a more complicated version of something already published, and the added complexity works against both manufacturability and patentability.
- Treat Cao et al. (2024) and the Cellresin family as the benchmark to beat, and define success quantitatively against them.
- The differentiator, if one exists, will have to be **demonstrated with numbers those documents do not report** — which is fortunate, because that is also what makes a useful product.

Systematic-search methodology

Design principle: retrieval-first

The brief forbids fabricated references, values, and conclusions. A language model asked to produce a 40-paper × 25-column table from memory will invent most of it, so this review was built so that fabrication is structurally impossible rather than merely discouraged:

1. **Identifiers never originate from a model.** DOIs, publication years, journal names, author lists and patent numbers come from OpenAlex, Crossref, Europe PMC and Google Patents. The extraction model is never given the opportunity to emit one.
2. **The model only ever reads retrieved text.** Extraction runs against a document already stored on disk — never against recall.
3. **Every asserted value is re-checked against that text.** An independent grounding step verifies each value; anything the document does not contain is blanked and flagged, and the row survives with reduced confidence.

The pipeline implementing this is in `scripts/harvest/` and is re-runnable.

Sources actually used

Source	Role	Status
OpenAlex	primary discovery	used; <code>title_and_abstract.search</code> with explicit boolean operators
Crossref	reference strings, DOI verification	used
Europe PMC	discovery + open-access JATS full text	used
Unpaywall	open-access location lookup	used
Tavily	grey literature, commercial/regulatory pages, patent retrieval	used
Google Patents	patent discovery and full documents	used indirectly (see 3.6)
Semantic Scholar	citation enrichment	not used — rate-limited without a key; OpenAlex + Europe PMC + Crossref already covered this literature
Web of Science, Scopus, CAB Abstracts, AGRICOLA	—	not accessible ; no subscription or API credentials available
EPO OPS, PatentsView	patent metadata cross-check	not accessible ; both require API keys

The brief lists Web of Science, Scopus, CAB Abstracts and AGRICOLA. None were reachable from this environment. Their absence is a genuine coverage limitation: CAB Abstracts and AGRICOLA in particular index horticultural and postharvest literature — including conference and extension material — that OpenAlex and Europe PMC cover unevenly. Any conclusion about the *absence* of prior work should be read with that gap in mind.

Query grid

30 literature queries and 18 patent queries were run, built from the concept list in the brief. The grid is in `scripts/harvest/queries.py` and each query is tagged with a band so the coverage report can show which part of the question a paper answers.

Bands span the core topic (1-MCP release, humidity triggering, cyclodextrin carriers, packaging formats, measurement, produce validation, scale-up, stability) and a deliberate **analogue band** that does not mention 1-MCP at all — humidity-triggered release of any volatile active, CNC-stabilised Pickering emulsion films, essential-oil packaging, roll-to-roll coating, and release kinetics. A grid anchored only on 1-MCP would have missed the materials-science half of the question.

Date range 2000–present, as specified.

Screening

Records were deduplicated on DOI, then on normalised-title overlap for records without one.

A **deterministic pre-filter** ran before any model time was spent. Crossref’s bibliographic search returns loosely related work for any query: of 1 570 unique harvested records, 278 contained no

topical term whatsoever (3D bioprinting, ischemic heart disease, urinary-tract infections) and 258 predated 2000. These were removed by regular expression, not by judgement, and the dropped set is recorded in `02_dropped.jsonl` for audit.

Remaining records were ranked by topical-anchor density and the top 700 screened by the local model against the brief’s inclusion and exclusion criteria, which were embedded verbatim in the prompt so each decision is auditable against a fixed text.

Two safeguards protect recall:

- An uninformative or missing abstract yields **“maybe”**, never **“exclude”** — absence of information is not evidence of irrelevance.
- Any record whose signals indicate 1-MCP **and** controlled release, humidity triggering, or a cyclodextrin carrier is forced into review even when the model voted to exclude.

Full-text acquisition and its limits

Full text was obtained **only where openly licensed** — Europe PMC JATS, then open-access PDF via Unpaywall, then open-access landing pages. Paywalled papers were left at abstract depth.

Open-access coverage of this literature is thin. Roughly a quarter of screened papers yielded usable full text. This matters directly for the table: process detail — drying temperature, 1-MCP loading, coat weight, package volume and headspace — is usually reported in methods sections and almost never in abstracts. Consequently many cells in the literature table are legitimately empty, and the table distinguishes two different reasons:

- **not stated** — the full text is held and does not report the quantity.
- **unverifiable** — too little of the paper could be obtained to know.

These are never merged. Collapsing them would let “we could not read the paper” masquerade as “the paper omits this”, which would be a fabricated claim about the literature. Per-column counts of each are in `tables/coverage.csv`.

A PDF ingest path (`--pdf-drop`) exists so specific papers can be upgraded to full-text depth later without re-running the rest of the pipeline. This is the single highest-leverage way to deepen the evidence base.

Patent retrieval

Google Patents’ search endpoint **blocked this IP** after roughly six rapid requests during method development and did not recover through four rounds of exponential backoff. Patent discovery therefore ran through site-restricted web search, and full patent documents — including complete claim text, priority and family tables — were retrieved through Tavily’s crawler, which reaches the same pages independently.

Patent metadata (assignee, priority date, inventors) is extracted from the retrieved document text and then grounded against it, so it is evidenced rather than recalled. Publication numbers are taken deterministically from URLs.

Two limitations follow and are carried into the patent table:

- **Families are clustered heuristically** from title, assignee and priority date. INPADOC family data was unavailable, so family membership is *observed* — the members we saw — not authoritative.
- **Legal status is not reported.** Google Patents’ own status field carries an explicit disclaimer that it is not a legal conclusion, and no independent source was reachable to verify it. A column of unverified statuses would invite exactly the misreading the brief warns against, so the table reports observed family members instead.

Grounding and verification

Each extracted value passes three deterministic checks before it may appear in a table cell:

1. The model’s supporting quote must be locatable in the retrieved document (exactly, or repaired by fuzzy match back to the exact source span, so what ships is always the source’s own wording).
2. For numeric fields, the magnitude must match a quantity in the document after unit canonicalisation, so 2.1 mg g^{-1} and 2.1 mg/g compare equal.

3. The matched number must sit near a keyword for that field. Without this, any paper containing "40" would appear to confirm a drying temperature of 40 °C.

For appraisal columns the value may paraphrase, but it may not introduce a number absent from the quote — that is where invented precision otherwise enters.

The grounding module was adversarially tested: **10/10 true values grounded, 7/7 fabrications caught** (wrong magnitude, correct number in the wrong context, number absent entirely, quote not present in source, fabricated material, invented number inside an appraisal).

A verification stage then re-resolves every DOI against Crossref, re-checks every stored quote against its stored source independently of the extraction path, detects duplicates, and reports per-column evidence coverage. Prose citations are checked against the retrieved corpus, so the report cannot cite a source the pipeline never fetched.

What this methodology cannot tell you

- It cannot establish that something has **never** been published. Four indexes named in the brief were inaccessible, and patent coverage rests on a single provider.
- It cannot assess **claim scope or validity**. The patent analysis is technical description only.
- Cells marked *unverifiable* are statements about our access, not about the paper.

Peer-reviewed literature table

How to read this table

Twenty-five columns per the brief. Every non-empty cell is backed by a quote from a document that was actually retrieved; the quotes are in `tables/literature_evidence.csv`, one row per cell, with the evidence level and the checks that were applied.

Empty cells are not all the same thing, and the distinction is recorded per cell in the evidence file:

Evidence level	Meaning
full_text_verified	Quote located in retrieved full text
abstract_only	Quote located in the abstract; full text was never openly available
metadata_only	From Crossref/OpenAlex (reference, DOI, publication type)
not_stated	We hold the full text and it does not report this — a positive finding
unverifiable	We could not obtain enough of the paper to know

The last two are never merged. Treating “we could not read it” as “the paper omits it” would be a fabricated claim about the literature.

Values that failed grounding were blanked, not published. Where the extraction model proposed a value that could not be located in the source, the cell is empty and flagged `ungrounded_blank` in the evidence file, and the row’s confidence is reduced. The row survives; the unsupported value does not.

Coverage — the honest quality statistic

Per-column counts of each evidence level:

Table 7: Evidence coverage per column of the literature table

Column	n	Full text	Abstract	Metadata	Not stated	Unverifiable	Blanked
Reference	82	0	0	82	0	0	0
DOI	82	0	0	82	0	0	0
Publication type	82	0	0	82	0	0	0
Packaging substrate / polymer	82	19	27	0	15	21	2
1-MCP carrier / inclusion	82	6	8	0	27	41	1
Antimicrobial active	82	17	23	0	17	25	0
Preparation method	82	16	15	0	18	33	0
Manufacturing temperature	82	10	1	0	21	50	3
Trigger mechanism	82	7	10	0	27	38	1
Relative-humidity conditions	82	7	7	0	27	41	0
Storage temperature	82	12	7	0	21	42	4
1-MCP loading	82	5	4	0	28	45	2
Release-measurement method	82	6	2	0	28	46	0
Release duration	82	6	6	0	28	42	1
Kinetic model	82	2	2	0	27	51	6
Main release result	82	6	16	0	28	32	0
Produce tested	82	20	27	0	14	21	0
Package volume & produce mass	82	4	0	0	29	49	1
Quality outcomes	82	17	19	0	12	34	6
Mechanical properties	82	4	5	0	25	48	10
Barrier properties	82	4	9	0	27	42	4
Antimicrobial outcomes	82	10	10	0	23	39	1
Major limitations	82	22	10	0	12	38	0
Scale-up relevance	82	31	12	0	0	39	30
Relevance to proposed invention	82	34	28	0	0	20	10

Columns describing process detail — manufacturing temperature, 1-MCP loading, package volume, coat weight — are sparse **by construction**. Roughly a quarter of screened papers yielded open full text, and this information lives in methods sections, not abstracts. The sparsity is a measurement of what is openly publishable, not an artefact of effort.

The single highest-leverage way to densify this table is to supply PDFs for specific papers via the pipeline's `--pdf-drop` ingest, which re-extracts them at full-text depth without re-running anything else.

The table

Table 8: Literature summary. Every field for every paper appears in Appendix A.

Reference	Substrate	1-MCP carrier	Antimicrobial	Trigger	Loading	Release result	Produce
Zhaoxin Cao et al. (2024). Preparation of carvacrol β -cyclodextrin inclusions and 1-methy...	filter paper	α -cyclodextrin	carvacrol	humidity_moisture		92%, 88%, 73%	peach
Aaruba Maqbool et al. (2024). Studies on extending the shelf life of golden delicious app...	polymeric film	SmartFresh cards					Golden Delicious apples
Prisma Espinal-Hernández et al. (2021). Postharvest effects of 1-MCP and Chitosan/Oleic a...	chitosan	sachet	chitosan	liquid_water			Pitaya (Stenocereus griseus H.)
Soesiladi Esti Widodo (2015). Effects of 1-Methylcyclopropene and Chitosan on the Fruit S...	chitosan						'Cavendish' banana
Dorota Wichrowska et al. (2026). Antioxidant potential and storage stability of sweet che...		Vidre+ stickers			1, 2, 4 uL of 1-MCP per liter uL/L		sweet cherries (Prunus avium L.)
Xukai Hou et al. (2026). The impact of 1-MCP treatment combined with polyethylene film ba...	polyethylene film bag				1 μ L/L	inhibited ethylene production	'Shannong Su' pears
Dorota Wichrowska et al. (2025). Antioxidant Potential and Storage Stability of Sweet Che...		sticker			1, 2, 4 uL/L uL/L		sweet cherries (Prunus avium L.)
H D Ariyanto et al. (2019). Effect of 1-Methylcyclopropene Release from Inclusion Complex...		α -cyclodextrin	1-MCP		5, 15 and 30 mg mg		tomatoes
Cheng Zhao et al. (2026). Combined application of 1-methylcyclopropene and silicon-window...							'Saiwaihong' apples
Liyang Yang et al. (2026). Efficient Preservation of Perishable Fruits by Erasable Metal-...	carboxymethyl chitosan (CMCS)		clove essential oil (CEO)				emperor bananas
Hongsheng Zhou et al. (2025). Improving the Surface Color and Delaying Softening of Peach...					1.5 μ L·L -1		Honey peach
Laras Putri Wigati et al. (2025). Influence of cellulose nanocrystal formulations on the ...	pregelatinized corn-starch		cornmint essential oil				Apples
Zhibin Pan et al. (2024). Effects of oregano essential oil Pickering emulsion and ZnO nan...	konjac glucomannan/carboxymethyl chitosan		oregano essential oil				
Anna Yu. Pnevskaya et al. (2026). In Situ Diffuse Reflectance Infrared Fourier-Transform ...			1-methylcyclopropene (1-MCP)	temperature		>90% of 1-MCP capture efficiency by Co-FA at room temperature	banana fruits
Yuxuan Ban et al. (2026). Preparation of a Pickering emulsion film loaded with essential ...	chitosan/polyvinyl alcohol		chili pepper leaf essential oil (CPLEO)				fresh broccoli
Kyoungook Kim et al. (2025). Effects of sachet-type 1-methylcyclopropene (1-MCP) treatmen...				liquid_water			Hongro apples
Jia Kan et al. (2025). Interfacial Engineering of High-Performance Pickering Emulsion-Gel...	gelatin		oregano essential oil	passive_diffusion		The Korsmeyer-Peppas model showed higher correlation coefficients ($R^2 = 0.9592$ for 50% e...	
Qiyuan Feng et al. (2025). Tannic Acid-Modified Sodium Caseinate Pickering Emulsion Coati...	carboxymethyl tara gum		bergamot essential oil			relatively stable and slower than SC	cherry tomatoes

Reference	Substrate	1-MCP carrier	Antimicrobial	Trigger	Loading	Release result	Produce
Andreas Romulo et al. (2024). Enhancing Antimicrobial Activity of Thyme Essential Oil Thr...	sodium alginate		thyme essential oil (TEO)				
Peng Liu et al. (2015). Research on SO₂ Controlled Release Packagi...							Kyoho grapes
Liufeng Wang et al. (2026). Chitosan-Based Composite Films Reinforced with Zein-Inulin-Th...	chitosan		thyme essential oil				blueberries
Niloofar Shirdam et al. (2026). Synthesis, characterization, and evaluation of antimicrob...	chitosan, poly (vinyl alcohol), cellulose acetate		nisin, thyme essential oil				
Xiuxiu Teng et al. (2025). 4D printed deformation labels with machine learning for monito...	konjac glucomannan grafted with methacrylic anhydride		garlic essential oil	mechanical		cumulative release of 1 upper layer-30%-60° labels was 42.32% higher than that of 4-upper...	kiwi fruits, green mangoes, and persimmons
Yiping Liu et al. (2025). Application of a Pickering Emulsion Stabilized by Zein and Cell...	zein colloidal particles (ZCPs) and cellulose nanocrystals (CNCs) paperboard		carvacrol				kiwifruit
Mariangela de Fátima Silva et al. (2025). Chitosan-Lemongrass Essential Oil on Paperboard...			lemongrass essential oil				
Tianyu Jiang et al. (2025). Sustainable active packaging developed using starch-based foa...	starch-based foam sheets	α-Cyclodextrin	1-MCP			significant reduction in ethylene production and respiration rates by up to 60.49 % and 5...	
Limeng Chang et al. (2026). Humidity-responsive starch foam pads for humidity-triggered r...	starch foam	α-cyclodextrin		humidity_moisture	18.5 g/m ² and 27.8 g/m ² g/m ²	18.5 g/m ² exhibited a more regular and sustained release profile; 27.8 g/m ² showed incomp...	litchi
Yanyi Xue et al. (2026). Morphology-regulated Pickering emulsions stabilized by soybean r...	chitosan		lemon essential oil				blueberry
Wei Zhou et al. (2021). 3D printed nanocellulose-based label for fruit freshness keeping ...	cellulose nanofibers (CNF)	chitosan	1-methylcyclopropene (1-MCP)	other		sustained release behavior	litchis
Syariful Mubarak et al. (2022). Impact of 1-Methylcyclopropene combined with chitosan on ...							Lady Finger banana
J.H. Hotchkiss et al. (2007). Release of 1-Methylcyclopropene from Heat-Pressed Polymer F...		alpha-cyclodextrin	1-methylcyclopropene		8/140 mg/mg polymer	rapid and high	
Hermawan Dwi Ariyanto et al. (2026). Effectiveness of 1-Methylcyclopropene Coated Paper-B...	paper	alpha-cyclodextrin, inclusion complex		humidity_moisture		320 µL L –1	'Orin' Apples
Jiarui Tong et al. (2026). Trans-2-hexenal-loaded gelatin/dialdehyde starch/nanocellulose...	gelatin/dialdehyde starch		trans-2-hexenal			sustained E2H release behavior	Chinese bayberry
Jiayan Zhang et al. (2025). An active packaging film prepared from aqueous two-phase Pick...	chitosan						bananas and tomatoes
Han Zhang (2016). Optimization of encapsulation methods and conditions to maximize 1-MCP ...		modified β-cyclodextrin (MβCD)			0.535 %		
Suliman Khan et al. (2026). Carvacrol-enriched chitosan/berry wax active films via Picker...	chitosan/berry wax		carvacrol essential oil				rabbit meat

Reference	Substrate	1-MCP carrier	Antimicrobial	Trigger	Loading	Release result	Produce
Lin Zhu et al. (2026). Construction of Pickering Emulsion of Amomum tsaoko Essential Oil ...	pullulan		Amomum tsaoko essential oil				Mango
Chongfeng He et al. (2025). Construction of magnetic response nanocellulose particles to ...	cellulose nanocrystal		oregano essential oil	other		The OEO released rapidly within 0-8 h followed by the slow release when the emulsion was ...	
Mahmoud Younis et al. (2025). Improving Barhi date shelf life with cellulose nanocrystals...	sodium alginate		cinnamon essential oil			over 60 %	Barhi dates
Zou X et al. (2025). Incorporation of Schizochytrium limacinum oil-based pickering emulsi...	chitosan		Schizochytrium limacinum (SL) oil				strawberry
Özge Horzum & Randolph M. Beaudry (2025). "In-box" 1-methylcyclopropene (1-MCP) treatment...	corrugated cardboard		1-methylcyclopropene	humidity_moisture		The release of 1-MCP from IBP was found to be highly dependent on the relative humidity o...	'Red Delicious' apple
Chang Shu et al. (2023). Effect of Humidity-Triggered Controlled-Release 1-Methylcyclopro...				humidity_moisture		inhibited fruit softening, and reduced weight loss and peel color deterioration	Papaya (Carica papaya L.)
Custódia Gago et al. (2022). 1-Methylcyclopropene and lemongrass essential oil nanocoatin...	sodium alginate		lemongrass essential oil		312 nL L-1		'Rocha' pear
Esmat F. Ali et al. (2022). The Efficacies of 1-Methylcyclopropene and Chitosan Nanoparti...							damask rose
Sofia M. Costa et al. (2021). Active natural-based films for food packaging applications:...	chitosan						chicken meat
Weina Wu et al. (2021). Novel packaging film for humidity-controlled manipulating of ethy...	polysaccharide	palladium on carbon		humidity_moisture		controlled by adjusting the humidity significantly extended storage life	Agaricus bisporus
Liping Kou et al. (2012). Extending the Shelf Life of Edible Flowers with Controlled Rele...							carnations and snapdragons
Sobia Naseem et al. (2026). Chlorogenic acid crosslinked cellulose nanocomposite film emb...	chitosan		Chlorogenic acid				strawberries
Ahn S et al. (2026). Controlled release system inducing change in stability of volatile e...	polyvinyl alcohol (PVA)		lemongrass essential oil (LO)	humidity_moisture		decreasing to 9.98 ± 1.42 % at 90 % RH	
Ke Liu et al. (2026). L-cysteine modified carboxymethyl chitosan composite film: sustaine...	oxidized hyaluronic acid and thiol-modified carboxymethyl chitosan		cinnamon essential oil (CEO)	other		sustained release over 42 h	shrimp
Yayue Wang et al. (2026). Moisture-triggered one-step encapsulation of cinnamaldehyde in ...	β -cyclodextrin		cinnamaldehyde	humidity_moisture		CIN release was humidity-responsive with sustained kinetics.	blueberries and cherry tomatoes
Tingting Zhang et al. (2026). TEMPO-oxidized cellulose nanofiber/chitosan composite films...			cumin essential oil				tomatoes and blueberries
Xiaotong Zou et al. (2025). Development and characterization of corn starch-based films e...	corn starch		Zanthoxylum bungeanum essential oil (ZBEO)				cherry tomatoes
Li M et al. (2025). Performance of konjac glucomannan/tragacanth gum-stabilized eugenol p...	konjac glucomannan/chitosan		eugenol				blueberries
Xiaoxue Wang et al. (2025). Preparation, Characterization and Application of Antimicrobia...	pectin and konjac glucomannan		clove essential oil			CCPE could further slow down the release rate of CEO from composite films	grape
Xiaoxue Wang et al. (2025). Preparation, characterization and application of antimicrobia...	pectin and konjac glucomannan		clove essential oil				grape

Reference	Substrate	1-MCP carrier	Antimicrobial	Trigger	Loading	Release result	Produce
Anna Yu. Pnevskaya & Aram L. Bugaev (2023). Theoretical screening of M3(btc)2 metal-organ...		MOFs					
Putri Amanda et al. (2022). Carrageenan functional film integrated with Pickering Emulsio...	carrageenan		Oregano essential oil				
Alana Gabrieli de Souza et al. (2021). Cinnamon Essential Oil Nanocellulose-Based Pick-eri...	nanocellulose		cinnamon essential oil				
Gregor Lavrič et al. (2021). Functional Nanocellulose, Alginate and Chitosan Nanocomposit...	chitosan, alginate						
Tao Yang et al. (2026). Hydrocolloid-based stimuli-responsive nanofiber films: Precision...							
Zhijun Hu et al. (2017). 1-Methylcyclopropene (MCP)-Containing Cellulose Paper Packaging ...	cellulosic paper			humidity_moisture			
Anand Kumar et al. (2026). Chitosan-Based Active Packaging for Shrimp Preservation: Devel...	chitosan		thymol				
Bruna Moura Bastos et al. (2026). Lipid-Enriched Biopolymer Films for Active Packaging: A...							
Reshma Krishnan et al. (2025). Emerging Trends and Application of Edible Coating as a Sus...							
Siphumle Owen Jama et al. (2026). Edible Coatings for Fresh Fruits: Functional Roles, Opt...							
W. M. C. B. Wasala et al. (2021). Application of 1-Methylcyclopropene (1-MCP) for Delayin...							banana
Amr Zaitoon et al. (2021). Triggered and controlled release of active gaseous/volatile co...							
Ana Beltrán Sanahuja & Arantzazu Valdés García (2021). New Trends in the Use of Volatile ...							
Donghyun Jang et al. (2026). Biopolymer coatings for barrier and functional enhancement o...							
Dimitrie Stoica et al. (2026). Protein-Polysaccharide Gel Systems for Antioxidant and Ant...							
Amanullah Sabir et al. (2026). Polysaccharides: Nature's Guardians of Freshness in Food P...							
Weina Jiang et al. (2025). Multifunctional Metal-Organic Frameworks for Enhancing Food Sa...							
Aleksandra Bibow & Wiesław Oleszek (2024). Essential Oils as Potential Natural Antioxidan...							
Sharath Kumar Nagaraja et al. (2026). A Comprehensive Review of Biochemical Insights and ...							
Mariana A. Andrade et al. (2025). Emerging Trends in Active Packaging for Food: A Six-Yea...							

Reference	Substrate	1-MCP carrier	Antimicrobial	Trigger	Loading	Release result	Produce
Antia G. Pereira et al. (2026). Natural Antimicrobials from Plants Used as Food Preservat... Mo Chen et al. (2020). Stabilization and controlled release of gaseous/volatile active co... Abu-Bakr Ali Abu-Goukh (2023). Review Article on: 1-Methylcyclopropene (1-MCP) a Break-thr... Afreen Sultana et al. (2022). Metal-organic frameworks for active food packaging. A review... C.B. Watkins & W.B. Miller (2005). 1-METHYLCYCLOPROPENE (1-MCP) BASED TECHNOLOGIES FOR ST... Ziao Zhang et al. (2026). Bio-Based Smart Packaging Materials for Next-Generation Food Sy...							

The full 25-column record for every paper — including manufacturing temperature, RH conditions, release duration, kinetic model, package volume, mechanical and barrier properties — is in **Appendix A**. A 25-column table cannot be set on any page width, so it is re-laid-out rather than truncated.

Notes on interpretation

- **Publication type** was assigned by the screening model from title and abstract, then used to split originals from reviews. Crossref's own type field is unreliable for this — most reviews are typed journal-article — so the split should be spot-checked before being quoted as a count.
- **Reviews are included to map the field**, per the brief, and are **not** treated as evidence that an experimental result was demonstrated. Where a review is the only source for a claim, that claim is weaker than a table cell makes it look.
- **Contradictions are not reconciled**. Where sources disagree, both values stand with their quotes; the disagreement is information.

Existing commercial technologies

Independently verifiable facts (EPA registration documents, patent numbers, product labels) are separated below from **company marketing claims**, which are labelled as such throughout and are *not* treated as evidence that a performance claim is true.

Verified regulatory anchor

1-MCP is a **registered pesticide active in the United States**, not an unregulated packaging additive. This is the single most consequential fact for the project.

Fact	Source
1-MCP registered with EPA's Biopesticides and Pollution Prevention Division under the trade name SmartFresh , 17 July 2002, EPA Reg. No. 71297-2	USDA AMS 1-MCP petition; EPA pesticide product label 071297-00002
SmartFresh label amended as recently as 2023 (071297-00002-20230419)	EPA PPLS
SmartFresh InBox : powder in a sachet, EPA Reg. No. 71297-5 , sachet size 1.25 g , 1 000 sachets per package	AgroFresh product label
SmartFresh InBox label cites US Patents 5,518,988 and 6,017,849	AgroFresh product label
In California, SmartFresh InBox is approved for ki-wifruit only	AgroFresh CA label

The last row is the important one for planning. Registration is **crop-specific and format-specific**. A new delivery format is not covered by an existing registration simply because the active is the same.

Product landscape

Product	Owner	Format	Activation	Sealed space required?	Status
SmartFresh	AgroFresh	powder released into a sealed room	water added to powder	yes — sealed room	EPA-registered since 2002
SmartFresh InBox	AgroFresh	sachet in the box	humidity/water	no	EPA Reg. 71297-5
HarvestHold Fresh	Verdant Technologies	a sheet applied in the box at packing	in-package	no	introduced late 2021; EPA label expansion granted
Vidre+	Fresh Inset	stickers / labels	in-package	no	commercial; entering the US market
Hazel 100	Hazel Technologies	sachet	in-package	no	commercial; registered in Peru among others
AnsiP-G / EasyFresh	various	cyclodextrin-based powder, effervescent tablets, water-sensitive	water/humidity	varies	commercial

The two findings that matter most

1. The coated-insert format is already commercial. HarvestHold Fresh is described in its own materials and in trade press as a **sheet placed in the box as produce is packed**. That is, in commercial substance, the architecture Section 9 recommends. The recommendation remains correct as *engineering* — it is the cheapest, most scalable, most regulatorily tractable format, and that is precisely why an incumbent already sells it — but it must not be presented as a novel format.

2. Delayed and gradual release is already claimed. Fresh Inset states that Vidre+ is “*the first (and only) product in the world that releases 1-MCP with a delay and gradually, without the need to apply it in a tightly closed*” space [**company claim, unverified**], describing a “*gradual & timed release mechanism*” [**company claim**]. Whatever the merits of the superlative, the *concept* of delayed, gradual, in-package 1-MCP release is commercially occupied and, per the company, patented.

Taken with the patent landscape (Section 6) and Cao et al. (2024) (Section 2), this closes off several of the differentiators the brief nominated as possible white space:

Candidate differentiator	Status after commercial review
Paper/sheet insert format	Occupied — HarvestHold Fresh
Label / sticker format	Occupied — Vidre+
Sachet format	Occupied — SmartFresh InBox, Hazel 100
Delayed-onset release	Occupied (claimed) — Vidre+
Cyclodextrin carrier	Occupied — SmartFresh, AnsiP-G, EasyFresh all cyclodextrin-based
Humidity/water activation	Occupied — essentially every product above
Dual 1-MCP + essential oil on coated paper	Published — Cao et al. (2024)

What is *not* evident in the commercial literature

Across all of the above, the public materials describe **what the products do**, not **how reproducibly they dose a package**. Specifically, no publicly available source retrieved here provides, for any commercial product:

- a headspace 1-MCP concentration-time curve in $\mu\text{L}\cdot\text{L}^{-1}$;
- a stated humidity activation threshold;
- a fitted release model or lag time;
- a dosing rule relating insert area or active mass to **package volume and fruit mass**, with a stated reproducibility.

This is unsurprising — it is commercially sensitive and not required for a label — but it defines where a credible technical contribution could still sit. It is also a *weaker* white space than a format or mechanism gap: absence from public marketing material is not absence from the companies' internal knowledge or from their patents.

Cost anchor

The InBox label gives a concrete unit: a **1.25 g sachet**, sold **1 000 per package**. Any new insert must be costed against that unit — a per-box consumable at commodity scale, from a supplier with a two-decade-old registration and an established distribution channel.

Published pricing was not retrievable for any of these products, so the techno-economic model (Section 15) is built on transparent assumptions with the cost drivers exposed, rather than on a claimed benchmark price. **Any statement of competitor price would be a fabrication and none is made.**

Implication for the project

The realistic positioning is **not** "a new format" and **not** "delayed release". Both are taken. What remains defensible is narrower and quantitative: a **specified, measured dose-control relationship** — a defined coat weight and activation threshold delivering a reproducible headspace concentration across a stated range of package volumes and fruit masses — supported by GC data and a fitted model that no public source currently provides.

If that cannot be demonstrated to be both real and better than the incumbents, the honest conclusion is that this project should not proceed as a product play. Section 17 states the kill criteria in those terms.

Patent-family landscape

This is a technical prior-art map, not a legal analysis. No conclusion about freedom to operate, infringement, validity or patentability is expressed or implied, and none should be inferred. Those determinations require claim-by-claim review by qualified patent counsel, including a search of unexamined applications that this review could not access.

Retrieval and its limits

Patent discovery ran through site-restricted web search after Google Patents' search endpoint blocked this IP; full documents, including complete claim text and priority/family tables, were retrieved through an independent crawler. Metadata was extracted from those documents and grounded against them.

Three limits are carried into the table and should be read with it:

- **Families are clustered heuristically** on title, assignee and priority date. INPADOC family data was not available, so the family column reports the members **observed**, not an authoritative family definition.
- **Legal status is not reported.** Google Patents' own status field carries an explicit disclaimer that it is not a legal conclusion, and no independent source (EPO OPS, PatentsView) was reachable. A column of unverified statuses would invite exactly the misreading this section warns against.
- **Only published documents are visible.** Applications not yet published are, by definition, absent — and in a commercially active field, that is a material gap.

Prior-art map — the shape of the field

Retrieved families cluster into five groups:

A. 1-MCP/cyclodextrin complexes and their articles. The core, and the most consolidated. Patents originally prosecuted by Cellresin Technologies now list **Verdant Technologies LLC** as current assignee — the company that also sells HarvestHold Fresh. AgroFresh holds the foundational complex claim (US6017849A, 1998, cited on the SmartFresh InBox label) and film/coating claims. This group contains the closest art to the recommended architecture.

B. Release-control of volatiles from polymer matrices, fibres, films and sheets. Broader than 1-MCP but frequently reciting it, including Chinese families explicitly covering ripening-inhibitor release from polymers, fibres, films, sheets or packaging.

C. 1-MCP formulations and generators. Powders, liquids, tablets, generators — the delivery chemistry rather than the packaging article.

D. Essential-oil and antimicrobial active packaging, including cyclodextrin-encapsulated oils. Heavily crowded and largely independent of the 1-MCP art.

E. Ethylene management generally — absorbers, scavengers, modified atmosphere. Adjacent context rather than direct prior art.

The proposed concept sits in the **intersection of A and B**, which is the most densely occupied region of the map.

Claim-element matrix

Elements of the proposed concept against where each is already disclosed:

Element	Disclosure status	Where
1-MCP complexed with α -cyclodextrin	Extensively disclosed	Cellresin family; Chinese families; commercial products (SmartFresh, AnsiP-G, EasyFresh)
Cyclodextrin composition coated on a substrate	Extensively disclosed	US10212931B2; US20210331990A1
Second substrate over the coating (laminated)	Disclosed	US10212931B2 claim 1
Moisture/humidity as release trigger	Extensively disclosed	Across group A and B; intrinsic to α -CD complexes
Paper / sheet / insert format	Extensively disclosed + commercial	Group B; HarvestHold Fresh

Element	Disclosure status	Where
Label / sticker format	Disclosed + commercial	Vidre+
Delayed / differential / sequential release	Disclosed	US20210331990A1; commercially claimed by Vidre+
Essential oil + ethylene inhibitor together	Disclosed in a different form	Cao et al. (2024) in the literature; group D patents
CNC-stabilised Pickering emulsion packaging	Disclosed in a different material system	Group D; not combined with 1-MCP in retrieved art
Coated substrate + 1-MCP/ α -CD clathrate coating	Already claimed	US20210331990A1 (Verdant, 2020-04-27)
Film containing a cyclopropene/cyclodextrin complex	Already claimed	US8603524B2 (AgroFresh, 2007)
Coatings/films for vapour-phase 1-MCP release	Already claimed	PH12016500070A1 (AgroFresh, 2013)
1-MCP carrier with optional essential oils	Already claimed	US20190037839A1 (Hazel, 2016)
Specified RH activation threshold with a defined release window	Uncertain — requires further searching	Not found in retrieved art; absence is weak evidence
Defined coat weight ($\text{g}\cdot\text{m}^{-2}$) tied to delivered dose	Uncertain — requires further searching	Not found in retrieved art
Dose reproducibility across package volume / produce mass	Uncertain — requires further searching	Not found in retrieved art

Novelty-risk classification

Per the brief's requested categories:

Feature	Classification
1-MCP/CD on paper	Already extensively disclosed
Humidity trigger	Already extensively disclosed
Insert / label / sachet format	Already extensively disclosed — and commercial
Delayed-onset release	High patent-overlap risk
Two-stage / sequential release	High patent-overlap risk
Bilayer with a moisture-control topcoat	High patent-overlap risk — approaches the US10212931B2 laminate
Spatial separation of 1-MCP and essential oil	Disclosed but not well demonstrated
CNC Pickering + 1-MCP in one article	Disclosed only in a different material system
Low-temperature / water-free manufacture	Disclosed but not well demonstrated — Cao et al. use it; the quantitative retention relationship is not reported
Defined binder:cyclodextrin ratio	Uncertain — requires further searching
Defined coat weight tied to dose	Potentially differentiating
Specified RH threshold + release window	Potentially differentiating
Dose control across variable package volume	Potentially differentiating
Repulpable paper-based design	Disclosed but not well demonstrated

Design-around analysis (technical, not legal)

- **The laminate design-around is not sufficient on its own.** Avoiding the *second substrate* of US10212931B2 does nothing about US20210331990A1 (Verdant, 2020-04-27), which claims a coated substrate whose coating consists essentially of a 1-MCP clathrate of α -cyclodextrin — no second substrate required. Any design-around must clear that claim first.
- **Claim a specification and a method, not an article.** The article space is occupied. A method of achieving a specified headspace concentration by selecting coat weight and binder to a defined RH threshold is a different kind of claim.
- **Do not build the differentiator on delay.** It is commercially claimed and separately titled in a published application.
- **Keep the essential oil out of v1**, which also keeps the disclosure clear of the crowded group D art.

White-space analysis

The credible white space is not a material or a format. It is **quantitative specification**: threshold, coat weight, delivered dose, reproducibility.

Its weakness must be stated plainly: this is an *absence in retrieved, published* art. Incumbents plausibly hold this data internally, and unpublished applications were not searchable. A white space defined by what companies have not published is the weakest kind, and should be verified by a professional search before any resource is committed on the strength of it.

The table

Table 14: Patent families by priority date. Full records in Appendix B.

Publication	Priority	Assignee	Claim 1	Trigger	Carrier
EP1584234A2 PL346176A1 US6017849A	1998-08-20 1998-08-20 1998-08-20	Biotechnologies for Horticulture Inc; AgroFresh Inc	A complex formed from a molecular encapsulation agent and a cyclopropene derivative compo...	humidity_moisture	Cyclodextrin (specifically alpha-cyclodextrin)
KR100791251B1 US6426319B1 US6953540B2	2000-09-29 2000-09-29 2000-09-29	로템 앤드 하스 캄파니 / Rohm & Haas Company AgroFresh Inc (Current); Rohm and Haas Co (Original) AgroFresh Inc	A method of purifying a cyclopropene gas prepared by reaction of an allyl compound with a...		alpha-cyclodextrin
CN1325446C US8129450B2	2002-05-14 2002-12-10	Rohm and Haas Co Verdant Technologies LLC (Current); Cellresin Technologies LLC (Original)	Claim 1 requires an article comprising a thermoplastic polymer fiber made from an acid an...		alpha-cyclodextrin (alpha-cyclodextrin)
CR9490A US8603524B2	2006-11-09 2007-01-17		A composition comprising a cyclopropene molecular encapsulation agent complex and a calci...		cyclopropene molecular encapsulation agent complex
CN101416658B CN101578978A	2008-11-07 2009-06-11	AgroFresh Inc China Agricultural University	A film comprising an ethylene blocking agent complex (cyclopropene/derivatives + cyclodex...	humidity_moisture	cyclodextrins or crown ethers modified stabilizer
CN101715760A	2009-12-01	Shandong Nutrient Source Food Technology Co Ltd	An ethylene absorbent prepared from a carrier substance, a metal promoter (1-5% of carrie...		1-methyl cyclopropene stable solution (alcohol/aldehyde mixture)
CN104304242B MX2012006797A US9642356B2	2009-12-14 2009-12-14 2009-12-14	Cellresin Technologies, Llc Cellresin Tech Llc Verdant Technologies LLC	A liquid preparation comprising a 1-methylcyclopropene stable solution as host, mixed wit...		
CN101990936A US8481127B2	2010-10-19 2011-03-27	Xian Jiaotong University Verdant Technologies LLC	A controlled release composition comprising a cyclodextrin modified thermoplastic polymer...	humidity_moisture humidity_moisture	cyclodextrin
CN102440237B US9381163B2	2011-09-30 2012-08-16		A cyclodextrin composition comprising one or more radiation polymerizable monomers, a pho...		cyclodextrin copper complex cyclodextrin inclusion complex (specifically α -cyclodextrin)
CN102823644A US20150272148A1	2012-09-04 2012-10-10	NANYANG TECHNOLOGICAL UNIVERSITY, SINGAPORE THE UNIVERSITY OF QUEENSLAND, AUSTRALIA		humidity_moisture	α -cyclodextrin (α -CD)
AU2013387471A1 CA2910267A1 US20170101250A1	2013-04-26 2013-04-26 2013-05-05				
PH12016500070A1	2013-07-11	Mirtech Inc Agrofresh Inc	Delivery systems in the form of coatings or films that provide controlled release of vola...	humidity_moisture	cyclodextrin or a cyclodextrin derivative

Publication	Priority	Assignee	Claim 1	Trigger	Carrier
US20160330952A1	2014-01-15	University of Guelph	A method for encapsulating a VOC comprising preparing a stabilized system of biopolymer, ...		
EP3161558B1	2014-06-26	VERDANT TECHNOLOGIES, LLC			cyclodextrin
US9332751B2	2014-07-09	Sabancı Universitesi	Claim 1 requires a method for producing a polymeric film for food packaging by incorporat...		halloysite nanotubes (HNTs)
US10449156B2	2014-07-23	UNIVERSIDAD ANDRES BELLO, CHILE			
US10351490B2	2014-11-10	AgroFresh Inc		humidity_moisture	metal coordination polymer network (MCPN)
US20190037839A1	2016-02-19	Hazel Technologies Inc	A composition comprising a carbon or silicate delivery material and a volatile/gaseous ac...	passive_diffusion	carbon materials or silicate materials
US20170318804A1	2016-05-04	AgroFresh Inc	An aqueous-in-oil emulsion composition comprising an oil phase (25-70% by weight) and an ...	humidity_moisture	
US20190166861A1	2016-12-05	Pace International LLC	Deploys a portable treatment system with a compound generator and sampling subsystem, ope...		
CN112218529A	2018-04-27				alpha-cyclodextrin
KR20210030899A	2018-04-27	프레쉬 인셋 에스.에이.			α -cyclodextrin complex
US20200296978A1	2019-03-15	NANOTEC SA	A stable nanobubble comprising an ethylene inhibitor and at least one food grade sugar.		food-grade sugar-stabilized nanobubbles in aqueous solution
US11033870B1	2019-11-16	Individual			α -cyclodextrin, metal organic framework, zeolites, activated carbons, cucurbit[6]uril, an...
US20210235692A1	2020-02-03	Fresh Inset SA	A composition comprising 1-methylcyclopropene and a stabilizer, wherein the concentration...		
US20210331990A1	2020-04-27	Verdant Technologies LLC	A coated substrate comprising a coating disposed on a substrate surface, the coating cons...		

All 19 fields for every family — jurisdictions, family members observed, claim limitations, overlap, design-around and remaining uncertainty — are in **Appendix B**.

Scientific compatibility assessment

The brief asks a specific question: should 1-MCP and an essential oil sit in the same layer, in separate layers, or should the oil be omitted at first? And does the CNC-stabilised Pickering emulsion add genuine functional value or just complexity?

The governing physical conflict

Everything else follows from one asymmetry:

The 1-MCP/ α -cyclodextrin complex is discharged by water. That is its release mechanism — it is not a side effect to be engineered around, it is the product's operating principle. Cao et al. (2024) state it directly from their own measurements:

"1-MCP- α -CD is sensitive to high moisture, which may result in the complete release of 1-MCP gas in a short period of time."

Therefore every processing step that introduces water is a step that partially discharges the product before it ships. This single fact disqualifies most of the proposed starting formulation.

Proposed element	Interaction with that fact
Aqueous film casting	Immerses the complex in bulk water — the maximal-release condition
Chitosan matrix	Hygroscopic; draws in the very water that triggers release, in storage as well as in use
Chitosan processing	Requires acidic aqueous solution (typically acetic acid) — water plus a reactive environment
CNC-stabilised Pickering emulsion	An oil-in-water emulsion: its continuous phase is water
Drying to remove that water	Heat, applied to a thermally labile complex, for a prolonged period

The Pickering emulsion is the sharpest case. Its whole function is to stabilise an oil phase *in water*. Adding it to a formulation whose central engineering problem is excluding water is working directly against the design intent.

Does the Pickering emulsion earn its place?

Assessed on its merits rather than dismissed:

Genuine benefits, which are real: it stabilises a volatile oil without surfactants, reduces oil coalescence, can slow essential-oil release, and CNC can improve film mechanics and tortuosity.

Costs in this specific application:

- It requires an aqueous continuous phase (Section 7.1).
- It adds two materials (CNC, oil) and at least two process steps (emulsification, then stabilisation) before coating.
- It requires its own stability programme — droplet size, zeta potential, creaming, coalescence on storage — none of which relates to 1-MCP performance.
- CNC at commercial scale is expensive relative to the alternatives, and this application does not need its distinctive properties.
- It complicates drying, regulatory review, and the patent position (CNC Pickering active-packaging films are themselves a crowded area).

Verdict: not justified in v1. The emulsion solves a problem — delivering an essential oil in a stable aqueous coating — that the product does not have if the essential oil is deferred. It is a well-engineered answer to the wrong question. It should be reconsidered only if (a) the antimicrobial function proves necessary, **and** (b) a water-based coating proves acceptable for 1-MCP retention, **and** (c) simpler oil-delivery routes (oil/ β -CD inclusion powder, as Cao et al. used) prove inadequate.

Note that Cao et al. achieved the dual-active coated paper **without** a Pickering emulsion, by using a separate β -CD inclusion for carvacrol. That is the simpler precedent, and it works.

Same layer or separate layers?

If an essential oil is used at all, four mechanisms argue against co-locating it with the 1-MCP complex:

1. **Cavity competition.** Cyclodextrin inclusion is competitive. Thymol, carvacrol and eugenol are all well-documented cyclodextrin guests. A free oil molecule adjacent to a loaded 1-MCP/ α -CD complex is a plausible displacement risk. α -CD's small cavity affords partial protection — it fits 1-MCP well and bulky phenolics poorly — but the risk is real enough not to design into v1 without data. *This is a mechanistic argument; no retrieved study measured direct displacement of 1-MCP by an essential oil, which is itself a gap.*
2. **Partitioning into the oil phase.** 1-MCP is a small non-polar hydrocarbon. A lipophilic oil domain in the same layer is a sink it can dissolve into, attenuating headspace delivery — the quantity the product is sold on.
3. **Divergent release requirements.** Antimicrobial action wants early, sustained vapour output. Ethylene inhibition wants a controlled dose delivered inside a defined window. Optimising one layer for both compromises both.
4. **Plasticisation.** Essential oils commonly increase film free volume and water-vapour permeability. In this system that directly accelerates 1-MCP release — a coupled failure where the antimicrobial degrades the primary function.

If both actives are required: separate layers, oil on the outside, 1-MCP underneath. That keeps the oil away from the complex and puts the antimicrobial nearer the headspace. This is Architecture B, and it carries its own patent concern (Section 9.4).

Is chitosan's antimicrobial contribution sufficient?

Chitosan is genuinely antimicrobial, but by **contact**, requiring protonation at acidic pH. In a **non-contact insert** there is no contact and no wet acidic interface, so the mechanism cannot operate. Chitosan therefore contributes essentially nothing antimicrobial in the recommended architecture, while contributing hygroscopicity that actively harms it.

This is a clean argument for omitting chitosan — not on cost, but on mechanism.

Does the essential oil justify its complexity at all?

Costs: sensory taint risk on peach and apple at effective vapour doses; a second regulatory conversation; oxidation and volatility during drying and storage; batch-to-batch variability in a natural product; accelerated 1-MCP release via plasticisation; and additional patent crowding.

Benefit: decay control, which for stone fruit (brown rot, *Monilinia*) is commercially significant — Cao et al. demonstrated activity against *Monilinia fructicola*.

Recommendation: omit in v1, keep as a planned v2 module. The brief explicitly asks that this be tested rather than assumed, and the evidence supports omission: the antimicrobial does not address the project's central unsolved problem (reproducible 1-MCP dose control), while adding the sensory and regulatory risks most likely to kill a produce product. Establish the 1-MCP system, then add the oil as a separate layer with its own evidence — including a sensory panel before any scale-up.

Summary of material decisions

Material	v1	Reason
1-MCP/ α -CD complex	Keep	The active; α -CD's small cavity is also the best protection against guest competition
Paper / paperboard	Keep	Cheap, repulpable, coatable on existing lines
Binder (PEG, shellac, or latex)	Keep	The actual release-control lever
Chitosan	Drop	Hygroscopic; acidic processing; antimicrobial mechanism inoperative in a non-contact insert
CNC	Drop	Solves a problem the simplified design does not have; expensive at scale

Material	v1	Reason
Pickering emulsion	Drop	Requires the aqueous phase the design exists to avoid
Essential oil	Defer to v2	Real benefit, but wrong problem first, and carries the highest sensory/regulatory risk

Seven materials become three. Every removal is justified by a mechanism, not by preference.

Antimicrobial component selection

Section 7 recommends deferring the essential oil to v2. This section answers the brief's question of *which* one, so that the v2 decision is pre-made rather than reopened.

Comparison

Criterion	Thymol	Carvacrol	Eugenol	Cinnamon EO / cin-namaldehyde	Oregano EO	Citral	Hexanal
Chemical status	pure compound	pure compound	pure compound	mixture	mixture	pure compound	pure compound
Vapour-phase activity	strong	strong	moderate	strong	strong	moderate	moderate
Typical produce targets	<i>Botrytis</i> , <i>Penicillium</i> , <i>Rhizopus</i>	<i>Monilinia</i> , <i>Botrytis</i>	broad, weaker	broad, strong	broad	moulds, some bacteria	<i>Botrytis</i> ; also a ripening-related volatile
Odour threshold / sensory risk	high risk — thyme note	high risk — oregano note	moderate — clove note	very high risk — unmitigable	very high risk	moderate — citrus, better matched to fruit	lower — "green" note, native to fruit
Batch variability	low (pure)	low (pure)	low (pure)	high (natural mixture)	high	low	low
Cyclodextrin inclusion	well documented	well documented	well documented	documented	mixture — inconsistent	documented	less documented
Effect on film properties	plasticising; raises WVP	plasticising; raises WVP	plasticising	plasticising	plasticising	plasticising	volatile; hard to retain
Interaction risk with 1-MCP	CD guest competition; oil-phase partitioning	same	same	same	same	same	same
Patent crowding	heavy	heavy	heavy	heavy	heavy	moderate	moderate
Cost	low-moderate	moderate	low	low	moderate	moderate	low

Recommendation

If an antimicrobial is added, use carvacrol — and prefer a β -cyclodextrin inclusion powder over a Pickering emulsion.

Reasons, in order:

1. **Direct comparability.** Cao et al. (2024) used carvacrol/ β -CD alongside 1-MCP/ α -CD on coated paper and demonstrated activity against *Monilinia fructicola*. Using the same active means the closest prior art becomes a benchmark to measure against rather than an unknown.
2. **Pure compound, not a mixture.** Cinnamon and oregano essential oils are natural mixtures with batch-to-batch variability that would sit directly in the dose-reproducibility claim this project is built on. A product whose differentiator is *reproducible dosing* should not carry a variable-composition ingredient.
3. **Right target.** Carvacrol's documented activity against *Monilinia* matches the secondary commodity (peach, brown rot).
4. **Two cyclodextrins, two cavities.** Keeping carvacrol in β -CD and 1-MCP in α -CD exploits the cavity-size difference and reduces — without eliminating — the guest-competition risk of Section 7.3.

Thymol is a close second and is the brief's default; it is chemically similar to carvacrol with comparable activity. The tiebreak is the direct precedent, not a performance difference.

Hexanal deserves a mention the brief invites: it is a naturally occurring fruit volatile with a lower sensory penalty, and it is the most interesting option if sensory testing kills the phenolics. It is less developed as a packaging active, which is both its risk and its opportunity.

Avoid cinnamon and oregano essential oils for this product, despite strong antimicrobial data: variable composition plus the highest sensory risk is the worst combination for a fruit application.

The dose-versus-sensory question decides this

Every one of these actives works *in vitro* at concentrations that may be unacceptable on fruit. The binding constraint is not MIC, it is **the gap between the minimum inhibitory vapour concentration in a real package and the sensory detection threshold on that commodity**.

That gap is the single number that determines whether the antimicrobial arm is viable, and it is commodity-specific. It should be measured early in any v2 work — before formulation optimisation, not after — because a negative result there ends the antimicrobial arm regardless of how good the formulation is (FMEA FM10, FM16).

No MIC values are quoted here. MICs are highly method-dependent (vapour-phase versus contact, medium, inoculum, headspace volume), and quoting numbers from abstracts without verified full-text methods would be exactly the fabrication this review is built to avoid. Any MIC used for design must come from a vapour-phase study whose method has been read in full.

Architecture comparison

Six architectures were scored 1–5 against 20 criteria. Scores and the full weighting rationale are in `assets/architecture_matrix.csv`; the scoring script is reproducible from that file.

Weighting assumptions

Weights are 1–3 and reflect a single principle stated in the brief: **prioritise a working, inexpensive, scalable design over novelty or material complexity.**

Weight 3 (project-critical) — scientific feasibility; patent-overlap risk; 1-MCP retention through manufacture; trigger reliability; dose controllability; shelf stability before activation; regulatory difficulty; roll-to-roll compatibility; industrial scalability.

These are the criteria on which the project *fails outright* rather than performs poorly. 1-MCP retention through manufacture is weighted 3 because the payload is a gas held in a moisture-labile complex: any aqueous or heated step risks discharging it before the product ships. Patent-overlap risk is weighted 3 because Cellresin/Kimberly-Clark hold coated-substrate and laminate claims in precisely this space.

Weight 2 (materially important) — novelty potential; ingredient count; step count; food-contact safety; sensory risk; raw-material cost; equipment availability; recyclability; time to prototype.

Novelty is deliberately weight 2, not 3. The brief is explicit that an architecture should not be recommended because it sounds innovative.

Weight 1 (secondary) — antimicrobial performance; mechanical performance. The essential oil is optional in the brief, and the product is an insert, not a load-bearing package.

Criteria phrased as risks are scored so that **5 is always good** (e.g. "patent-overlap risk, 5 = low risk"), so higher totals are uniformly better.

Results

Architecture	Weighted	/100	Unweighted /100
C — paper / paperboard insert	206 / 235	87.7	87.0
D — label, patch or patterned coating	181 / 235	77.0	76.0
E — coated absorbent pad	172 / 235	73.2	72.0
B — bilayer coating	148 / 235	63.0	63.0
F — compartmentalised dual-active insert	125 / 235	53.2	54.0
A — single homogeneous chitosan/CNC film	102 / 235	43.4	45.0

The ranking is unchanged when all weights are set to 1. The conclusion is therefore not an artefact of the weighting scheme — a sensitivity check worth stating explicitly, because weighted scoring otherwise invites the suspicion that the weights were chosen to produce the answer.

Why the proposed starting materials score worst

Architecture A — a chitosan/CNC film carrying both a 1-MCP/cyclodextrin complex and a CNC-stabilised essential-oil Pickering emulsion — is the combination the brief starts from, and it finishes last. The reasons are specific, not aesthetic:

- **1-MCP retention through manufacture scores 1.** Aqueous casting exposes a moisture-labile inclusion complex to bulk water, which is the one condition known to discharge it. Cao et al. (2024) chose a *water-free* route precisely to avoid this, and they state the reason plainly: 1-MCP- α -CD is "sensitive to high moisture, which may result in the complete release of 1-MCP gas in a short period of time."
- **Shelf stability before activation scores 1** for the same reason: chitosan is hygroscopic, so the matrix that carries the complex also draws the water that triggers it.

- **Ingredient and step counts are the worst of the six.** Chitosan, CNC, acid, plasticiser, cyclodextrin complex, essential oil, and a separately prepared Pickering emulsion — each an input to source, qualify, cost and defend.
- **Novelty potential scores only 2.** Adding materials to an already-published concept does not create protectable novelty; it creates a more complicated version of the prior art.

Architecture F (compartmentalised dual-active) scores second-worst for related reasons: it is the most complex to build, the slowest to prototype, and its headline feature — sequential or differential release — collides directly with US20210331990A1, a published application titled for differential release of 1-MCP.

Why C wins

Architecture C — a low-cost paper or paperboard insert carrying a 1-MCP coating, with the antimicrobial deferred — wins because it scores well on every criterion weighted 3:

- **Retention (5):** a non-aqueous or low-moisture coating on paper, dried at ambient or mildly warm temperature, never immerses the complex in bulk water.
- **Roll-to-roll compatibility (5) and scalability (5):** paper coating is a mature converting operation. No new capital equipment is implied.
- **Regulatory difficulty (4):** an insert need not contact the fruit, which materially simplifies the food-contact position relative to a direct coating.
- **Recyclability (5):** a paper-based, repulpable insert is a real and defensible sustainability claim rather than a marketing one.
- **Cost (5) and time to prototype (5):** fewest materials, fewest steps.

Its weakest score is patent-overlap risk (3), and that is the honest concern: US10212931B2 claims a cyclodextrin composition coated on a substrate with a second substrate over it. A single-layer coated insert is not that laminate, but the neighbourhood is occupied and the distinction must be drawn deliberately rather than assumed.

D (label/patch) is close behind and scores *higher* than C on dose controllability (5 vs 4), because a fixed-area patch meters dose per package directly. It loses on patent crowding among slow-release label technologies and on adhesive compatibility with a moisture-triggered system.

What the matrix does not settle

Scores are informed judgements against retrieved evidence, not measurements. Three in particular should be revisited once data exists:

- **Trigger reliability** for E (coated pad) is scored 5 on the reasoning that a pad already sits in the condensate path. If pads prove to over-wet locally and dump the payload, that score is wrong and E collapses.
- **Patent-overlap risk** for every architecture is a technical impression, not a legal opinion.
- **Dose controllability** distinguishes C from D by one point, and that single point is the difference between the primary and backup recommendation. It is the first thing an experiment should test.

Recommended architecture

Primary recommendation — Architecture C-1: single-layer 1-MCP coated paper insert

Format. A rectangular coated-paper insert, placed in the package but not required to contact the fruit, sized to the package by area.

Layer structure.

Layer	Function
Protective overwrap (removed at packing)	Keeps the insert below its activation humidity until use. This is a component of the product, not packaging — the shelf-stability problem is otherwise unsolved.
Coating (one layer, one side)	Binder + 1-MCP/ α -cyclodextrin complex. The binder sets how fast water reaches the complex; the complex holds the payload.
Paper / paperboard base	Structure, repulpability, coat-weight control.

Deliberate omissions, and why each is a decision rather than an oversight:

- **No essential oil in v1.** Section 12 sets out the case; briefly, it adds a sensory risk on peach and apple, a second regulatory conversation, a possible competition for cyclodextrin cavities, and — critically — it does not fix the problem the product exists to solve. Establish 1-MCP dose control first. The brief explicitly asks that this option be tested rather than assumed.
- **No chitosan.** It is hygroscopic, which is exactly wrong for a payload discharged by moisture, and it requires acidic processing. Its antimicrobial contribution is not needed in a non-contact insert.
- **No CNC and no Pickering emulsion.** With no oil phase to stabilise, a Pickering emulsion has nothing to do. CNC's tortuosity benefit is real but is a solution to a problem (barrier control) that coat weight and binder choice address more cheaply.
- **No bilayer in v1.** A moisture-control topcoat is the first upgrade if — and only if — single-layer release proves too fast. It is Architecture B, held in reserve.

This reduces the brief's seven starting materials to **three**: paper, binder, 1-MCP/ α -CD complex.

Composition — proposed starting ranges

These are experimental starting points, not literature-established values. They are hypotheses for a screening design, and are labelled as such. Where the literature does anchor a number, it is marked.

Parameter	Proposed starting range	Basis
Base paper grammage	80–200 g·m ⁻²	Proposed. Spans label stock to light board.
Coat weight (dry)	4–20 g·m ⁻²	Proposed. No retrieved source reports coat weight for a 1-MCP paper — this is itself a gap worth owning.
1-MCP/ α -CD complex in dry coating	10–40 % w/w	Proposed.
Binder in dry coating	60–90 % w/w	Proposed.
Nominal 1-MCP payload	0.5–5 mg per insert	Proposed, back-calculated from the dose logic below.
Drying temperature	ambient–50 °C	Constrained: the payload is heat- and moisture-labile. Cao et al. (2024) adopted a water-free route for this reason.

Binder candidates, in the order they should be screened:

1. **PEG** — a literature-anchored choice: Cao et al. (2024) used polyethylene glycol specifically to moderate 1-MCP release from α -CD. Starting with the published solution is the fastest route to a working baseline.

2. **Shellac or an ethanolic system** — enables a genuinely water-free coating, protecting the payload during manufacture.
3. **Water-based latex / HPMC** — cheapest and most industrially standard, but introduces the water the design is trying to avoid. Screen it, but expect it to be the retention risk.

Dose logic. 1-MCP is dosed to headspace concentration, not to insert mass:

“ required 1-MCP mass (μg) = $C_{\text{target}} (\mu\text{L}\cdot\text{L}^{-1}) \times V_{\text{headspace}} (\text{L}) \times \rho \times 1/f_{\text{release}}$ “

where ρ converts μL to μg for 1-MCP (MW 54.09) at the storage temperature, and f_{release} is the measured fraction released within the treatment window. The whole point of the release-kinetics programme (Section 11) is to measure f_{release} so this equation stops being a guess. Commodity-specific target concentrations C_{target} come from the postharvest literature, not from this formula.

Trigger and diffusion pathway

Water vapour from fruit respiration raises package RH; vapour permeates the binder; the α -CD complex hydrates; the lattice releases 1-MCP; the gas diffuses back out through the coating into the headspace.

Release rate is therefore controlled by the binder’s water-vapour permeability and the coat weight — not by the cyclodextrin. That is the design lever, and it is the one this project can actually specify. It is also why coat weight is worth measuring precisely: it is a manufacturing parameter that maps directly onto a release parameter, and no retrieved source reports it.

9.3a The format itself is not novel — and that is fine

Commercial review (Section 5) establishes that **Verdant Technologies’ HarvestHold Fresh is a 1-MCP sheet applied inside the box at packing**, sold since late 2021 with an EPA label expansion. In commercial substance that is this architecture.

This does not invalidate the recommendation — it corroborates it. The format wins the matrix because it is the cheapest, most scalable and most regulatorily tractable option, which is exactly why an incumbent already sells it. But the format must be presented as **the right engineering choice, not as an invention**. Any disclosure that describes “a 1-MCP-coated paper insert” as novel will meet HarvestHold immediately.

The consequence is that the differentiator has to move down a level of abstraction — from *what the product is* to *what it is specified to do*.

Proposed technical differentiator

*A coated-paper 1-MCP insert with a **specified humidity activation threshold and release window**, achieved by binder selection and a defined coat weight, delivering a **reproducible headspace 1-MCP concentration** across a stated range of package volumes and fruit masses — demonstrated by headspace GC quantification in $\mu\text{L}\cdot\text{L}^{-1}$ and a fitted release model.*

This is deliberately a claim about **dose control and measurement**, not about materials or format, because both are taken (Sections 2 and 5). It is testable, it is what a commercial user actually needs, and it is the gap visible in the closest prior art — Cao et al. (2024) report no kinetic model, no GC quantification of 1-MCP, no coat weight, and no defined activation threshold.

Honest assessment of its strength. This is a *weak-to-moderate* differentiator, and it should be described that way to a technology-transfer office rather than oversold. Its weaknesses:

- Delayed and gradual release is already claimed commercially by Fresh Inset (Vidre+), so “controlled release” alone is not available.
- The gap identified is an absence in *public* material. Incumbents may hold the same data internally and in unpublished or unexamined applications.
- A specification range can often be designed around by choosing a different binder or threshold.

Its strength is that it is the one thing in this space that is **demonstrable with a modest lab programme** and that a buyer would actually pay for: reproducible dose across real, variable packages. If the data show sharp threshold behaviour and low replicate variance where incumbents show neither, that is a genuine result — and if they do not, that is a clear signal to stop.

Principal patent-overlap concern — and it is severe.

US20210331990A1 (Verdant Technologies, priority 2020-04-27) claims "A coated substrate comprising a coating disposed on a substrate surface, the coating consisting essentially of a 1-methylcyclopropene clathrate of α -cyclodextrin."

That is this architecture. Substrate, coating, 1-MCP/ α -CD — with **no second substrate required**, so the "we are a single layer, not a laminate" distinction that would separate this design from Kimberly-Clark's US10212931B2 does not help here.

The position is worse than a single document suggests. AgroFresh's US8603524B2 (2007) claims a film containing a cyclopropene/cyclodextrin complex in a water-soluble polymer; PH12016500070A1 (2013) claims coatings and films for vapour-phase 1-MCP release; and the Cellresin estate now sits with Verdant, the company selling the commercial 1-MCP sheet.

This must be put to patent counsel before any formulation work begins, not after. The honest reading is that the *article* — a paper coated with 1-MCP/ α -CD — is not available as an invention, and quite possibly not available to practise without a licence. What may remain is a **method and specification** claim (Section 9.4), and even that needs professional clearance. A project that spends a year optimising coat weight before asking this question is spending it badly.

Backup — simplest thing that could work

Architecture C·0: 1-MCP/ α -CD complex adsorbed onto uncoated absorbent paper, no binder, protective sachet until use.

Two materials. It exists to answer one question quickly: does a paper-borne complex release measurable 1-MCP into a package headspace at realistic RH, at all? If C·0 releases everything in the first hour, that is the strongest possible evidence that binder control is the whole invention — which is useful, and cheap to learn.

High-risk / high-reward — Architecture E·1

A condensate-managing coated pad that releases 1-MCP only when local moisture exceeds a threshold, marrying an existing commodity product (absorbent produce pads) with the active.

The reward is a genuinely differentiated function: the pad *manages* the condensation that would otherwise cause both decay and premature release, and uses it as the trigger. The risk is that the same wetness that makes the trigger reliable makes it uncontrollable — local saturation could dump the payload in minutes. It scored 4/5 on trigger reliability but that score is the least evidence-backed in the matrix (Section 8.5), and it should not be pursued until C has produced a release curve to compare against.

Preparation method and manufacturing feasibility

The constraint that selects the process

The payload is discharged by water and degraded by heat. So process selection is dominated by one question: **how much water and heat does the 1-MCP complex see, and for how long?**

Cao et al. (2024) reached the same conclusion independently and built their coated paper by a **water-free method** for exactly this reason. That is a literature-anchored precedent for a low-moisture route, not a preference.

Method comparison

Method	Max temp	Water/solvent exposure	Expected 1-MCP loss	Line speed	Uniformity	Capital	Industrial analogue	Verdict
Solvent casting (lab)	low	high, prolonged	high	n/a	poor	trivial	none	Reject as a commercial route
Knife/blade coating	low	medium	medium	low-med	medium	low	yes	Lab only
Meyer-rod coating	low	medium	medium	low	good, metered	low	yes → slot-die	Recommended lab method
Spray coating	low	high (atomised)	high	med	medium	medium	yes	Poor: maximises water contact area
Dip coating	low	very high	very high	low	poor	low	limited	Reject
Flexographic printing	low	medium	medium	high	good	high	yes	Viable at scale; good for patterned coatings
Gravure coating	low	medium	medium	high	excellent	high	yes	Viable at scale
Slot-die coating	low	medium	medium	high	excellent, precise coat weight	high	yes	Recommended scale-up route
Extrusion coating	high (>200 °C)	none	near-total	high	good	high	yes	Reject: thermal destruction
Lamination	medium	none	low	high	good	high	yes	Only if a bi-layer proves necessary — and see the patent concern
Adhesive transfer	low	low	low	high	good	medium	yes	Interesting for label formats

Recommended path

Laboratory: Meyer-rod (wire-wound bar) coating.

Chosen specifically because it has a **direct industrial analogue**. A Meyer rod meters coat weight by wire diameter, so lab work is expressed in the same units ($\text{g}\cdot\text{m}^{-2}$) that a production slot-die is set to. The brief warns against selecting solvent casting as the final commercial method; the deeper problem with casting is that it produces data in units nobody can scale — a cast film's thickness is not a coat-weight specification.

Pilot and production: slot-die coating, with gravure or flexo as alternatives where patterned coverage is wanted.

Drying: ambient to mildly warm, forced air. Minimise both peak temperature and residence time. Treat drying as a controlled variable and measure retained 1-MCP as a function of it — that relationship is itself a candidate result worth protecting (Section 17).

Binder screen (the actual experiment)

Three binders, in ascending order of water exposure:

Binder	Carrier	1-MCP retention risk	Cost	Note
Shellac / ethanolic	ethanol	lowest	medium	Genuinely water-free; requires solvent handling and recovery
PEG	melt or minimal water	low	low	Literature-anchored: used by Cao et al. specifically to moderate 1-MCP release
Water-based latex / HPMC	water	highest	lowest	Cheapest and most standard; the retention risk the design is trying to avoid

Run all three. The expected result — that retention rises as water falls — is worth *quantifying*, because a measured retention-versus-water-content relationship is more useful (and more patentable) than a formulation that merely works.

Worker exposure and solvent recovery

An ethanolic route needs solvent recovery and explosion-proof drying at scale; this is routine in converting but is a real capital consideration. The complex must be handled as a pesticide product throughout (Section 14). Dust control matters when handling dry complex powder.

Scale-up gate

The transition that must be demonstrated in Phase 5 is **lab Meyer-rod → pilot slot-die at the same coat weight**, with release curves compared directly. If pilot-coated material does not reproduce lab release behaviour, the formulation is not manufacturable and the project stops there (FMEA FM17). Because both methods are specified in $\text{g}\cdot\text{m}^{-2}$, that comparison is meaningful — which is the whole reason for choosing Meyer rod over casting at the start.

Release-kinetics protocol

The differentiator proposed in Section 9 is a measurement claim, so this protocol is the invention’s evidence base rather than a supporting detail. It is also the clearest gap in the closest prior art: Cao et al. (2024) followed release gravimetrically and report no kinetic model and no headspace 1-MCP concentration.

Analytical method

Detector: GC-FID. 1-MCP (C₄H₆, MW 54.09) is a small hydrocarbon; FID is sensitive, linear over several decades, cheap to run, and is what the postharvest literature uses. GC-MS is worth one confirmatory run to prove peak identity against the α -CD blank, then FID for throughput.

Calibration is the hard part and must not be improvised. 1-MCP is a gas supplied as a complex; there is no bottled liquid standard. Generate standards by releasing a *weighed* mass of complex into a sealed vessel of known volume with excess water, assume complete release (verify by exhaustive re-extraction of the spent complex), and construct a multi-point curve by dilution. Every calibration must carry its own mass balance. Report LOD and LOQ, determined from the calibration curve, not assumed.

Sampling losses are a first-order error, not a detail. Each headspace draw removes gas from a closed system. Either (a) use individual sacrificial vials — one vial per timepoint, destructively sampled, which eliminates the problem entirely and is strongly preferred — or (b) correct explicitly for cumulative withdrawal and state the correction. Repeated sampling of one vial without correction produces a decay curve that is partly an artefact of sampling.

Test system

- Sealed glass vials or jars of accurately known volume; gas-tight septa.
- Insert area scaled to headspace volume so that area-to-volume ratio is a controlled variable, not an accident of vial size. Report it explicitly (cm²·L⁻¹) — this is what makes results transferable to a real package.
- RH controlled by saturated salt solutions in a separate well within the vessel, physically isolated from the insert so liquid water never contacts the coating.
- Leak testing before every run; blank vials (paper + binder, no complex) on every run to exclude co-eluting volatiles from the binder or the paper itself.

Produce-free tests first. Fruit respire, transpire and absorb 1-MCP, so a produce-containing vessel confounds release with uptake. Establish the release curve in a clean system, then validate in the presence of fruit.

Condition matrix

RH and temperature levels must be justified against the commodity and package (Section 13), not chosen for roundness. The *structure* of the matrix:

RH level	Rationale
Low	Storage/transport condition. Tests shelf stability before activation — the product must not release here.
Intermediate	Partial activation; locates the threshold.
Typical produce-package RH	The design point.
Near-saturation	Condensation risk; tests worst-case dumping.

Temperature level	Rationale
Refrigerated	Normal cold-chain storage for the chosen commodity.
Cool-chain interruption	The realistic abuse case; a product that dumps its payload during a transport excursion is unsafe to sell.
Ambient / retail	Terminal market conditions.

Saturated-salt RH values are themselves temperature-dependent, so the achieved RH must be measured and reported at each temperature rather than taken from a table at 25 °C.

Models — and the trap to avoid

Fit only physically defensible models, and report the *physics*, not just the R^2 . Several of these models are near-collinear over short time windows, so a good fit is weak evidence for a mechanism.

Model	Equation	Parameters mean	Appropriate when	Limitation here
Zero-order	$Mt/M_\infty = k \cdot t$	constant rate	a maintained reservoir at constant activity	rarely true once the complex depletes
First-order	$Mt/M_\infty = 1 - e^{(-k \cdot t)}$	k = rate constant	release proportional to remaining load	assumes no diffusion barrier — i.e. assumes away the binder
Higuchi	$Mt/M_\infty = k \cdot \sqrt{t}$	matrix diffusion	planar matrix, constant D	derived for dissolved drug in a matrix; a gas from a hydrated lattice is a different physics
Korsmeyer-Peppas	$Mt/M_\infty = k \cdot t^n$	n diagnoses mechanism	first 60 % of release	n is only interpretable for the geometry it was derived for ; quoting n without stating geometry is meaningless
Weibull	$Mt/M_\infty = 1 - e^{(-(t-T)/a)^b}$	T = lag, a = scale, b = shape	empirical shape fitting, including a lag time	empirical — b is not a mechanism
Peppas-Sahlin	$Mt/M_\infty = k_1 t^m + k_2 t^{2m}$	splits Fickian vs relaxation	swelling-controlled systems	needs dense early data
Dual-resistance	series resistances	separates hydration from diffusion	two-step control	more parameters than short curves support
Arrhenius	$k = A \cdot e^{(-E_a/RT)}$	E_a	temperature dependence of k	only valid if mechanism is unchanged across T — check before applying

The honest recommendation is Weibull as the primary descriptive model, because the quantity this product is actually claiming — a *lag* before onset — is an explicit parameter (T) rather than something inferred. Korsmeyer-Peppas should be fitted over the first 60 % as a mechanism diagnostic, with geometry stated. Arrhenius applies to the temperature series only after confirming the mechanism does not change between refrigerated and ambient.

Do not force a model. “No fitted model described these data” is a publishable and honest result, and is better evidence of unusual behaviour — which is what a patent application would want — than a forced fit.

Performance targets

Targets are stated as the shape of the required result. Absolute values must be set against the commodity’s known effective 1-MCP dose, which comes from the postharvest literature, not from this protocol.

Metric	Target shape	Why it matters
Lag time at low RH	Long — no meaningful release	Shelf stability; the core commercial requirement
Lag time at package RH	Short and reproducible	Treatment must start when the package closes
Time to effective headspace concentration	Well inside the treatment window	Efficacy
Duration above effective concentration	Spans the intended treatment period	Efficacy
Maximum headspace concentration	Below any level causing off-flavour or disorder	Safety and quality
Total fractional release	High and, above all, reproducible	Dose control
Residual 1-MCP in spent insert	Measured, not assumed	Mass balance; disposal

Metric	Target shape	Why it matters
Replicate CV	Low	The differentiator stands or falls here
RH sensitivity ($\partial\text{release}/\partial\text{RH}$)	Sharp at threshold, flat above	This is what "humidity-triggered" has to mean quantitatively
Temperature sensitivity	Modest across the cold chain	Robustness to excursions

Mass balance closes the argument. Report released + residual + unaccounted for every run. A study that reports "82 % released" without accounting for the other 18 % has not measured release; it has measured one number.

Pre-activation storage stability

Run in parallel from day one, because it gates commercial viability independently of everything above: inserts stored in the intended protective overwrap at realistic warehouse conditions, assayed for retained 1-MCP at intervals, against unwrapped controls.

If retained payload falls materially over a few weeks in the overwrap, the product does not have a route to market in that form, and that should be known early rather than after formulation optimisation.

Characterisation programme

Split by what actually gates a decision. The most common failure in this literature is a full materials-characterisation battery on a system whose release behaviour was never measured properly — FTIR, XRD, SEM, TGA and tensile data on a film nobody quantified the headspace of.

Essential for proof of concept (Phases 1-3)

Test	What decision it gates	Why essential
Retained 1-MCP after coating and drying	Go/no-go on the process (FMEA FM01)	If the payload does not survive manufacture, nothing else matters
Headspace 1-MCP by GC over time	The entire differentiator	The primary measurement; everything else is context
Coat weight (gravimetric, g·m⁻²)	Dose specification	The manufacturing parameter that maps to release; no retrieved source reports it
Water-vapour permeability of the coating	Binder selection	The actual release-control mechanism
Water uptake / moisture sorption	Trigger design	Defines the activation threshold
Storage stability of retained payload	Commercial viability (FM02)	Gates the product independently of performance
Mass balance (released + residual)	Data integrity	Without it, "82 % released" is one number, not a measurement
Blank-substrate GC	Analytical validity	Excludes binder/paper volatiles masquerading as signal

That is eight tests, and only two require instrumentation beyond a GC and a balance.

Valuable but deferrable (Phase 4+)

- **DSC / TGA** — thermal stability of the complex; informs the drying window. Useful once a binder is chosen.
- **SEM, surface and cross-section** — coating continuity and penetration into the paper; explains anomalous release rather than predicting it.
- **FTIR / XRD** — confirm the inclusion complex survived processing. Genuinely useful *if* retention is poor and the cause is unclear; premature before that.
- **Contact angle, Cobb value** — surface energy and water absorbency of the base paper; matters for coating quality.
- **Coating adhesion, folding resistance** — converting robustness.
- **Air permeability, optical properties** — largely cosmetic for an insert.
- **Tensile strength, elongation, modulus** — **low priority for this architecture**. An insert carries no load. These belong to a standalone-film product, which is not what is recommended. Measure them only if the insert is physically failing in handling.

Only if the essential oil is reinstated (Phase 4+)

Droplet size, Pickering-emulsion stability, zeta potential, and emulsion creaming/coalescence on storage. **All of these are deferred with the emulsion itself** (Section 7.2) — they characterise a component the recommended design does not contain. Listing them now would be characterising a decision already made against.

If an oil is reinstated as a separate layer using a β -CD inclusion powder (as Cao et al. did) rather than an emulsion, most of this list is unnecessary.

What to measure that others have not

The prior art gap is not in materials characterisation — that literature is well served. It is in **dose and reproducibility**:

- Coat weight versus release rate, as a designed relationship.
- Release fraction versus RH, resolved finely enough to show whether a threshold exists.
- Replicate-to-replicate coefficient of variation in delivered headspace concentration — reported, not omitted.
- Delivered concentration across a deliberately varied package volume and fruit mass.
- Retained payload versus drying condition and versus storage time.

These are cheap measurements that no retrieved source reports, and they are precisely the evidence the proposed differentiator requires.

Commodity selection and produce-validation experiment

Commodity ranking

Candidates ranked against the brief's criteria: ethylene sensitivity, documented 1-MCP response, typical package humidity, respiration rate, commercial loss, decay relevance, package size, availability, experiment duration, ease of measuring effects, risk of adverse 1-MCP responses, and industrial value.

Rank	Commodity	Case for	Case against
1	Tomato	Textbook climacteric; large, unambiguous, <i>fast</i> 1-MCP response; year-round availability; cheap; effects visible in days; firmness and colour both easy to quantify	Lower per-unit value; decay pressure less commercially central
2	Peach / nectarine	High commercial loss; strong industry pull; brown rot (<i>Monilinia</i>) makes the antimicrobial arm meaningful; directly comparable to Cao et al. (2024)	Seasonal; risk of internal breakdown/mealiness confounding results; 1-MCP response more variable
3	Banana	Dramatic, easily staged ripening response	Chilling-sensitive; supply chain hard to replicate in a lab
4	Avocado	High value; strong 1-MCP literature	Long, variable ripening; harder to score
5	Apple / pear	The commercial home of 1-MCP; excellent literature	Long experiments (months); incumbent-dominated ; risk of superficial scald/disorder confounds
6	Kiwifruit	InBox is registered for it in CA	Slow; specialised
7	Mango / papaya	Commercially important	Availability and variability outside producing regions
8	Plum	Reasonable model	Less distinct endpoints

Primary: tomato. Not because it is the most commercially valuable, but because it is the fastest, cheapest, most reproducible way to answer the question that gates everything — *does this insert deliver an effective, reproducible 1-MCP dose into a real package?* A commodity with a slow, variable response would confound a delivery-system failure with a physiological one. Pick the commodity that makes the delivery system's behaviour easiest to see.

Secondary: peach. Adds commercial relevance, the decay pressure that makes the optional antimicrobial arm worth testing, and — importantly — direct comparability with the closest prior art.

Explicitly not first: apple. It is where 1-MCP lives commercially, which is exactly why it is the wrong first experiment: months-long trials against an entrenched incumbent, before the delivery system is even characterised.

Experimental parameters

Values below are the *structure* of the design. Absolute figures (dose, temperature, RH, duration) must be set from the commodity's postharvest literature and from the release data of Section 11, not chosen here.

Parameter	Tomato (primary)	Peach (secondary)
Cultivar	One determinate cultivar, single source	One freestone cultivar
Maturity at treatment	Mature green to breaker — before the ethylene climacteric , or 1-MCP cannot act	Commercial harvest maturity
Package	Vented corrugated carton or sealed chamber of measured volume	Standard tray/carton
Produce mass	Fixed per replicate; recorded individually	Fixed per replicate

Parameter	Tomato (primary)	Peach (secondary)
Headspace volume	Measured, not assumed	Measured
Storage temperature	Above chilling threshold for tomato	Commercial cold-chain temperature
RH	Measured continuously, logged	Measured continuously
Treatment duration	Set from release curve	Set from release curve
Target 1-MCP dose	From published effective dose for the commodity	From published effective dose
Target organism (peach arm)	—	<i>Monilinia fructicola</i>

Treatment groups

All eight groups from the brief, each earning its place:

#	Group	Question it answers
1	Untreated produce	Baseline ripening
2	Blank substrate (uncoated paper)	Does the paper itself do anything (moisture buffering)?
3	Matrix-only (binder-coated, no active)	Does the binder alone affect the package?
4	1-MCP insert	The primary test
5	Essential-oil-only insert	Isolates antimicrobial effect (peach arm)
6	Dual-active insert	Tests interaction — only meaningful against 4 and 5
7	Commercial benchmark	Are we better than what exists?
8	Conventional gaseous 1-MCP	Positive control: does the commodity respond at all?

Groups 2 and 3 are the ones most often dropped and most often needed: without them, a moisture-buffering or barrier effect of the insert is indistinguishable from 1-MCP action.

Group 8 is non-negotiable. If the fruit does not respond to a conventional 1-MCP treatment, the experiment cannot interpret a null result from the insert — a delivery failure and an unresponsive fruit batch look identical.

Group 7 caveat: commercial products are sold under label restrictions. Confirm that comparative use is legally permissible before planning it, and document the product, lot and use rate.

Measurements

Delivery (the part unique to this project): headspace 1-MCP by GC at intervals; RH and temperature logged continuously; residual 1-MCP in the spent insert; mass balance.

Physiology: ethylene production rate; respiration rate (CO₂).

Quality: firmness (instrumental, consistent probe/geometry); weight loss; colour (instrumental L*a*b|, not visual); soluble solids; titratable acidity; pH.

Decay and microbiology: decay incidence and severity on a defined scale; microbial load; for peach, inoculated challenge with the target organism.

Sensory: off-odour screening on every arm; a triangle test for any essential-oil arm. **If sensory testing involves human panellists, ethics approval and informed consent are required, and treated fruit must be confirmed safe for consumption before any tasting — for a pesticide-regulated active this is a genuine constraint, not a formality.** Where that is not feasible, restrict to instrumental and trained-assessor odour evaluation and say so.

Disorders: internal breakdown, mealiness, chilling injury where relevant — 1-MCP can *cause* disorders in some commodities, and a project that measures only firmness will miss the harm it does.

Design and analysis

- **Randomised complete block**, blocking on the real nuisance factors: position within the storage room (temperature and airflow gradients are not negligible) and fruit source lot.
- **Experimental unit is the package, not the fruit.** Fruit within a package share a headspace and are not independent. Treating individual fruit as replicates inflates n and is the most common

statistical error in this literature. Use package means, or a mixed model with package as a random effect.

- At least 4 replicate packages per treatment per sampling time; destructive sampling means replicates must be planned per timepoint, not reused.
- Analysis: mixed-effects model for repeated measures; ANOVA with an appropriate post-hoc test for single timepoints; survival/ordinal methods for decay incidence rather than treating percentages as normal.
- **Pre-register the primary endpoint** (proposed: firmness retention at the commercial end-of-shelf-life point) before unblinding. With ~15 measured variables, something will be significant by chance.

Screening design first, response surface later. A factorial or RSM design over binder $\times$ coat weight $\times$ loading is premature until the release programme (Section 11) shows which factors actually move the response. Use a simple screening design first; escalate only if the factors interact.

Regulatory and safety assessment

No legal or regulatory assurance is given here. This section identifies the questions that must go to a regulatory specialist and flags which ones are capable of ending the project.

The fact that dominates everything

1-MCP is a registered pesticide active ingredient in the United States, not a food additive and not a generic packaging component.

Verified: EPA registered 1-MCP through the Biopesticides and Pollution Prevention Division under the trade name SmartFresh on **17 July 2002, EPA Reg. No. 71297-2**; the label has been amended as recently as 2023. SmartFresh InBox carries its own registration, **EPA Reg. No. 71297-5**.

Three consequences follow immediately:

1. **The product is a pesticide product.** It is regulated under FIFRA. Its packaging format, crops, use directions and claims all live on a pesticide label.
2. **Registrations are crop- and format-specific.** Verified example: SmartFresh InBox is approved **for kiwifruit only in California**. A new format does not inherit an existing registration merely by using the same active.
3. **A new delivery format almost certainly requires regulatory action** — at minimum an amendment, plausibly a new product registration. Whether it is an amendment to an existing registration or a fresh one is **the single most important question to put to a regulatory specialist, and it should be asked in Phase 1, not after formulation is fixed**.

An academic project can generate data under research conditions, but it cannot sell or commercially distribute a 1-MCP product without going through this. Any plan that treats regulation as a late-stage formality is wrong.

Questions for a regulatory specialist

Priority 1 — capable of ending the project:

1. Does a coated-paper 1-MCP insert require a new FIFRA registration, or can it be added to an existing registration? By whom — the complex supplier, or us?
2. Is a registrant partner (a company already holding a 1-MCP registration) effectively a precondition for commercialisation?
3. What data package would EPA expect for a new format — efficacy, residue, worker exposure, environmental fate?
4. Does the intended crop have an existing 1-MCP tolerance/exemption, and does the intended use pattern fall inside it?

Priority 2 — shapes the design:

1. Does an insert that does not touch the fruit still constitute a food-contact material, and does that change with incidental contact in a vibrating carton?
2. If an essential oil is added, what is its status in this context — GRAS as a flavouring does **not** automatically translate to an antimicrobial use in a pesticide-regulated article, and an antimicrobial claim may itself be a pesticidal claim.
3. What migration testing applies, and to which simulant?
4. Do repulpability/compostability claims require third-party certification?

Priority 3 — operational:

1. Worker exposure and ventilation requirements for handling the complex in the lab and on a coating line.
2. Disposal route for spent inserts and for off-spec coated stock.

How the EU differs

The pathway diverges structurally and should not be assumed to parallel the US:

- 1-MCP is regulated in the EU as a **plant protection product** under Regulation (EC) No 1107/2009, with authorisations granted **member state by member state** rather than centrally.
- Maximum residue levels sit under Regulation (EC) No 396/2005.
- Food-contact materials fall under Regulation (EC) No 1935/2004, with **active and intelligent materials specifically addressed by Regulation (EC) No 450/2009** — directly relevant, since this product is by definition an active material.
- Paper and board food-contact materials are **not** harmonised at EU level in the way plastics are (Regulation 10/2011); national rules and industry frameworks apply.

These citations identify the applicable frameworks; their application to this specific product must be confirmed by an EU regulatory specialist. Do not treat the list as a compliance route.

Safety and handling in the research phase

- Treat the complex as a pesticide product in the laboratory: engineering controls, ventilation, documented handling, restricted access.
- 1-MCP is active at very low concentrations — cross-contamination between treatment and control samples is both a safety and a **data-integrity** issue. Physically separate control storage; do not rely on timing alone.
- Essential oils at antimicrobial vapour doses are irritants and require their own handling assessment.
- Fruit from treatment trials must not enter the food chain.

Environmental and disposal claims

A repulpable paper insert is a genuine advantage, but the claim must be qualified honestly: the insert carries a **pesticide residue and a synthetic binder**. Whether it is acceptable in a recycled paper stream is an empirical and regulatory question, not a marketing one. Do not make a recyclability claim without a repulpability test and confirmation that residual active is acceptable to the receiving mill.

Bottom line

The regulatory pathway is **the largest non-technical risk in the project** (FMEA FM13, severity 5). It is also the one most often discovered too late. The single most valuable action in Phase 1 is not an experiment — it is a conversation with a FIFRA specialist about whether this product can reach market at all without a registrant partner.

Techno-economic assessment

The model is a live-formula workbook: `assets/cost_model.xlsx`, generated by `assets/build_cost_model.py`. Yellow cells are editable assumptions with a stated basis; blue cells are formulas. Nothing below should be trusted further than its inputs.

Base case

Using the proposed starting specification (120 g·m⁻² paper, 10 g·m⁻² coat weight, 25 % w/w complex in the coating, 0.02 m² insert, 85 % yield, 1.35× overhead):

Output	Value
Complex in coating	2.50 g·m ⁻²
Subtotal before yield	\$0.704 /m ²
Yield-adjusted	\$0.828 /m ²
Materials + conversion per insert	\$0.0166
Fully loaded (incl. overwrap, QC)	\$0.0416
With overhead and margin	\$0.056 per insert
Cost per kg produce protected	\$0.011 /kg
1-MCP delivered per insert	~1.5 mg

The delivered payload of ~1.5 mg falls inside the 0.5–5 mg range proposed in Section 9.2, so the cost case and the dose case are at least mutually consistent — which is a check, not a validation.

Cost structure

At base-case assumptions the **1-MCP/α-CD complex is ~36 % of pre-yield cost** (\$0.300 of \$0.704 per m²), with coating and drying next at ~21 %. Paper is almost negligible (~2 %).

This is the important structural finding: **the active dominates, and the substrate does not**. Two consequences:

- Optimising the paper is not worth effort. Choosing the cheapest coatable stock is fine.
- Every gram of coat weight is a direct cost, so the coat-weight optimisation in Section 11 is simultaneously a *release-control* and a *cost* exercise. If a lower coat weight gives the same delivered dose, it is doubly valuable.

What the model deliberately does not claim

No competitor price is asserted. Published pricing for SmartFresh, HarvestHold Fresh, Vidre+ and Hazel 100 could not be retrieved. Inventing a benchmark to declare this product “cheaper” would be a fabrication and would invalidate the comparison. The benchmark cell is left at zero for the reader to fill with a real quote.

So the honest statement is: **this model produces a defensible internal cost, and no verified comparison to incumbents**. The comparison is an open question (Section 19).

Registration and compliance costs are excluded from the per-unit model. For a FIFRA-regulated product these are potentially larger than manufacturing cost and behave as a programme cost, not a unit cost. A registration requirement could dominate the entire economic picture (Section 14), and no credible figure can be put on it without specialist advice.

Sensitivity

Driver	Low	Base	High	Why it matters
Complex price (\$/kg)	40	120	400	The dominant uncertainty. Unverified; a supplier quote is the highest-value single data point in this model

Driver	Low	Base	High	Why it matters
Coat weight ($\text{g}\cdot\text{m}^{-2}$)	4	10	20	Scales complex usage linearly; also the release lever
Complex fraction (w/w)	0.10	0.25	0.40	Scales complex usage linearly
Insert area (m^2)	0.01	0.02	0.04	Scales material per package
Coating + drying ($\$/\text{m}^2$)	0.05	0.15	0.50	Falls sharply with volume; dominant at pilot scale
Yield	0.70	0.85	0.95	Waste of a speciality active is expensive

Because complex price, coat weight and complex fraction all enter linearly and multiply, the plausible range on cost per insert is roughly an order of magnitude — **from under \$0.02 to over \$0.30**. That spread is too wide to support any commercial decision, and narrowing it does not require lab work: it requires **a supplier quotation and an assay of the complex's actual 1-MCP content**. Both are Phase 1 actions.

Are chitosan and CNC needed at commercial scale?

No — and the cost model shows why the question is moot rather than marginal. Since the active dominates cost, the argument against chitosan and CNC is not primarily economic; it is that they **add water, hygroscopicity and process steps** (Section 7) while contributing nothing the design needs. Their cost is a secondary reason to omit them.

Of the cheaper alternatives the brief lists, the binder screen (Section 10.4) covers the relevant span: shellac/ethanolic (water-free, protects the payload), PEG (literature-anchored for this exact purpose), and water-based latex/HPMC (cheapest, highest retention risk). Starch and alginate are aqueous and hygroscopic — the same objection as chitosan. Commercial water-based coatings are worth screening precisely because they are cheapest and most industrially standard, but they must be measured for retention, not assumed adequate.

Scale

Minimum viable volume is set by **coating-line economics, not materials**. A slot-die or gravure line has a minimum practical run length; below it, setup and cleaning dominate and the $\$0.15/\text{m}^2$ conversion assumption is optimistic by an order of magnitude. For early work, toll-coating on an existing line is far more sensible than dedicated equipment — which also means the design must fit *someone else's* line, reinforcing the Section 10 recommendation to use a lab method (Meyer rod) with a real industrial analogue.

Failure modes and kill criteria

Full FMEA in `assets/fmea.csv` — 20 failure modes with cause, consequence, detection method, mitigation, severity, likelihood, RPN, kill criterion and the phase in which each is testable.

The three that dominate

All score RPN 20 (severity 5 × likelihood 4):

FM01 — premature 1-MCP release during manufacture. Aqueous coating or drying heat discharges the complex before the product exists. This is the *defining* technical risk: the release mechanism and the manufacturing hazard are the same physics. Detection is a retention assay against charged mass. Mitigation is a non-aqueous or low-water binder and mild drying. *Kill: retention below 50 % of charged 1-MCP after coating and drying, with no low-water route doing better.*

FM02 — premature release during storage. A product that cannot sit in a warehouse cannot be sold, regardless of how well it performs when fresh. This risk is frequently discovered late because storage studies take real time — hence starting it in Phase 2, not Phase 4. *Kill: payload loss materially degrades efficacy within a realistic warehouse window even in the intended overwrap.*

FM14 — patent overlap. Cellresin/Kimberly-Clark hold coated-substrate and laminate claims; Fresh Inset claims delayed release commercially. This risk is unaffected by any experimental result. *Kill: counsel finds no practicable design-around of a blocking claim.*

That two of the top three are non-experimental (FM02 is slow, FM14 is legal) is the central scheduling message of Section 18.

The differentiator's own failure modes

FM05 (no engineerable RH threshold) and FM06 (dose not reproducible) deserve separate emphasis because they do not merely degrade the product — **they delete the reason for the project**. Everything else being fine while these fail leaves a working insert that is indistinguishable from HarvestHold Fresh.

They are also the cheapest failures to detect: both fall out of the Phase 3 release programme, before any produce trial or scale-up spend.

Coupled failures

Two pairs where fixing one worsens the other, worth watching explicitly:

- **FM03 ↔ FM04.** Increasing coat weight to slow dumping can push release below the effective concentration. The optimum is a window, not a direction, and it may be narrow.
- **FM10 ↔ FM11 ↔ FM16.** The essential oil must exceed MIC in the package vapour, stay below the sensory threshold, and not plasticise the layer enough to accelerate 1-MCP release. Three constraints on one variable is the strongest argument for deferring the oil entirely (Section 7.5).

Kill criteria, consolidated

The project should **stop or be redesigned** if any of the following holds:

1. No coating route retains a usable fraction of the 1-MCP charge (Phase 1).
2. Payload cannot be held through realistic storage in the intended overwrap (Phase 2).
3. Release versus RH is smooth and shallow across every binder — no threshold exists to specify (Phase 3).
4. Delivered-dose CV stays high after optimisation — no specification can be written (Phase 3).
5. The insert cannot match conventional gaseous 1-MCP within cost and sensory limits (Phase 4).
6. Pilot-coated material does not reproduce lab release behaviour (Phase 5).
7. Counsel identifies a blocking claim with no practicable design-around (Phase 6).
8. Regulatory advice indicates a new registration whose cost and timeline exceed project resources, with no registrant partner available (Phase 1/6).

What "redesign" means rather than "stop"

Not every failure is terminal, and the distinction is worth stating so the criteria are not applied mechanically:

- Failing 3 or 4 kills *the IP case* but not necessarily a useful product — it becomes a commodity insert competing on cost, which is a different and much weaker business.
- Failing 1 with a water-based binder but passing with an ethanolic one is not a failure at all; it is the answer.
- Failing 5 on tomato while passing on peach would indicate a commodity-fit problem, not a delivery-system problem.

The criteria above are written to stop *self-deception*, not to stop thinking.

Patent-oriented evidence plan

No opinion on patentability or freedom to operate is offered. This section sets out what evidence would be worth putting in front of a technology-transfer office, and — equally important — what would not.

Separating the layers of the claim

Layer	Status
Broad idea — humidity-triggered 1-MCP release from packaging, optionally with an essential oil, for climacteric fruit	Not novel. Disclosed in the literature, claimed in patents, and sold commercially
Format — coated paper insert / sheet / label	Not novel. HarvestHold Fresh is a sheet applied in the box; Vidre+ is a sticker
Materials — 1-MCP/ α -CD on paper with an essential-oil/ β -CD co-active	Not novel. Cao et al. (2024)
Delayed / gradual release	Claimed commercially (Vidre+) and titled in US20210331990A1
Potentially differentiating feature	A specified humidity activation threshold and release window , set by defined binder and coat weight, delivering a reproducible headspace 1-MCP concentration across varying package volume and produce mass
Evidence required	Sections 17.2–17.3
Closest prior art	Cao et al. 2024 (10.21203/rs.3.rs-4251760/v1); US10212931B2 (Kimberly-Clark/Cellresin); US20210331990A1; HarvestHold Fresh; Vidre+
Proposed design-around	Avoid the second-substrate laminate of US10212931B2; claim a <i>specification and method</i> rather than an article format
Remaining legal uncertainty	High. Requires claim-by-claim review, an FTO search by counsel, and a search of unexamined applications this review could not access

Results that would strengthen a disclosure

Ranked by how much they would actually add, given what is already public:

Strong — these are genuinely absent from the retrieved prior art:

1. **A sharp RH activation threshold.** A step change in release fraction over a narrow RH band, reproducible across batches. This is the difference between “humidity-triggered” as a description and as a specification. No retrieved source demonstrates it.
2. **A quantitative coat-weight $\rightarrow$ release-rate relationship.** A designed, monotonic, predictive relationship in $\text{g}\cdot\text{m}^{-2}$. No retrieved source reports coat weight at all for a 1-MCP paper.
3. **Dose reproducibility across varying package volume and fruit mass**, reported with a coefficient of variation. This is what a commercial buyer needs and what nobody publishes.
4. **A measured retention-versus-processing-water relationship** — retained 1-MCP as a function of binder water content and drying condition. Supports a low-temperature/low-water manufacturing claim with numbers rather than an assertion.

Moderate:

1. A reproducible, tunable lag time before onset (weaker than it looks — Vidre+ claims delayed release commercially).
2. Improved pre-activation shelf stability versus an uncoated or unwrapped control, quantified.
3. Equivalent efficacy at lower 1-MCP loading than a commercial benchmark.

Weak — do not build a case on these:

1. “It works on tomato/peach.” Efficacy of 1-MCP on climacteric fruit is thoroughly established; demonstrating it again adds nothing.

2. "Antimicrobial plus ethylene control." Cao et al. did this. Only a *demonstrated interaction* — measured against both single-active controls — would add anything, and Section 7 argues the interaction is more likely to be negative than synergistic.
3. Materials characterisation (FTIR/XRD/SEM confirming an inclusion complex). Standard, expected, and not distinguishing.

Minimum evidence package for a TTO conversation

Deliberately small — enough to make a decision, not a thesis:

1. **Release curves** at ≥ 3 RH levels $\times$ ≥ 2 temperatures, ≥ 4 replicates, with full mass balance and a stated analytical method including LOD/LOQ.
2. **A threshold plot:** release fraction versus RH, resolved finely enough to show whether a step exists. *This is the single most important figure.*
3. **A coat-weight series:** ≥ 3 coat weights, same formulation, release rate versus $\text{g}\cdot\text{m}^{-2}$.
4. **A dose-reproducibility table:** delivered headspace concentration across ≥ 3 package volumes and ≥ 2 fruit masses, with CV.
5. **A retention table:** retained 1-MCP after each process route (water-based, PEG, ethanolic) and after storage in the intended overwrap.
6. **One produce trial** with all essential controls, including conventional gaseous 1-MCP as a positive control.
7. **This prior-art map**, with Cao et al. and the Cellresin family named explicitly rather than omitted.

Point 7 matters. Presenting a disclosure that does not name its closest prior art wastes the TTO's time and damages credibility when they find it — and they will.

What would tell you to stop

- Release versus RH is smooth and shallow across every binder → there is no engineerable threshold, and the differentiator does not exist.
- Replicate CV in delivered concentration stays high after optimisation → no dose specification can be written, which was the whole claim.
- Counsel identifies a blocking claim with no practicable design-around.

Any of these should end the patent track. The engineering may still be worth doing as a product-development or teaching exercise, but not as an IP play.

A note on the preprint

Cao et al. (2024) is a **Research Square preprint**, not a peer-reviewed article. That affects how much weight to give its *data*, but **not** its status as prior art — public disclosure is public disclosure regardless of peer review. It should be treated as fully citable prior art and checked for a subsequent journal version before any filing.

Six-phase development roadmap

Sequenced so the cheapest experiment that can kill the project runs first. Each phase has an explicit exit test; failing it means stop or redesign, not proceed with caveats.

Phase 1 — Feasibility screening (2-4 weeks)

Question: does a paper-borne 1-MCP complex survive manufacture and release measurable gas into a headspace?

Minimum experiments:

1. Assay the as-received complex for actual 1-MCP content. *This single number determines whether any of the dose arithmetic is real.*
2. Establish the GC-FID method: calibration with mass balance, LOD/LOQ, blank substrate, leak test.
3. Make Architecture C-0 (complex on absorbent paper, no binder) and measure headspace release at high and low RH.
4. Coat three binders (ethanolic/shellac, PEG, water-based) by Meyer rod; assay retained 1-MCP after drying.

Run in parallel, not as an experiment: a conversation with a FIFRA regulatory specialist about whether this format needs a new registration and whether a registrant partner is a precondition. This is the highest-value action in Phase 1 and it is not a lab activity.

Exit test: measurable, quantifiable headspace 1-MCP from a coated sample, with retention after coating and drying materially better than the water-based control. **Kill if** no route retains a usable fraction of the charge (FM01).

Phase 2 — Formulation optimisation (4-8 weeks)

Optimise binder, coat weight (≥ 3 levels), complex loading, and the protective overwrap. Begin the pre-activation storage-stability study **on day one of this phase** — it runs longest and gates commercialisation independently.

Exit test: a formulation with reproducible coat weight and acceptable payload retention over a realistic storage window in its overwrap. **Kill if** payload cannot be held in storage (FM02).

Phase 3 — Release-kinetics validation (6-10 weeks)

The full protocol of Section 11: RH matrix $\times$ temperature matrix, replicates, mass balance, sacrificial-vial sampling, model fitting.

This phase decides whether the project has a differentiator. The two figures that matter are release-fraction-versus-RH (is there a threshold?) and delivered concentration across varying package volume (is dose reproducible?).

Exit test: a defensible release model, a demonstrated RH threshold, and a stated dose-reproducibility CV. **Kill if** release versus RH is smooth and shallow across all binders, or CV stays high after optimisation (FM05, FM06) — in that case there is no specification to claim and no product to differentiate.

Phase 4 — Produce validation (one season, or 6-12 weeks on tomato)

The design of Section 13 on tomato, with all eight treatment groups including conventional gaseous 1-MCP as a positive control. Peach follows in season if Phase 3 succeeded and the antimicrobial arm is being pursued.

Exit test: quality retention significantly better than untreated and than matrix-only controls, with no off-odour or disorder. **Kill if** the insert underperforms the conventional positive control by a margin that a dose increase cannot close within cost and sensory limits.

Phase 5 — Scale-up demonstration (8-12 weeks)

Lab Meyer rod $\rightarrow$ pilot slot-die at matched coat weight, ideally toll-coated on an existing line. Compare release curves lab versus pilot directly — the reason both are specified in $\text{g}\cdot\text{m}^{-2}$.

Exit test: pilot material reproduces lab release behaviour within the tolerance implied by the dose specification. **Kill if** it does not (FM17); a formulation that only works at bench scale is not a product.

Phase 6 — Patent and commercialisation assessment (parallel from Phase 3)

Invention disclosure with the claim-element matrix and the prior-art map naming Cao et al. and the Cellresin family explicitly; updated prior-art search including unexamined applications; FTO review by counsel; cost model refreshed with a real complex quotation; regulatory consultation concluded.

Exit test: counsel identifies a claimable position with a practicable design-around, and regulatory advice identifies a viable route to market. **Kill if** either fails (FM13, FM14).

Critical-path note

Phases 1 and 6 both start immediately. The regulatory and patent questions are capable of ending the project independently of every experimental result, and they have the longest lead times. Running them last — the usual instinct — is how projects spend a year on formulation before discovering they cannot sell the product.

Honest assessment of the schedule

The lab work is roughly 6–9 months to the end of Phase 5 with one full-time researcher, assuming the complex is available and the GC is accessible. The regulatory and IP path is likely longer and is the binding constraint on commercialisation, not the science.

The ten most important unanswered questions

Ordered by how much the answer would change what happens next. The first three can end the project and none of them is a laboratory experiment.

1. Does a coated-paper 1-MCP insert require a new FIFRA registration, and is a registrant partner a precondition for commercialisation? Nothing else matters if the answer is unfavourable. 1-MCP is a registered pesticide active (EPA Reg. 71297-2, 2002) and registrations are crop- and format-specific — SmartFresh InBox is verified as approved for kiwifruit only in California. *Resolve by: regulatory specialist consultation, Phase 1.*

2. Is there a practicable design-around of the Cellresin / Kimberly-Clark claims, and what exactly does Fresh Inset's Vidre+ patent cover? US10212931B2 claims a cyclodextrin composition coated on a substrate with a second substrate over it. Vidre+ is stated by its owner to be patented for delayed, gradual 1-MCP release. Both sit directly across the proposed position. *Resolve by: claim-by-claim review and FTO search by counsel.*

3. What is the actual 1-MCP content of the α -CD complex, and what does it cost? The techno-economic model spans roughly an order of magnitude on cost per insert almost entirely because of this one unverified input, and the dose arithmetic in Section 9.2 rests on an assumed 3 % w/w payload. *Resolve by: supplier quotation plus assay. Cheapest high-value action in the project.*

4. Does release fraction versus relative humidity show a genuine threshold, or only a smooth gradient? This is the technical question the differentiator stands on. A smooth gradient means "humidity-triggered" can be described but not *specified*, and the product collapses into an undifferentiated slow-release insert. *Resolve by: Phase 3 release programme.*

5. How reproducible is the delivered headspace concentration across realistic variation in package volume, produce mass and temperature? A dose specification is the proposed contribution. No retrieved source — academic or commercial — reports this. *Resolve by: Phase 3, reported as a coefficient of variation.*

6. How much 1-MCP survives each coating route, quantitatively? Cao et al. adopted a water-free method on the reasoning that moisture discharges the complex, but the *relationship* between process water content and retained payload appears nowhere. It is both a design input and a candidate patentable result. *Resolve by: Phase 1 binder screen with retention assay.*

7. What is the gap between the minimum inhibitory vapour concentration of carvacrol in a real package and its sensory detection threshold on the target fruit? This single number decides whether the antimicrobial arm is viable at all. It is commodity-specific and no verified value was obtained here. *Resolve by: paired in-package challenge test and sensory panel, before any v2 formulation work.*

8. What do the incumbent products actually cost per package? No published pricing for SmartFresh InBox, HarvestHold Fresh, Vidre+ or Hazel 100 was retrievable, so the cost model has an internal figure and **no verified comparison**. Any claim of being cheaper is currently unsupported. *Resolve by: distributor quotation.*

9. Does an essential oil measurably displace 1-MCP from cyclodextrin, or partition it into an oil phase? Section 7.3 argues both are plausible from first principles, but no retrieved study measured either directly. If the effect is negligible, single-layer dual-active designs become viable and the architecture calculus changes. *Resolve by: a small headspace experiment — complex with and without oil contact.*

10. What lies in the literature the four inaccessible indexes cover? Web of Science, Scopus, CAB Abstracts and AGRICOLA were unavailable. CAB Abstracts and AGRICOLA in particular index postharvest, extension and conference material that OpenAlex and Europe PMC cover unevenly — exactly where a 1-MCP paper insert might have been described without appearing here. *Resolve by: a search through an institutional subscription. Until then, every statement in this report about the absence of prior work is weaker than it appears.*

A note on questions 1, 2 and 8

Three of the ten are not scientific questions, and two of those three are capable of ending the project on their own. That distribution is itself a finding: **the principal risks to this project are regulatory, legal and commercial, not technical**. The technical work is tractable and cheap by comparison. Resourcing the project as though the hard part is the chemistry would be a mistake.

Complete verified references

What "verified" means here

Every entry below was **retrieved, not recalled**. Specifically:

- Bibliographic details — authors, year, title, journal, DOI — come from Crossref, OpenAlex or Europe PMC, never from a language model.
- Every DOI was **re-resolved against Crossref** during verification, and the returned title was checked for agreement with the stored record. Entries whose DOI failed to resolve, or whose title disagreed, are reported as issues in `outputs/1mcp/20_issues.csv` and excluded from the counts.
- Each entry is annotated with whether **full text was retrieved** or only the abstract was available, because that determines how much of the literature table that source could support.
- Patent entries carry the publication number and priority date **as stated in the retrieved document**. Priority dates that could not be established from the document are marked rather than guessed.

No reference was added by hand. If a work is not in the harvested corpus, it does not appear — which also means this list is not a complete bibliography of the field, but a complete list of what this review actually read.

Known gaps

- Four indexes named in the brief were inaccessible (Web of Science, Scopus, CAB Abstracts, AGRI-COLA), so postharvest, extension and conference literature is under-represented relative to a full systematic review.
- Patent coverage rests on a single provider; unpublished applications are by definition absent.
- Commercial and regulatory sources cited in Sections 5 and 14 (EPA registration documents, product labels, company materials) are cited inline at their point of use rather than listed here, and company performance claims are labelled as claims throughout.

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60. Syariful Mubarak et al. (2022). Impact of 1-Methylcyclopropene combined with chitosan on postharvest quality of tropical banana 'Lady Finger'. *International Journal of Food Properties*, 25, 1171-1185. [10.1080/10942912.2022.2074028](https://doi.org/10.1080/10942912.2022.2074028) [original research; abstract only]
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63. Tingting Zhang et al. (2026). TEMPO-oxidized cellulose nanofiber/chitosan composite films stabilized by cumin essential oil-loaded Pickering emulsions for multifunctional food preservation. *International Journal of Biological Macromolecules*, 362, 152106. [10.1016/j.ijbiomac.2026.152106](https://doi.org/10.1016/j.ijbiomac.2026.152106) [original research; abstract only]
64. W. M. C. B. Wasala et al. (2021). Application of 1-Methylcyclopropene (1-MCP) for Delaying the Ripening of Banana: A Review. *Asian Research Journal of Agriculture*, 44-56. [10.9734/arja/2021/v14i130118](https://doi.org/10.9734/arja/2021/v14i130118) [review; abstract only]
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68. Xiaotong Zou et al. (2025). Development and characterization of corn starch-based films enhanced with *Chlorella vulgaris* nanocellulose-stabilized Pickering emulsion of *Zanthoxylum bungeanum* essential oil for cherry tomato preservation. *International Journal of Biological Macromolecules*, 304, 140837. [10.1016/j.ijbiomac.2025.140837](https://doi.org/10.1016/j.ijbiomac.2025.140837) [original research; abstract only]
69. Xiaoxue Wang et al. (2025). Preparation, Characterization and Application of Antimicrobial Pectin-Konjac Glucomannan Composite Films Incorporating Cellulose Nanocrystals Stabilized Clove Essential Oil Pickering Emulsion. [10.2139/ssrn.5102163](https://doi.org/10.2139/ssrn.5102163) [original research; abstract only]
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71. Xiuxiu Teng et al. (2025). 4D printed deformation labels with machine learning for monitoring and preservation of respiring climacteric fruits. *Nature Communications*, 16, 11525. [10.1038/s41467-025-66554-6](https://doi.org/10.1038/s41467-025-66554-6) [original research; full text retrieved]

72. Xukai Hou et al. (2026). The impact of 1-MCP treatment combined with polyethylene film bag packaging on the quality of 'Shannong Su' pears during storage. *Food Chemistry: Molecular Sciences*, 12, 100358. [10.1016/j.fochms.2026.100358](https://doi.org/10.1016/j.fochms.2026.100358) [original research; full text retrieved]
73. Yanyi Xue et al. (2026). Morphology-regulated Pickering emulsions stabilized by soybean residue-derived nanocellulose for tunable-release bio-based food packaging. *International Journal of Biological Macromolecules*, 376, 153411. [10.1016/j.ijbiomac.2026.153411](https://doi.org/10.1016/j.ijbiomac.2026.153411) [original research; abstract only]
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75. Yiping Liu et al. (2025). Application of a Pickering Emulsion Stabilized by Zein and Cellulose Nanocrystalline Composite Particles to Preserve Kiwifruit. *Molecules*, 30, 3478. [10.3390/molecules30173478](https://doi.org/10.3390/molecules30173478) [original research; full text retrieved]
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77. Zhaoxin Cao et al. (2024). Preparation of carvacrol β -cyclodextrin inclusions and 1-methylcyclopropene- α -CD coated paper by water-free method for postharvest preservation of peach. [10.21203/rs.3.rs-4251760/v1](https://doi.org/10.21203/rs.3.rs-4251760/v1) [original research; full text retrieved]
78. Zhibin Pan et al. (2024). Effects of oregano essential oil Pickering emulsion and ZnO nanoparticles on the properties and antibacterial activity of konjac glucomannan/carboxymethyl chitosan nanocomposite films. *RSC Advances*, 14, 6548-6556. [10.1039/d3ra07845k](https://doi.org/10.1039/d3ra07845k) [original research; full text retrieved]
79. Zhijun Hu et al. (2017). 1-Methylcyclopropene (MCP)-Containing Cellulose Paper Packaging for Fresh Fruit and Vegetable Preservation: A Review. *BioResources*, 12. [10.15376/biores.12.1.2234-2248](https://doi.org/10.15376/biores.12.1.2234-2248) [review; full text retrieved]
80. Ziao Zhang et al. (2026). Bio-Based Smart Packaging Materials for Next-Generation Food Systems. *Materials*, 19, 2393. [10.3390/ma19112393](https://doi.org/10.3390/ma19112393) [review; abstract only]
81. Zou X et al. (2025). Incorporation of Schizochytrium limacinum oil-based pickering emulsion in chitosan active composite films for strawberry preservation. [original research; abstract only]
82. Özge Horzum & Randolph M. Beaudry (2025). "In-box" 1-methylcyclopropene (1-MCP) treatments improve quality retention of 'Red Delicious' apple fruit. *The Journal of Horticultural Science and Biotechnology*, 101, 527-536. [10.1080/14620316.2025.2574319](https://doi.org/10.1080/14620316.2025.2574319) [original research; abstract only]

Patent documents (40 families)

1. **EP1584234A2** — Plant ethylene response inhibition complexes. Priority: 1998-08-20. patents.google.com/EP1584234A2
2. **PL346176A1** — Plant ethylene response inhibition compounds and complexes. Priority: 1998-08-20. patents.google.com/PL346176A1
3. **US6017849A** — Synthesis methods, complexes and delivery methods for the safe and convenient storage, transport and application of compounds for inhibiting the ethylene respo... (Biotechnologies for Horticulture Inc; AgroFresh Inc). Priority: 1998-08-20. patents.google.com/US6017849A
4. **KR100791251B1** — Cyclopropene delivery system requiring less water (로엠 앤드 하스 캄파니 / Rohm & Haas Company). Priority: 2000-09-29. patents.google.com/KR100791251B1
5. **US6426319B1** — Delivery systems for cyclopropenes requiring less water (AgroFresh Inc (Current); Rohm and Haas Co (Original)). Priority: 2000-09-29. patents.google.com/US6426319B1
6. **US6953540B2** — Continuous process for the preparation of encapsulated cyclopropenes (AgroFresh Inc). Priority: 2000-09-29. patents.google.com/US6953540B2
7. **CN1325446C** — Method and device for producing cyclopropenyl compound (Rohm and Haas Co). Priority: 2002-05-14. patents.google.com/CN1325446C
8. **US8129450B2** — Articles having a polymer grafted cyclodextrin (Verdant Technologies LLC (Current); Cell-resin Technologies LLC (Original)). Priority: 2002-12-10. patents.google.com/US8129450B2
9. **CR9490A** — CYCLOPROPEN COMPOSITIONS. Priority: 2006-11-09. patents.google.com/CR9490A
10. **US8603524B2** — Delivery of ethylene blocking and/or promoting agents (AgroFresh Inc). Priority: 2007-01-17. patents.google.com/US8603524B2
11. **CN101416658B** — 1-MCP/modified stabilizer coating material, production method and use thereof. Priority: 2008-11-07. patents.google.com/CN101416658B
12. **CN101578978A** — Ethylene absorbent and preparation method thereof (China Agricultural University). Priority: 2009-06-11. patents.google.com/CN101578978A

13. **CN101715760A** — 1-methylcyclopropene liquid preparation (Shandong Nutrient Source Food Technology Co Ltd). Priority: 2009-12-01. patents.google.com/CN101715760A
14. **CN104304242B** — Maturation or curing inhibitor release from polymer, fiber, film, thin slice or packaging (Cellresin Technologies, Llc). Priority: 2009-12-14. patents.google.com/CN104304242B
15. **MX2012006797A** — Maturation or ripening inhibitor release from polymer, fiber, film, sheet or packaging. (Cellresin Tech Llc). Priority: 2009-12-14. patents.google.com/MX2012006797A
16. **US9642356B2** — Maturation or ripening inhibitor release from polymer, fiber, film, sheet or packaging (Verdant Technologies LLC). Priority: 2009-12-14. patents.google.com/US9642356B2
17. **CN101990936A** — 1-methyl-3-(2-methyl cyclopropane)-1-cyclopropylene stable inclusion compound and preparation method thereof (Xian Jiaotong University). Priority: 2010-10-19. patents.google.com/CN101990936A
18. **US8481127B2** — Cyclodextrin compositions, articles, and methods (Verdant Technologies LLC). Priority: 2011-03-27. patents.google.com/US8481127B2
19. **CN102440237B** — 1-methylcyclopropene preparation and preparation method thereof. Priority: 2011-09-30. patents.google.com/CN102440237B
20. **US9381163B2** — Floating capsules encapsulating particles loaded with one or more drugs (NANYANG TECHNOLOGICAL UNIVERSITY, SINGAPORE). Priority: 2012-08-16. patents.google.com/US9381163B2
21. **CN102823644A** — Fresh-keeping method for muskmelon. Priority: 2012-09-04. patents.google.com/CN102823644A
22. **US20150272148A1** — Encapsulation of gases within cyclodextrins (THE UNIVERSITY OF QUEENSLAND, AUSTRALIA). Priority: 2012-10-10. patents.google.com/US20150272148A1
23. **AU2013387471A1** — Methods and compositions for polymer matrix synthesized by polycondensation. Priority: 2013-04-26. patents.google.com/AU2013387471A1
24. **CA2910267A1** — Gel formulations for extended release of volatile compounds. Priority: 2013-04-26. patents.google.com/CA2910267A1
25. **US20170101250A1** — Active compound formulation package and its subsequent release for use on plant and plant parts (Mirtech Inc). Priority: 2013-05-05. patents.google.com/US20170101250A1
26. **PH12016500070A1** — Humidity activated formulation for volatile compounds (Agrofresh Inc). Priority: 2013-07-11. patents.google.com/PH12016500070A1
27. **US20160330952A1** — Encapsulation and controlled release of volatile organic compounds (University of Guelph). Priority: 2014-01-15. patents.google.com/US20160330952A1
28. **EP3161558B1** — Electrostatic printing of cyclodextrin compositions (VERDANT TECHNOLOGIES, LLC). Priority: 2014-06-26. patents.google.com/EP3161558B1
29. **US9332751B2** — Food packaging material with antibacterial, ethylene scavenging and barrier properties (Sabanci Universitesi). Priority: 2014-07-09. patents.google.com/US9332751B2
30. **US10449156B2** — Controlled release system including a gas or volatile encapsulated in a polymeric support and a matrix system, a method of preparing the system, and their use (UNIVERSIDAD ANDRES BELLO, CHILE). Priority: 2014-07-23. patents.google.com/US10449156B2
31. **US10351490B2** — Complexes of 1-methylcyclopropene with metal coordination polymer networks (AgroFresh Inc). Priority: 2014-11-10. patents.google.com/US10351490B2
32. **US20190037839A1** — Compositions for controlled release of active ingredients and methods of making same (Hazel Technologies Inc). Priority: 2016-02-19. patents.google.com/US20190037839A1
33. **US20170318804A1** — Preparation and dispersion of stable emulsion formulations (AgroFresh Inc). Priority: 2016-05-04. patents.google.com/US20170318804A1
34. **US20190166861A1** — Method and system for the treatment of agricultural products (Pace International LLC). Priority: 2016-12-05. patents.google.com/US20190166861A1
35. **CN112218529A** — Compositions and articles comprising complexes of 1-methylcyclopropene with alpha-cyclodextrin. Priority: 2018-04-27. patents.google.com/CN112218529A
36. **KR20210030899A** — Compositions and articles comprising complexes of 1-methylcyclopropene and alpha-cyclodextrin (프레쉬 인셋 에스.에이.). Priority: 2018-04-27. patents.google.com/KR20210030899A
37. **US20200296978A1** — Modified 1-methylcyclopropene (1-MCP) nanobubbles (NANOTEC SA). Priority: 2019-03-15. patents.google.com/US20200296978A1
38. **US11033870B1** — Generators for 1-methylcyclopropene release from carrier complex (Individual). Priority: 2019-11-16. patents.google.com/US11033870B1
39. **US20210235692A1** — Stable 1-Methylcyclopropene Compositions and Uses Thereof (Fresh Inset SA). Priority: 2020-02-03. patents.google.com/US20210235692A1
40. **US20210331990A1** — Compositions and Methods for Differential Release of 1-Methylcyclopropene (Verdant Technologies LLC). Priority: 2020-04-27. patents.google.com/US20210331990A1

Appendix A — Literature records (all 25 fields)

One block per paper, carrying every field of the 25-column table. Fields that are absent were either not stated in the source or could not be verified from what was retrievable; the per-cell evidence level and supporting quote for each is in tables/literature_evidence.csv.

1. Zhaoxin Cao et al. (2024). Preparation of carvacrol β -cyclodextrin inclusions and 1-methylcyclopropene- α -CD coated paper by water-free method for postharvest preservation of peach.

DOI	10.21203/rs.3.rs-4251760/v1
Year	2024
Type	original_research
Substrate	filter paper
1-MCP carrier	α -cyclodextrin
Antimicrobial	carvacrol
Preparation	coating
Mfg. temp.	45 °C
Trigger	humidity_moisture
RH conditions	30, 98 %
Storage temp.	4, 25, 45 °C
Release	gravimetric
Retention	8 days
Retention result	92%, 88%, 73%
Produce	peach
Package	6.9 L; 1200 $\pm$ 50 g L; g
Quality	decay index, weight loss rate, hardness, total soluble solid (TSS), L* and a*
Antimicrobial	inhibition of <i>M. fructicola</i> mycelial growth
Scale-up	low
Relevance	core
Full text	oa_pdf: https://www.researchsquare.com/article/rs-4251760/latest.pdf
Confidence	high

2. Aaruba Maqbool et al. (2024). Studies on extending the shelf life of golden delicious apples through the application of 1-methylcyclopropene (1-MCP) and KMnO₄ and using various packaging materials. Internation...

DOI	10.33545/26174693.2024.v8.i6h.1390
Year	2024
Venue	International Journal of Advanced Biochemistry Research
Type	original_research
Substrate	polymeric film
1-MCP carrier	SmartFresh cards
Produce	Golden Delicious apples
Quality	colour
Scale-up	high
Relevance	core
Full text	oa_pdf: https://www.biochemjournal.com/archives/2024/vol8issue6/Parth/8-6-71-432.pdf
Confidence	medium

3. Prisma Espinal-Hernández et al. (2021). Postharvest effects of 1-MCP and Chitosan/Oleic acid coating in Pitaya (*Stenocereus griseus* H.). Journal of the Professional Association for Cactus Development, 23, 43-5...

DOI	10.56890/jpacd.v23i.322
Year	2021
Venue	Journal of the Professional Association for Cactus Development
Type	original_research
Substrate	chitosan
1-MCP carrier	sachet
Antimicrobial	chitosan
Preparation	coating
Trigger	liquid_water
Produce	Pitaya (<i>Stenocereus griseus</i> H.)
Quality	weight loss, skin firmness, total soluble solids (TSS), ethylene production, rotting incidence
Limitations	lab-only conditions and manual processing; requires further optimization for industrial scale
Scale-up	low
Relevance	adjacent_material

Full text oa_pdf:<https://jpacd.org/jpacd/article/download/322/312>
Confidence high

4. Soesiladi Esti Widodo (2015). Effects of 1-Methylcyclopropene and Chitosan on the Fruit Shelf-Life and Qualities of Two Different Ripening Stages of 'Cavendish' Banana. Journal of Food and Nutrition Sciences, ...

DOI 10.11648/j.jfns.s.2015030102.20
Year 2015
Venue Journal of Food and Nutrition Sciences
Type original_research
Substrate chitosan
Produce 'Cavendish' banana
Quality outcomes slow color development, fruit quality deterioration, decrease of fruit firmness, decrease of soluble solid content, increase of acidity
Scale-up not_assessable
Relevance adjacent_material
Full text oa_html:<https://doi.org/10.11648/j.jfns.s.2015030102.20>
Confidence medium

5. Dorota Wichrowska et al. (2026). Antioxidant potential and storage stability of sweet cherries (Prunus avium L.) depending on the use of 1- methylcyclopropene in the innovative form of a sticker. Scientific Re...

DOI 10.1038/s41598-026-42369-3
Year 2026
Venue Scientific Reports
Type original_research
1-MCP carrier Vidre+ stickers
RH conditions 90-95 %
Storage temp. 2-3 °C
1-MCP loading 1, 2, 4 uL of 1-MCP per liter uL/L
Release 24 h h
Duration sweet cherries (Prunus avium L.)
Package One Liter clamshells, 6 kg batches L, kg
Quality outcomes weight loss, firmness, stem browning, visual index, dry matter, vitamin C, total polyphenols, anthocyanins, total soluble solids, reducing and total sugars, antioxidant capacity
Limitations Lack of actual concentration measurement
Scale-up low
Relevance low
Full text epmc_jats:PMC13111586
Confidence high

6. Xukai Hou et al. (2026). The impact of 1-MCP treatment combined with polyethylene film bag packaging on the quality of 'Shannong Su' pears during storage. Food Chemistry: Molecular Sciences, 12, 100358.

DOI 10.1016/j.fochms.2026.100358
Year 2026
Venue Food Chemistry: Molecular Sciences
Type original_research
Substrate polyethylene film bag
RH conditions 90-95 %
Storage temp. 4.00 ± 0.50 °C
1-MCP loading 1 µL/L
Release gc_fid
Release 120 days
Release result inhibited ethylene production
Produce 'Shannong Su' pears
Quality outcomes Firmness, ethylene release rate, total sugar content, titratable acid content, total antioxidant capacity, starch content, protopectin content, cellulose content
Limitations The study is limited to a single pear cultivar and requires validation across other cultivars with different physiological traits.
Scale-up high
Relevance core
Full text epmc_jats:PMC12870802
Confidence high

7. Dorota Wichrowska et al. (2025). Antioxidant Potential and Storage Stability of Sweet Cherries (*Prunus Avium* L.) Depending on the Use of 1- Methylcyclopropene.

DOI	10.21203/rs.3.rs-6546955/v1
Year	2025
Type	original_research
1-MCP carrier	sticker
RH conditions	90-95 %
Storage temp.	2-3 °C
1-MCP loading	1, 2, 4 uL/L uL/L
Produce	sweet cherries (<i>Prunus avium</i> L.)
Package	1 Liter clamshells L
Quality outcomes	weight loss, stem browning, firmness, fruit quality, dry matter, vitamin C, total polyphenols, anthocyanins, total soluble solids, reducing sugars, total sugars, antioxidant capacity (FRAP)
Limitations	The study relies on theoretical dosing without verifying actual internal concentrations, and uses a small-scale laboratory setup with plastic bags rather than industrial packaging conditions.
Scale-up	medium
Relevance	adjacent_mechanism
Full text	oa_pdf: https://www.researchsquare.com/article/rs-6546955/latest.pdf
Confidence	high

8. H D Ariyanto et al. (2019). Effect of 1-Methylcyclopropene Release from Inclusion Complexes Powder on the Quality of Tomato by Short-Term Treatment. IOP Conference Series: Earth and Environmental Science, 309,...

DOI	10.1088/1755-1315/309/1/012065
Year	2019
Venue	IOP Conference Series: Earth and Environmental Science
Type	conference_paper
1-MCP carrier	α -cyclodextrin
Antimicrobial	1-MCP
1-MCP loading	5, 15 and 30 mg mg
Produce	tomatoes
Quality	red colour index
Limitations	Small sample size and lab-only conditions; short-term storage evaluation.
Scale-up	low
Relevance	core
Full text	oa_html: https://doi.org/10.1088/1755-1315/309/1/012065
Confidence	medium

9. Cheng Zhao et al. (2026). Combined application of 1-methylcyclopropene and silicon-window modified atmosphere packaging delays cell wall degradation and softening in cold-stored 'Saiwaihong' apples. Food Chemi...

DOI	10.1016/j.foodchem.2026.150131
Year	2026
Venue	Food Chemistry
Type	original_research
Produce	'Saiwaihong' apples
Quality	firmness loss 21.50%, weight loss 6.23%
Relevance	core
Full text	oa_html: https://research.wur.nl/en/publications/combined-application-of-1-methylcyclopro
Confidence	medium

10. Liying Yang et al. (2026). Efficient Preservation of Perishable Fruits by Erasable Metal-Organic Frameworks. Advanced Science, 13, e19222.

DOI	10.1002/advs.202519222
Year	2026
Venue	Advanced Science
Type	original_research
Substrate	carboxymethyl chitosan (CMCS)
Antimicrobial	clove essential oil (CEO)
Preparation	coating
RH conditions	75 %
Storage temp.	25 °C
Produce	emperor bananas

Mechanical	tensile strength: 30–35 MPa MPa
Barrier	water-vapour transmission rate: $1104.99 \pm 5.37 \text{ g m}^{-2} \text{ day}^{-1}$
Limitations	high energy consumption and reliance on organic solvents
Scale-up	low
Relevance	core
Full text	epmc_jats:PMC12970200
Confidence	high

11. Hongsheng Zhou et al. (2025). Improving the Surface Color and Delaying Softening of Peach by Minimizing the Harmful Effects of Ethylene in the Package. Foods, 14, 2472.

DOI	10.3390/foods14142472
Year	2025
Venue	Foods
Type	original_research
Storage temp.	$22 \pm 2 \text{ }^{\circ}\text{C}$
1-MCP loading	$1.5 \mu\text{L}\cdot\text{L}^{-1}$
Release	8 d
Food type	Honey peach
Quality	firmness, color change, anthocyanin content, weight loss
Scale-up	low
Relevance	background
Full text	epmc_jats:PMC12295221
Confidence	medium

12. Laras Putri Wigati et al. (2025). Influence of cellulose nanocrystal formulations on the properties of pregelatinized cornstarch and cornmint essential oil films. Scientific Reports, 15, 18997.

DOI	10.1038/s41598-025-03318-8
Year	2025
Venue	Scientific Reports
Type	original_research
Substrate	pregelatinized cornstarch
Antimicrobial	cornmint essential oil
Preparation	solution_casting
Mfg. temp.	$60 \text{ }^{\circ}\text{C}$
Produce	Apples
Barrier	water vapor transmission rate, water vapor permeability $\text{g h}^{-1} \text{ m}^{-2}$, $\text{g mm h}^{-1} \text{ m}^2 \text{ kPa}$
Limitations	Reduced mechanical strength at higher CNC concentrations and lack of real-world food application testing.
Scale-up	medium
Relevance	adjacent_material
Full text	epmc_jats:PMC12125276
Confidence	medium

13. Zhibin Pan et al. (2024). Effects of oregano essential oil Pickering emulsion and ZnO nanoparticles on the properties and antibacterial activity of konjac glucomannan/carboxymethyl chitosan nanocomposite films...

DOI	10.1039/d3ra07845k
Year	2024
Venue	RSC Advances
Type	original_research
Substrate	konjac glucomannan/carboxymethyl chitosan
Antimicrobial	oregano essential oil
Preparation	solution_casting
Mfg. temp.	$45 \text{ }^{\circ}\text{C}$
AM outcomes	inhibition zones against <i>E. coli</i> and <i>S. aureus</i>
Scale-up	low
Relevance	adjacent_material
Full text	epmc_jats:PMC10882515
Confidence	medium

14. Anna Yu. Pnevskaya et al. (2026). In Situ Diffuse Reflectance Infrared Fourier-Transform Spectroscopy Investigation of 1-Methylcyclopropene Adsorption in Cobalt-Formate Metal-Organic Framework. ChemistryOpen, ...

DOI	10.1002/open.202500559
Year	2026
Venue	ChemistryOpen
Type	original_research
Antimicrobial	1-methylcyclopropene (1-MCP)
Preparation	other
Trigger	temperature
Storage temp.	ca. 55 °C
Release	other
Release duration	about 20 min
Performance result	>90% of 1-MCP capture efficiency by Co-FA at room temperature
Produce	banana fruits
Quality	yellow color, solid texture, brown spotting, darker color, soft composition, mold
Limitations	Lack of humidity control in validation tests led to mold growth after 12 days across all groups.
Scale-up	not_assessable
Relevance	adjacent_mechanism
Full text	epmc_jats:PMC12949449
Confidence	high

15. Yuxuan Ban et al. (2026). Preparation of a Pickering emulsion film loaded with essential oils extracted from chili pepper leaves and its application in the post-harvest preservation of fresh broccoli. Current ...

DOI	10.1016/j.crfs.2026.101411
Year	2026
Venue	Current Research in Food Science
Type	original_research
Substrate	chitosan/polyvinyl alcohol
Antimicrobial	chili pepper leaf essential oil (CPLEO)
Preparation	solution_casting
Mfg. temp.	40 °C
RH conditions	90 ~ 95 %
Storage temp.	4 ± 2 °C
Produce	fresh broccoli
Mechanical	18.5 MP tensile strength, 280% elongation at break MP, %
Barrier	2.0 g/m ² /h water vapor permeability g/m ² /h
AM outcomes	inhibition zone diameter increased by 50% against <i>Listeria monocytogenes</i> and 158% against <i>Erwinia carotovora</i>
Limitations	Preservation efficacy validated only for broccoli under short-term refrigerated conditions; lacks industrial scalability data and safety assessments of CPLEO migration.
Scale-up	low
Relevance	adjacent_material
Full text	epmc_jats:PMC13138034
Confidence	high

16. Kyoungook Kim et al. (2025). Effects of sachet-type 1-methylcyclopropene (1-MCP) treatment on fruit quality during ambient and cold storage of Hongro apples. Food Science and Preservation, 32, 595-602.

DOI	10.11002/fsp.2025.32.4.595
Year	2025
Venue	Food Science and Preservation
Type	original_research
Trigger	liquid_water
Storage temp.	20±2°C, 0±0.5°C °C
Produce	Hongro apples
Package	10 kg box kg
Quality outcomes	firmness (64.0-65.2 N), colour (L*, a*, b*), soluble solids (SSC), weight loss, ethylene concentration (IEC: 0.4-14.4 µL/L)
Limitations	Requires further studies to determine optimal SFIS concentration for practical grower application
Scale-up	low
Relevance	core
Full text	oa_html: https://doi.org/10.11002/fsp.2025.32.4.595
Confidence	medium

17. Jia Kan et al. (2025). Interfacial Engineering of High-Performance Pickering Emulsion-Gelatin Composite Films for Active Packaging. Foods, 14, 3978.

DOI	10.3390/foods14223978
Year	2025
Venue	Foods
Type	original_research
Substrate	gelatin
Antimicrobial	oregano essential oil
Preparation	solution_casting
Mfg. temp.	35 °C
Trigger	passive_diffusion
Storage temp.	25 °C
Release	other
Release model	fickian
Release result	The Korsmeyer-Peppas model showed higher correlation coefficients ($R^2 = 0.9592$ for 50% ethanol and 0.9518 for 10% ethanol) than the Higuchi model ($R^2 = 0.8941$ for 50% ethanol and 0.8879 for 10% ethanol). The release exponent (n) values of 0.2389 and 0.1991 (<0.45) confirmed that the OEO release followed a Fickian diffusion mechanism
Mechanical	tensile strength: 41.2 MPa, elongation at break: 97.5%, modulus: 744.9 MPa, puncture: exceptional resistance MPa, %
Barrier	WVP: $3.98 \times 10^{-14} \text{ g}\cdot\text{cm}\cdot\text{cm}^{-2}\cdot\text{s}^{-1}\cdot\text{Pa}^{-1}$, OP: $1.89 \times 10^{-17} \text{ cm}^3\cdot\text{cm}\cdot\text{cm}^{-2}\cdot\text{s}^{-1}\cdot\text{Pa}^{-1}$ $\text{g}\cdot\text{cm}\cdot\text{cm}^{-2}\cdot\text{s}^{-1}\cdot\text{Pa}^{-1}$, cm $3\cdot\text{cm}\cdot\text{cm}^{-2}\cdot\text{s}^{-1}\cdot\text{Pa}^{-1}$
AM outcomes	inhibition zones: E. coli 22.32 ± 0.24 mm, S. aureus 39.58 ± 0.74 mm
Scale-up	medium
Relevance	adjacent_material
Full text	epmc_jats:PMC12652791
Confidence	high

18. Qiyuan Feng et al. (2025). Tannic Acid-Modified Sodium Caseinate Pickering Emulsion Coatings: Characterization, Enhanced Mechanical/Antibacterial Properties, and Application in Cherry Tomato Preservation. Food...

DOI	10.3390/foods14183190
Year	2025
Venue	Foods
Type	original_research
Substrate	carboxymethyl tara gum
Antimicrobial	bergamot essential oil
Preparation	coating
Release	other
Release result	relatively stable and slower than SC
Produce	cherry tomatoes
Quality	weight loss, firmness, TSS (soluble solids), lycopene, vitamin C, MDA
AM outcomes	inhibition zones against Escherichia coli and Staphylococcus aureus
Limitations	Lab-only conditions without supply chain simulations.
Scale-up	not_assessable
Relevance	adjacent_mechanism
Full text	epmc_jats:PMC12469039
Confidence	low

19. Andreas Romulo et al. (2024). Enhancing Antimicrobial Activity of Thyme Essential Oil Through Cellulose Nano Crystals-Stabilized Pickering Emulsions. Foods, 13, 3706.

DOI	10.3390/foods13223706
Year	2024
Venue	Foods
Type	original_research
Substrate	sodium alginate
Antimicrobial	thyme essential oil (TEO)
Preparation	solution_casting
Mfg. temp.	35 °C
AM outcomes	MIC50 of 4096 µg/mL against Staphylococcus aureus and Escherichia coli; MIC50 > 8192 µg/mL for 5% TEO
Limitations	Formulation requires further optimization and broader application testing
Scale-up	low
Relevance	adjacent_material
Full text	epmc_jats:PMC11593629
Confidence	medium

20. Peng Liu et al. (2015). Research on SO₂ Controlled Release Packaging on the Preservation Performance of 'Kyoho' Grapes. Applied Mechanics and Materials, 731, 369-373.

DOI 10.4028/www.scientific.net/amm.731.369
 Year 2015
 Venue Applied Mechanics and Materials
 Type conference_paper
 Storage temp. 23±1 °C
 Produce Kyoho grapes
 Quality appearance quality, water loss, stem browning, solidity, TSS and TA, rot rate, stem browning index
 Antimicrobial core
 Full text oa_html:<https://doi.org/10.4028/www.scientific.net/amm.731.369>
 Confidence medium

21. Liufeng Wang et al. (2026). Chitosan-Based Composite Films Reinforced with Zein-Inulin-Thyme Essential Oil Pickering Emulsion for Enhanced Structural Integrity and Preservation Capacity. Foods, 15, 484.

DOI 10.3390/foods15030484
 Year 2026
 Venue Foods
 Type original_research
 Substrate chitosan
 Antimicrobial thyme essential oil
 Preparation solution_casting
 Mfg. temp. 35 °C
 Produce blueberries
 Quality weight loss, firmness decline, pH increase, total soluble solids content, reducing sugar content, appearance/decay
 outcomes
 AM outcomes inhibition zone diameter, total plate count (TPC), yeasts and molds (Y&M) counts
 Limitations biosafety and migration behavior not evaluated; practical applicability requires further study
 Scale-up low
 Relevance low
 Full text epmc_jats:PMC12896857
 Confidence low

22. Niloofar Shirdam et al. (2026). Synthesis, characterization, and evaluation of antimicrobial double-layer mat incorporating nisin and thyme essential oil to enhance food safety. Scientific Reports, 16, 10265.

DOI 10.1038/s41598-025-34848-w
 Year 2026
 Venue Scientific Reports
 Type original_research
 Substrate chitosan, poly (vinyl alcohol), cellulose acetate
 Antimicrobial nisin, thyme essential oil
 Preparation electrospinning
 Mfg. temp. 90 °C
 Mechanical elongation at break of 21.66 ± 0.62% %
 AM outcomes inhibition zones reaching 27.26 ± 1.60 mm for L. monocytogenes and 25.13 ± 1.40 mm for E. coli; reduced L. monocytogenes by 2 logs
 Limitations Evaluated only in broth-based model systems with two reference strains, lacking testing in real food matrices, variable storage conditions, migration behavior, release kinetics, sensory impacts, barrier properties, and techno-economic assessments.
 Scale-up high
 Relevance adjacent_mechanism
 Full text epmc_jats:PMC13031788
 Confidence medium

23. Xiuxiu Teng et al. (2025). 4D printed deformation labels with machine learning for monitoring and preservation of respiring climacteric fruits. Nature Communications, 16, 11525.

DOI 10.1038/s41467-025-66554-6
 Year 2025
 Venue Nature Communications
 Type original_research
 Substrate konjac glucomannan grafted with methacrylic anhydride
 Antimicrobial garlic essential oil

Preparation	printing
Mfg. temp.	25 °C
Trigger	mechanical
RH conditions	93 %
Storage temp.	25 °C
Release	other
Release duration	10 days
Release model	korsmeyer_peppas
Release result	cumulative release of 1 upper layer-30%-60° labels was 42.32% higher than that of 4-upper layer-30%-90° labels on 10 th day
Produce	kiwi fruits, green mangoes, and persimmons
Quality	hardness, titratable acid content, soluble solid content, weight loss rate, sensory evaluation
Antimicrobial	total bacterial count
Limitations	Requires dedicated packaging space and is restricted to climacteric fruits
Scale-up	low
Relevance	core
Full text	epmc_jats:PMC12749378
Confidence	high

24. Yiping Liu et al. (2025). Application of a Pickering Emulsion Stabilized by Zein and Cellulose Nanocrystalline Composite Particles to Preserve Kiwifruit. *Molecules*, 30, 3478.

DOI	10.3390/molecules30173478
Year	2025
Venue	Molecules
Type	original_research
Substrate	zein colloidal particles (ZCPs) and cellulose nanocrystals (CNCs)
Antimicrobial	carvacrol
Preparation	other
Produce	kiwifruit
Quality	decay
Antimicrobial	inhibition rate against Botryosphaeria dothidea
Relevance	adjacent_material
Full text	epmc_jats:PMC12430640
Confidence	medium

25. Mariangela de Fátima Silva et al. (2025). Chitosan-Lemongrass Essential Oil on Paperboard for Active Food Packaging Applications. *Polymers*, 17, 473.

DOI	10.3390/polym17040473
Year	2025
Venue	Polymers
Type	original_research
Substrate	paperboard
Antimicrobial	lemongrass essential oil
Preparation	coating
Mfg. temp.	120 °C
Limitations	lab-only extrapolation from medical device cytotoxicity standards using drastic extraction conditions not reflective of real packaging use
Scale-up	low
Relevance	adjacent_material
Full text	epmc_jats:PMC11859360
Confidence	medium

26. Tianyu Jiang et al. (2025). Sustainable active packaging developed using starch-based foam incorporating 1-Methylcyclopropene@ α -Cyclodextrin. *Carbohydrate Polymers*, 347, 122696.

DOI	10.1016/j.carbpol.2024.122696
Year	2025
Venue	Carbohydrate Polymers
Type	original_research
Substrate	starch-based foam sheets
1-MCP carrier	α -Cyclodextrin
Antimicrobial	1-MCP
Release result	significant reduction in ethylene production and respiration rates by up to 60.49 % and 57.50 % respectively, leading to an extension of the shelf life of climacteric fruits by 5-10 days
Quality	ethylene production reduction (60.49 %), respiration rate reduction (57.50 %), shelf life extension (5-10 days)
Relevance	core

Confidence medium

27. Limeng Chang et al. (2026). Humidity-responsive starch foam pads for humidity-triggered release of 1-methylcyclopropene in litchi fruit preservation. Food Wellness, 2, 100060.

DOI 10.1016/j.foodw.2026.100060
 Year 2026
 Venue Food Wellness
 Type original_research
 Substrate starch foam
 1-MCP carrier α -cyclodextrin
 Preparation extrusion
 Trigger humidity_moisture
 RH conditions 93 %
 Storage temp. 25 °C
 1-MCP loading 18.5 g/m² and 27.8 g/m² g/m²
 Release result 18.5 g/m² exhibited a more regular and sustained release profile; 27.8 g/m² showed incomplete and less regular release
 Produce litchi
 Quality pericarp browning, ethylene production, respiration rate, membrane stability, quality deterioration
 Limitations limited to litchi fruit; requires validation on other species before broader application
 Relevance core
 Confidence medium

28. Yanyi Xue et al. (2026). Morphology-regulated Pickering emulsions stabilized by soybean residue-derived nanocellulose for tunable-release bio-based food packaging. International Journal of Biological Macromole...

DOI 10.1016/j.ijbiomac.2026.153411
 Year 2026
 Venue International Journal of Biological Macromolecules
 Type original_research
 Substrate chitosan
 Antimicrobial lemon essential oil
 Produce blueberry
 Quality weight loss, firmness, titratable acidity
 Mechanical 20.38 MPa
 Barrier 7.30×10^{-7} g/m·s·Pa
 Scale-up high
 Relevance core
 Confidence medium

29. Wei Zhou et al. (2021). 3D printed nanocellulose-based label for fruit freshness keeping and visual monitoring. Carbohydrate Polymers, 273, 118545.

DOI 10.1016/j.carbpol.2021.118545
 Year 2021
 Venue Carbohydrate Polymers
 Type original_research
 Substrate cellulose nanofibers (CNF)
 1-MCP carrier chitosan
 Antimicrobial 1-methylcyclopropene (1-MCP)
 Preparation printing
 Trigger other
 Release result sustained release behavior
 Produce litchis
 Quality shelf life
 Relevance core
 Confidence medium

30. Syariful Mubarak et al. (2022). Impact of 1-Methylcyclopropene combined with chitosan on postharvest quality of tropical banana 'Lady Finger'. International Journal of Food Properties, 25, 1171-1185.

DOI 10.1080/10942912.2022.2074028
 Year 2022
 Venue International Journal of Food Properties
 Type original_research

Produce Lady Finger banana
Quality pH, TA, TSS, total sugar, color index, weight loss, shelf life
Confidence medium

31. J.H. Hotchkiss et al. (2007). Release of 1-Methylcyclopropene from Heat-Pressed Polymer Films. Journal of Food Science, 72.

DOI 10.1111/j.1750-3841.2007.00391.x
Year 2007
Venue Journal of Food Science
Type original_research
1-MCP carrier alpha-cyclodextrin
Antimicrobial 1-methylcyclopropene
Preparation other
Mfg. temp. above 100 degrees C
RH conditions >= 75 %
1-MCP loading 8/140 mg/mg polymer
Release gc_fid
Release 36 to 48 h
Release result rapid and high
Confidence low

32. Hermawan Dwi Ariyanto et al. (2026). Effectiveness of 1-Methylcyclopropene Coated Paper-Based Shellac Solution During Postharvest Treatment of 'Orin' Apples as Functional Packaging for Extending Shelf Life. Pa...

DOI 10.1002/pts.70101
Year 2026
Venue Packaging Technology and Science
Type original_research
Substrate paper
1-MCP carrier alpha-cyclodextrin, inclusion complex
Preparation coating
Trigger humidity_moisture
Storage temp. 4, 20 °C
Release 15 days days
Release model other
Release result 320 µL L -1
Produce 'Orin' Apples
Confidence medium

33. Jiarui Tong et al. (2026). Trans-2-hexenal-loaded gelatin/dialdehyde starch/nanocellulose cushioning aerogel: synergistic postharvest preservation of Chinese bayberry with 1-MCP. Food Research International, 2...

DOI 10.1016/j.foodres.2026.118612
Year 2026
Venue Food Research International
Type original_research
Substrate gelatin/dialdehyde starch
Antimicrobial trans-2-hexenal
Preparation lyophilization
Storage temp. 4 °C
Kinetic model first_order
Release result sustained E2H release behavior
Produce Chinese bayberry
Quality decay, firmness
Antifungal Penicillium citrinum
Confidence medium

34. Jiayan Zhang et al. (2025). An active packaging film prepared from aqueous two-phase Pickering emulsion and chitosan for fruit preservation. Food Chemistry, 493, 146033.

DOI 10.1016/j.foodchem.2025.146033
Year 2025
Venue Food Chemistry
Type original_research

Substrate chitosan
Produce bananas and tomatoes
Scale-up not_assessable
Relevance adjacent_material
Confidence medium

35. Han Zhang (2016). Optimization of encapsulation methods and conditions to maximize 1-MCP loading in modified beta-cyclodextrins.

DOI 10.7282/t39z970v
Year 2016
Type original_research
1-MCP carrier modified β -cyclodextrin (M β CD)
Preparation other
1-MCP loading 0.535 %
Relevance core
Confidence medium

36. Suliman Khan et al. (2026). Carvacrol-enriched chitosan/berry wax active films via Pickering emulsion: a sustainable solution for extending meat shelf-life. Food Chemistry, 512, 148945.

DOI 10.1016/j.foodchem.2026.148945
Year 2026
Venue Food Chemistry
Type original_research
Substrate chitosan/berry wax
Antimicrobial carvacrol essential oil
Preparation other
Storage temp. 4 °C
Produce rabbit meat
AM outcomes Escherichia coli, Staphylococcus aureus, and Pseudomonas aeruginosa
Confidence low

37. Lin Zhu et al. (2026). Construction of Pickering Emulsion of Amomum tsaoko Essential Oil Based on Cellulose Nanocrystals to Enhance Pullulan Film: Structure, Antibacterial Property, and Preservation Effect of ...

DOI 10.3390/foods15132282
Year 2026
Venue Foods
Type original_research
Substrate pullulan
Antimicrobial Amomum tsaoko essential oil
Produce Mango
Quality weight loss, soluble solids
Scale-ups not_assessable
Relevance adjacent_material
Confidence medium

38. Chongfeng He et al. (2025). Construction of magnetic response nanocellulose particles to realize smart antibacterial of Pickering emulsion. International Journal of Biological Macromolecules, 294, 139408.

DOI 10.1016/j.ijbiomac.2024.139408
Year 2025
Venue International Journal of Biological Macromolecules
Type original_research
Substrate cellulose nanocrystal
Antimicrobial oregano essential oil
Preparation other
Trigger other
Release 0-8 h
Release result The OEO released rapidly within 0-8 h followed by the slow release when the emulsion was stimulated by 0.2 T, 0.4 T and 0.6 T magnetic field.
AM outcomes close to 100 %
Scale-up not_assessable
Relevance adjacent_mechanism
Confidence medium

39. Mahmoud Younis et al. (2025). Improving Barhi date shelf life with cellulose nanocrystals/cinnamon oil Pickering emulsion incorporated into sodium alginate-based films. International Journal of Biological Macromolecules, 101,...

DOI 10.1016/j.ijbiomac.2025.143058
 Year 2025
 Venue International Journal of Biological Macromolecules
 Type original_research
 Substrate sodium alginate
 Antimicrobial cinnamon essential oil
 Release result over 60 %
 Produce Barhi dates
 Quality firmness
 Butcher 42.23 % %
 AM outcomes over 60 %
 Confidence medium

40. Zou X et al. (2025). Incorporation of Schizochytrium limacinum oil-based pickering emulsion in chitosan active composite films for strawberry preservation.

Year 2025
 Type original_research
 Substrate chitosan
 Antimicrobial Schizochytrium limacinum (SL) oil
 Preparation solution_casting
 Produce strawberry
 Quality weight loss, firmness
 Butcher $1.354 \times 10^{-12} \text{ g}\cdot\text{cm}/\text{cm}^2\cdot\text{s}\cdot\text{Pa}$
 AM outcomes 28.84, 14.90, 95.19, 88.19
 Confidence low

41. Özge Horzum & Randolph M. Beaudry (2025). “In-box” 1-methylcyclopropene (1-MCP) treatments improve quality retention of ‘Red Delicious’ apple fruit. The Journal of Horticultural Science and Biotechnology, 101,...

DOI 10.1080/14620316.2025.2574319
 Year 2025
 Venue The Journal of Horticultural Science and Biotechnology
 Type original_research
 Substrate corrugated cardboard
 Antimicrobial 1-methylcyclopropene
 Trigger humidity_moisture
 RH conditions ~98.5 %RH
 Release 1 day
 Release result The release of 1-MCP from IBP was found to be highly dependent on the relative humidity of the environment but was essentially complete within 1 day at near saturation (~98.5%RH) at 0°C.
 Produce ‘Red Delicious’ apple
 Quality outcomes respiration rate, ethylene production, colour, firmness, volatiles, soluble solids, acidity, weight loss, and sensory attributes
 Limitations Current efficacy could be improved by adjusting box venting, construction materials, release kinetics, and active ingredient dosage.
 Relevance core
 Confidence medium

42. Chang Shu et al. (2023). Effect of Humidity-Triggered Controlled-Release 1-Methylcyclopropene (1-MCP) on Postharvest Quality of Papaya Fruit. Horticulturae, 9, 1062.

DOI 10.3390/horticulturae9101062
 Year 2023
 Venue Horticulturae
 Type original_research
 Trigger humidity_moisture
 RH conditions 85 %
 Release result inhibited fruit softening, and reduced weight loss and peel color deterioration
 Produce Papaya (Carica papaya L.)
 Quality fruit softening, weight loss, peel color deterioration
 Relevance core

Confidence medium

43. Custódia Gago et al. (2022). 1-Methylcyclopropene and lemongrass essential oil nanocoatings effect on the preservation of cold stored 'Rocha' pear. Postharvest Biology and Technology, 192, 111992.

DOI 10.1016/j.postharvbio.2022.111992
 Year 2022
 Venue Postharvest Biology and Technology
 Type original_research
 Substrate sodium alginate
 Antimicrobial lemongrass essential oil
 RH conditions 90–95 %
 Storage temp. 0 °C
 1-MCP loading 312 nL L⁻¹
 Produce 'Rocha' pear
 Quality color CIE (L*, hue), firmness, soluble solids content (SSC), titratable acidity (TA), weight loss
~~Relevance~~
 core
 Confidence medium

44. Esmat F. Ali et al. (2022). The Efficacies of 1-Methylcyclopropene and Chitosan Nanoparticles in Preserving the Postharvest Quality of Damask Rose and Their Underlying Biochemical and Physiological Mechanisms....

DOI 10.3390/biology11020242
 Year 2022
 Venue Biology
 Type original_research
 Storage temp. 4 or 20 °C
 Produce damask rose
 Quality weight loss, relative water content, volatile oil, total anthocyanins, total carotenoids, total phenolics, antioxidant activity, ethylene evolution, H₂O₂ generation, malondialdehyde content, cell membrane functions, catalase activity, ascorbate peroxidase activity, polyphenol oxidase activity, lipoxygenase activity
 outcomes
 Relevance adjacent_material
 Confidence medium

45. Sofia M. Costa et al. (2021). Active natural-based films for food packaging applications: The combined effect of chitosan and nanocellulose. International Journal of Biological Macromolecules, 177, 241-251.

DOI 10.1016/j.ijbiomac.2021.02.105
 Year 2021
 Venue International Journal of Biological Macromolecules
 Type original_research
 Substrate chitosan
 Preparation solution_casting
 Produce chicken meat
 Confidence medium

46. Weina Wu et al. (2021). Novel packaging film for humidity-controlled manipulating of ethylene for shelf-life extension of Agaricus bisporus. LWT, 145, 111331.

DOI 10.1016/j.lwt.2021.111331
 Year 2021
 Venue LWT
 Type original_research
 Substrate polysaccharide
 1-MCP carrier palladium on carbon
 Preparation lamination
 Trigger humidity_moisture
 RH conditions 50 to 100 % RH
 Release other
~~Relevance~~
 controlled by adjusting the humidity
 Produce Agaricus bisporus
 Quality softening, browning, weight loss, soluble solid content
~~Relevance~~
 core
 Confidence medium

47. Liping Kou et al. (2012). Extending the Shelf Life of Edible Flowers with Controlled Release of 1-Methylcyclopropene and Modified Atmosphere Packaging. Journal of Food Science, 77.

DOI 10.1111/j.1750-3841.2012.02683.x
Year 2012
Venue Journal of Food Science
Type original_research
Storage temp. 5 °C
Release result significantly extended storage life
Produce carnations and snapdragons
Quality flower quality, overall quality, electrolyte leakage, abscission, dehydration
Confidence low

48. Sobia Naseem et al. (2026). Chlorogenic acid crosslinked cellulose nanocomposite film embedded with chitosan for improved mechanical and barrier properties. International Journal of Biological Macromolecules, ...

DOI 10.1016/j.ijbiomac.2026.151570
Year 2026
Venue International Journal of Biological Macromolecules
Type original_research
Substrate chitosan
Antimicrobial Chlorogenic acid
Produce strawberries
Barrier water-vapour permeability g/m³ s Pa
Limitations increased CNC content reduces tensile strength
Scale-up low
Relevance adjacent_material
Confidence medium

49. Ahn S et al. (2026). Controlled release system inducing change in stability of volatile essential oil emulsion incorporated in a hydrogel using a responsive polymer for practical active food packaging.

Year 2026
Type original_research
Substrate polyvinyl alcohol (PVA)
Antimicrobial lemongrass essential oil (LO)
Preparation solution_casting
Trigger humidity_moisture
RH conditions 30, 50, 70, and 90 %
Release 72 h
Release result decreasing to 9.98 ± 1.42 % at 90 % RH
Relevance core
Confidence medium

50. Ke Liu et al. (2026). L-cysteine modified carboxymethyl chitosan composite film: sustained release of cinnamon essential oil, enhanced antibacterial properties, and application in shrimp preservation. Inter-nat...

DOI 10.1016/j.ijbiomac.2025.149604
Year 2026
Venue International Journal of Biological Macromolecules
Type original_research
Substrate oxidized hyaluronic acid and thiol-modified carboxymethyl chitosan
Antimicrobial cinnamon essential oil (CEO)
Trigger other
Release 42 h h
Release result sustained release over 42 h
Produce shrimp
Quality outcomes weight loss 10.54 %, TVB-N content 27.55 mg/100 g, MDA content 0.51 mg/100 g, TVC 4.88 log CFU/g, ORP -49.51 mV
Mechanical elongation at break 59.83 %, puncture strength 5.89 kg/mm² %, kg/mm²
Barrier water vapor permeability 7.14 × 10⁻¹² g/m·s·Pa, oxygen permeability 16 × 10⁻⁹ g/m² s g/m·s·Pa, g/m² s
Relevance adjacent_mechanism
Confidence medium

51. Yayue Wang et al. (2026). Moisture-triggered one-step encapsulation of cinnamaldehyde in β -cyclodextrin for humidity responsive release and food preservation. Food Chemistry, 516, 149345.

DOI 10.1016/j.foodchem.2026.149345
 Year 2026
 Venue Food Chemistry
 Type original_research
 Substrate β -cyclodextrin
 Antimicrobial cinnamaldehyde
 Preparation spray_drying
 Trigger humidity_moisture
 Release result CIN release was humidity-responsive with sustained kinetics.
 Produce blueberries and cherry tomatoes
 AM outcomes Staphylococcus aureus, Escherichia coli, Botrytis cinerea, and Penicillium expansum
 Scale-up high
 Relevance core
 Confidence medium

52. Tingting Zhang et al. (2026). TEMPO-oxidized cellulose nanofiber/chitosan composite films stabilized by cumin essential oil-loaded Pickering emulsions for multifunctional food preservation. International Journ...

DOI 10.1016/j.ijbiomac.2026.152106
 Year 2026
 Venue International Journal of Biological Macromolecules
 Type original_research
 Antimicrobial cumin essential oil
 Produce tomatoes and blueberries
 Quality weight loss, firmness decline, pH changes
 Mechanical mechanical strength up to 45.2 MPa MPa
 Barrier water vapor transmission ~60%, oxygen permeability lower %
 AM outcomes inhibition zone diameters 19.5 mm for Staphylococcus aureus and 24.00 mm for Escherichia coli
 Confidence medium

53. Xiaotong Zou et al. (2025). Development and characterization of corn starch-based films enhanced with Chlorella vulgaris nanocellulose-stabilized Pickering emulsion of Zanthoxylum bungeanum essential oil for c...

DOI 10.1016/j.ijbiomac.2025.140837
 Year 2025
 Venue International Journal of Biological Macromolecules
 Type original_research
 Substrate corn starch
 Antimicrobial Zanthoxylum bungeanum essential oil (ZBEO)
 Produce cherry tomatoes
 Quality weight loss, acidity, total soluble solids
 Confidence medium

54. Li M et al. (2025). Performance of konjac glucomannan/tragacanth gum-stabilized eugenol particles in konjac glucomannan/chitosan films and their application in blueberry preservation.

Year 2025
 Type original_research
 Substrate konjac glucomannan/chitosan
 Antimicrobial eugenol
 Produce blueberries
 Quality decay rate, weight loss, firmness, and soluble solids
 Confidence low

55. Xiaoxue Wang et al. (2025). Preparation, Characterization and Application of Antimicrobial Pectin-Konjac Glucomannan Composite Films Incorporating Cellulose Nanocrystals Stabilized Clove Essential Oil Pickerin...

DOI 10.2139/ssrn.5102163
 Year 2025
 Type original_research

Substrate	pectin and konjac glucomannan
Antimicrobial	clove essential oil
Release result	CCPE could further slow down the release rate of CEO from composite films
Produce	grape
Mechanical	tensile strength: 23.58 MPa MPa
Barrier	water-vapour permeability: 1.05×10^{-10} g/m·s·Pa g/m·s·Pa
AM outcomes	inhibition zone diameters: 38.77 mm (S. aureus), 29.03 mm (E. coli)
Scale-up	low
Relevance	core
Confidence	medium

56. Xiaoxue Wang et al. (2025). Preparation, characterization and application of antimicrobial pectin-konjac glucomannan composite films incorporating cellulose nanocrystals stabilized clove essential oil pickerin...

DOI	10.1016/j.lwt.2025.117855
Year	2025
Venue	LWT
Type	original_research
Substrate	pectin and konjac glucomannan
Antimicrobial	clove essential oil
Produce	grape
Mechanical	23.58 MPa MPa
Barrier	1.05×10^{-10} g/m·s·Pa g/m·s·Pa
AM outcomes	38.77 mm and 29.03 mm
Relevance	core
Confidence	medium

57. Anna Yu. Pnevskaya & Aram L. Bugaev (2023). Theoretical screening of M3(btc)2 metal-organic frameworks for ethylene and 1-methylcyclopropene storage. Results in Chemistry, 5, 100831.

DOI	10.1016/j.rechem.2023.100831
Year	2023
Venue	Results in Chemistry
Type	original_research
1-MCP carrier	MOFs
Limitations	The study relies solely on theoretical/computational screening without experimental validation, limiting its direct utility as practical evidence.
Scale-up	low
Relevance	adjacent_material
Confidence	medium

58. Putri Amanda et al. (2022). Carrageenan functional film integrated with Pickering Emulsion of Oregano Oil Stabilized by Cationic Nanocellulose for Active Packaging. Food Science and Technology International, 3...

DOI	10.1177/10820132221132912
Year	2022
Venue	Food Science and Technology International
Type	original_research
Substrate	carrageenan
Antimicrobial	Oregano essential oil
Preparation	other
Relevance	adjacent_material
Confidence	medium

59. Alana Gabrieli de Souza et al. (2021). Cinnamon Essential Oil Nanocellulose-Based Pickering Emulsions: Processing Parameters Effect on Their Formation, Stabilization, and Antimicrobial Activity. Polysaccharide...

DOI	10.3390/polysaccharides2030037
Year	2021
Venue	Polysaccharides
Type	original_research
Substrate	nanocellulose
Antimicrobial	cinnamon essential oil
Preparation	other

AM outcomes inhibition halos
Scale-up low
Relevance adjacent_material
Confidence medium

60. Gregor Lavrič et al. (2021). Functional Nanocellulose, Alginate and Chitosan Nanocomposites Designed as Active Film Packaging Materials. *Polymers*, 13, 2523.

DOI 10.3390/polym13152523
Year 2021
Venue Polymers
Type original_research
Substrate chitosan, alginate
Barrier WVTR reduced by 15-45%, OTR up to 45% %
Confidence low

61. Tao Yang et al. (2026). Hydrocolloid-based stimuli-responsive nanofiber films: Precision-controlled release of essential oils for sustainable food packaging. *Current Research in Food Science*, 12, 101260.

DOI 10.1016/j.crfs.2025.101260
Year 2026
Venue Current Research in Food Science
Type review
Limitations The document is a review summarizing existing literature rather than presenting original experimental data; it explicitly notes unresolved challenges in scalability and regulatory compliance.
Scale-up low
Relevance core
Full text ePMC_jats:PMC12743375
Confidence medium

62. Zhijun Hu et al. (2017). 1-Methylcyclopropene (MCP)-Containing Cellulose Paper Packaging for Fresh Fruit and Vegetable Preservation: A Review. *BioResources*, 12.

DOI 10.15376/biores.12.1.2234-2248
Year 2017
Venue BioResources
Type review
Substrate cellulosic paper
Trigger humidity_moisture
Limitations The document is a review that proposes a conceptual packaging system without original experimental validation, explicitly stating that future research is required to develop optimized application technologies.
Scale-up high
Relevance core
Full text oa_html:<https://doi.org/10.15376/biores.12.1.2234-2248>
Confidence medium

63. Anand Kumar et al. (2026). Chitosan-Based Active Packaging for Shrimp Preservation: Development, Functionalization, and Industrial Prospects. *Foods*, 15, 1043.

DOI 10.3390/foods15061043
Year 2026
Venue Foods
Type review
Substrate chitosan
Antimicrobial thymol
Preparation solution_casting
Limitations The study is constrained by a non-meta-analytic approach, reliance on English language publications, and the rapid pace of discovery in the field.
Scale-up low
Relevance low
Full text ePMC_jats:PMC13025528
Confidence medium

64. Bruna Moura Bastos et al. (2026). Lipid-Enriched Biopolymer Films for Active Packaging: A Review of Structure, Properties, and Preservation Performance. *Polymers*, 18, 870.

DOI 10.3390/polym18070870
 Year 2026
 Venue *Polymers*
 Type review
 Limitations lab-only conditions / limited to laboratory scale
 Scale-up low
 Relevance core
 Full text epmc_jats:PMC13074946
 Confidence medium

65. Reshma Krishnan et al. (2025). Emerging Trends and Application of Edible Coating as a Sustainable Solution for Postharvest Management in Stone Fruits: A Comprehensive Review. *Comprehensive Reviews in Food Scie...*

DOI 10.1111/1541-4337.70179
 Year 2025
 Venue *Comprehensive Reviews in Food Science and Food Safety*
 Type review
 Scale-up low
 Relevance core
 Full text epmc_jats:PMC12079166
 Confidence medium

66. Siphumle Owen Jama et al. (2026). Edible Coatings for Fresh Fruits: Functional Roles, Optimization Strategies, and Analytical Perspectives. *Plants*, 15, 132.

DOI 10.3390/plants15010132
 Year 2026
 Venue *Plants*
 Type review
 Limitations Variability in barrier and mechanical properties, challenges in industrial-scale application, insufficient long-term validation under commercial cold-chain conditions, and regulatory uncertainty for active formulations.
 Scale-up low
 Relevance background
 Full text epmc_jats:PMC12788044
 Confidence medium

67. W. M. C. B. Wasala et al. (2021). Application of 1-Methylcyclopropene (1-MCP) for Delaying the Ripening of Banana: A Review. *Asian Research Journal of Agriculture*, 44-56.

DOI 10.9734/arja/2021/v14i130118
 Year 2021
 Venue *Asian Research Journal of Agriculture*
 Type review
 Produce banana
 Relevance core
 Confidence medium

68. Amr Zaitoon et al. (2021). Triggered and controlled release of active gaseous/volatile compounds for active packaging applications of agri-food products: A review. *Comprehensive Reviews in Food Science and Foo...*

DOI 10.1111/1541-4337.12874
 Year 2021
 Venue *Comprehensive Reviews in Food Science and Food Safety*
 Type review
 Confidence low

69. Ana Beltrán Sanahuja & Arantzazu Valdés García (2021). New Trends in the Use of Volatile Compounds in Food Packaging. *Polymers*, 13, 1053.

DOI 10.3390/polym13071053
 Year 2021
 Venue Polymers
 Type review
 Relevance background
 Confidence medium

70. Donghyun Jang et al. (2026). Biopolymer coatings for barrier and functional enhancement of sustainable food packaging: Recent advances and emerging trends. *International Journal of Biological Macromolecules*, 3...

DOI 10.1016/j.ijbiomac.2026.152492
 Year 2026
 Venue International Journal of Biological Macromolecules
 Type review
 Limitations performance variability under humid conditions, limited adhesion between biopolymers and cellulose fibers, cost-intensive processing, and uncertainties regarding biodegradation and environmental fate
 Scale-up low
 Relevance adjacent_material
 Confidence medium

71. Dimitrie Stoica et al. (2026). Protein-Polysaccharide Gel Systems for Antioxidant and Antimicrobial Delivery in Sustainable Food Packaging: A Review. *Gels*, 12, 297.

DOI 10.3390/gels12040297
 Year 2026
 Venue Gels
 Type review
 Limitations stability, scalability, and regulatory compliance
 Confidence medium

72. Amanullah Sabir et al. (2026). Polysaccharides: Nature's Guardians of Freshness in Food Preservation. *Molecules*, 31, 1545.

DOI 10.3390/molecules31091545
 Year 2026
 Venue Molecules
 Type review
 Limitations Challenges related to stability, sensory impact, and regulatory compliance are also discussed.
 Confidence medium

73. Weina Jiang et al. (2025). Multifunctional Metal-Organic Frameworks for Enhancing Food Safety and Quality: A Comprehensive Review. *Foods*, 14, 4111.

DOI 10.3390/foods14234111
 Year 2025
 Venue Foods
 Type review
 Confidence low

74. Aleksandra Bibow & Wiesław Oleszek (2024). Essential Oils as Potential Natural Antioxidants, Antimicrobial, and Antifungal Agents in Active Food Packaging. *Antibiotics*, 13, 1168.

DOI 10.3390/antibiotics13121168
 Year 2024
 Venue Antibiotics
 Type review
 Limitations The study identifies constraints and deficiencies in current investigations of essential oils for food packaging efficacy.
 Relevance adjacent_material
 Confidence medium

75. Sharath Kumar Nagaraja et al. (2026). A Comprehensive Review of Biochemical Insights and Advanced Packaging Technologies for Shelf-Life Enhancement of Temperate Fruits. Biosensors, 16, 94.

DOI 10.3390/bios16020094
 Year 2026
 Venue Biosensors
 Type review
 Scale-up low
 Relevance core
 Full text epmc_jats:PMC12938432
 Confidence medium

76. Mariana A. Andrade et al. (2025). Emerging Trends in Active Packaging for Food: A Six-Year Review. Foods, 14, 2713.

DOI 10.3390/foods14152713
 Year 2025
 Venue Foods
 Type review
 Limitations reliance on in vitro conditions and lack of standardized methodologies
 Scale-up low
 Relevance adjacent_material
 Full text epmc_jats:PMC12346170
 Confidence medium

77. Antia G. Pereira et al. (2026). Natural Antimicrobials from Plants Used as Food Preservatives. Foods, 15, 1309.

DOI 10.3390/foods15081309
 Year 2026
 Venue Foods
 Type review
 Scale-up low
 Relevance background
 Full text epmc_jats:PMC13114991
 Confidence medium

78. Mo Chen et al. (2020). Stabilization and controlled release of gaseous/volatile active compounds to improve safety and quality of fresh produce. Trends in Food Science & Technology, 95, 33-44.

DOI 10.1016/j.tifs.2019.11.005
 Year 2020
 Venue Trends in Food Science & Technology
 Type review
 Limitations The field lacks sufficient understanding of physical and chemical interactions needed to predict retention and release kinetics, alongside regulatory constraints that must be considered for practical application.
 Scale-up not_assessable
 Relevance core
 Confidence medium

79. Abu-Bakr Ali Abu-Goukh (2023). Review Article on: 1-Methylcyclopropene (1-MCP) a Breakthrough to Delay Ripening and Extend Shelf-Life of Horticultural Crops. University of Khartoum Journal of Agricultural Scie...

DOI 10.53332/uofkjas.v21i2.1842
 Year 2023
 Venue University of Khartoum Journal of Agricultural Sciences
 Type review
 Confidence low

80. Afreen Sultana et al. (2022). Metal-organic frameworks for active food packaging. A review. Environmental Chemistry Letters, 20, 1479-1495.

DOI 10.1007/s10311-022-01387-z
Year 2022
Venue Environmental Chemistry Letters
Type review
Confidence low

81. C.B. Watkins & W.B. Miller (2005). 1-METHYLCYCLOPROPENE (1-MCP) BASED TECHNOLOGIES FOR STORAGE AND SHELF-LIFE EXTENSION. Acta Horticulturae, 201-208.

DOI 10.17660/actahortic.2005.687.24
Year 2005
Venue Acta Horticulturae
Type review
Confidence low

82. Ziao Zhang et al. (2026). Bio-Based Smart Packaging Materials for Next-Generation Food Systems. Materials, 19, 2393.

DOI 10.3390/ma19112393
Year 2026
Venue Materials
Type review
Limitations The field faces challenges with long-term operational stability, production costs, scalable manufacturing, and life cycle assessments.
Scale-up not_assessable
Relevance adjacent_material
Confidence medium

Appendix B — Patent family records (all 19 fields)

One block per family. This is technical description only: no freedom-to-operate, infringement, validity or patentability conclusion is expressed or implied.

1. EP1584234A2 — Plant ethylene response inhibition complexes

Publication	EP1584234A2
Priority date	1998-08-20
Publication date	2005-10-12
Inventors	DALY, JAMES; KOURELIS, BOB
Jurisdictions	['US', 'EP', 'JP', 'KR', 'CN', 'AP', 'AT', 'AU', 'BG', 'BR', 'CA', 'CZ', 'DE', 'DK', 'ES', 'HK', 'HR', 'HU', 'IL', 'MX', 'NO', 'NZ', 'OA', 'PL', 'PT', 'SK', 'TR', 'WO', 'ZA']
Family seen	EP1584234A2
Overlap	The document discloses complexes for inhibiting ethylene response in plants, which aligns with 1-MCP delivery concepts. However, the provided text does not specify humidity-responsive packaging or antimicrobial components.
Confidence	medium

2. PL346176A1 — Plant ethylene response inhibition compounds and complexes

Publication	PL346176A1
Priority date	1998-08-20
Publication date	2002-01-28
Jurisdictions	US, EP, JP, KR, CN, AP, AT, AU, BG, BR, CA, CZ, DE, DK, ES, HK, HR, HU, IL, MX, NO, NZ, OA, PL, PT, SK, TR, WO, ZA
Family seen	PL346176A1
Overlap	The document covers ethylene response inhibition compounds and complexes, which aligns with the 1-MCP component of the project. However, the provided metadata lacks details on humidity-responsive release mechanisms, packaging formats, or antimicrobial additives.
Confidence	medium

3. US6017849A — Synthesis methods, complexes and delivery methods for the safe and convenient storage, transport and application of compounds for inhibiting the ethylene response in plants

Publication	US6017849A
Priority date	1998-08-20
Publication date	2000-01-25
Assignee	Biotechnologies for Horticulture Inc; AgroFresh Inc
Inventors	James Daly; Bob Kourelis
Jurisdictions	US, EP, CN, JP, AU, SK, PT, DK, HK, OA, ES, DE, CA, HR, KR, BR, AT, AP, NO, ZA, BG, IL, CZ, TR, MX, NZ
Family seen	US6017849A
Claim 1	A complex formed from a molecular encapsulation agent and a cyclopropene derivative compound.
Packaging	Package (polyvinyl alcohol)
1-MCP carrier	Cyclodextrin (specifically alpha-cyclodextrin)
Release	humidity_moisture
Preparation	Dissolution of the molecular encapsulation agent complex in a solvent (water) liberates the gaseous compound.
Control limits	Complex of molecular encapsulation agent + cyclopropene derivative where n=1-4 and R is H, C1-C4 alkyl, hydroxy, halogen, C1-C4 alkoxy, amino, or carboxy.
Confidence	low

4. KR100791251B1 — Cyclopropene delivery system requiring less water

Publication	KR100791251B1
Priority date	2000-09-29
Publication date	2008-01-04
Assignee	룸 앤드 하스 캄파니 / Rohm & Haas Company
Inventors	코스탄섹에드워드찰스
Jurisdictions	KR, US, EP, JP, CN, AR, AU, BR, CA, DE, ES, IL, MX, NZ, TW
Family seen	KR100791251B1
Manufacturing	contacting cyclopropene gas with cyclodextrin solution/slurry to form complex, precipitating/filtrating, then blending powder complex with water absorbent polymer or soluble compound
Claim limits	Composition comprises (a) molecular encapsulating agent encapsulating cyclopropene, (b) adjuvants, (c) water absorbent. Ratios: 3-97 wt% complex, 3-97 wt% water absorbent.
Confidence	low

5. US6426319B1 — Delivery systems for cyclopropenes requiring less water

Publication	US6426319B1
Priority date	2000-09-29
Publication	2002-07-30
Assignee	AgroFresh Inc (Current); Rohm and Haas Co (Original)
Inventors	Edward Charles Kostansek
Jurisdictions	['US', 'EP', 'CN', 'JP', 'KR', 'AR', 'AU', 'BR', 'CA', 'DE', 'ES', 'IL', 'MX', 'NZ', 'TW']
Family seen	US6426319B1
Claim limits	Comprising from 3 percent to 97 percent, by weight, cyclopropene/encapsulation agent complex and 3 percent to 97 percent, by weight, water absorbent material.
Confidence	low

6. US6953540B2 — Continuous process for the preparation of encapsulated cyclopropenes

Publication	US6953540B2
Priority date	2000-09-29
Publication	2005-10-11
Assignee	AgroFresh Inc
Inventors	Joshua Anthony Chong, Vincent John Farozic, Richard Martin Jacobson, Bret Alan Snyder, Randall Wayne Stephens, David Wayne Mosley
Jurisdictions	US
Family seen	US6953540B2
Claim 1	A method of purifying a cyclopropene gas prepared by reaction of an allyl compound with a non-nucleophilic strong base comprising scrubbing the cyclopropene gas in a scrubber with a reactive solvent mixture to remove residual allyl compound to give purified cyclopropene gas.
1-MCP carrier	alpha-cyclodextrin
Claim limits	Scrubbing the gas with a reactive solvent mixture to remove residual allyl compound.
Confidence	low

7. CN1325446C — Method and device for producing cyclopropenyl compound

Publication	CN1325446C
Priority date	2002-05-14
Publication	2007-07-11
Assignee	Rohm and Haas Co
Inventors	E·C·考斯坦赛克; L·A·威赛尔; C·L·沃兹
Jurisdictions	['US', 'EP', 'JP', 'KR', 'CN', 'AU', 'BR', 'CA', 'IL', 'MX', 'NZ', 'TW']
Family seen	CN1325446C
1-MCP carrier	alpha-cyclodextrin (alpha-cyclodextrin)
Release control	Gas bubbling through an aqueous suspension containing the inclusion complex to mechanically liberate the gas.
Manufacturing	Joining the cyclopropylene/carrier title complex to water in the container, then passing bubbled gas through the water with the device's gas production equipment to discharge the gas.
Overlap	Discloses using a 1-MCP/cyclodextrin inclusion complex in an aqueous suspension where bubbling gas through the water triggers rapid release of 1-MCP gas for treating stored plants/fruits/vegetables. Overlaps with moisture-triggered release concepts but uses active mechanical/aeration triggering rather than passive humidity absorption in packaging.
Confidence	low

8. US8129450B2 — Articles having a polymer grafted cyclodextrin

Publication	US8129450B2
Priority date	2002-12-10
Publication	2012-03-06
Assignee	Verdant Technologies LLC (Current); Cellresin Technologies LLC (Original)
Inventors	Willard E. Wood, Timothy H. Bohrer, Stanley R. Kellenberger, Neil J. Beaverson
Jurisdictions	US
Family seen	US8129450B2
Claim 1	Claim 1 requires an article comprising a thermoplastic polymer fiber made from an acid anhydride modified polyolefin resin that has covalently bonded cyclodextrin substituents. The cyclodextrin must be substantially free of compounds in its central pore, and the fiber must have a diameter between 10 nm and 2000 nm.
Manufacturing	melt grafting in an extruder; reactive extrusion; blending with unmodified polymer; coextrusion; coating; spun-bond/meltblown/electrospinning
Claim limits	fiber diameter of 10 nm to 2000 nm; cyclodextrin substantially free of a compound in its central pore; covalent bond via reaction product of cyclodextrin with acid anhydride
Overlap	Low overlap. The patent discloses cyclodextrin-grafted polymers for odor/flavor scavenging and barrier enhancement, not 1-MCP controlled release or humidity-responsive mechanisms.
Confidence	low

9. CR9490A — CYCLOPROPEN COMPOSITIONS

Publication	CR9490A
Priority date	2006-11-09
Publication date	2008-01-21
Jurisdictions	['US', 'EP', 'JP', 'KR', 'CN', 'AR', 'AU', 'BR', 'CA', 'CL', 'CO', 'CR', 'DK', 'ES', 'IL', 'MX', 'NZ', 'PL', 'PT', 'TW', 'UA', 'ZA']
Family seen	CR9490A
Claim 1	A composition comprising a cyclopropene molecular encapsulation agent complex and a calcium chloride salt in a specific dry weight ratio, with defined water content conditions.
1-MCP carrier	cyclopropene molecular encapsulation agent complex
Claim limits	Dry weight ratio of salt to complex is 0.03 to 500; water content is ≤30% with non-deliquescent salt, or >30% with salt-to-water dry weight ratio ≥0.005.
Overlap	Addresses cyclopropene carrier systems and moisture parameters relevant to controlled release, but does not claim packaging formats or essential oils.
Design-around	Use a salt other than calcium chloride, alter the dry weight ratio outside claimed ranges, or modify water content limits to fall outside the disclosed parameters.
Confidence	medium

10. US8603524B2 — Delivery of ethylene blocking and/or promoting agents

Publication	US8603524B2
Priority date	2007-01-17
Publication date	2013-12-10
Assignee	AgroFresh Inc
Inventors	Gretchen Baier, Mark D. Newsham, Irina V. Graf, Robert L. Schmitt, Lamy J. Chopin, III, Jacquelyn A. deGroot, Jeffrey J. Wooster, Tarlochan S. Dhadialla
Jurisdictions	US, EP, CN, JP, AU, BR, CA, CL, CR, ES, HU, IL, MX, NI, NZ, PL, PT, SI, WO, ZA
Family seen	US8603524B2
Claim 1	A film comprising an ethylene blocking agent complex (cyclopropene/derivatives + cyclodextrin/crown ether host) intermingled with a water-soluble polymer.
Packaging	film
1-MCP carrier	cyclodextrins or crown ethers
Release	humidity_moisture
Release control	Dissolution or swelling of the intermingled water-soluble polymer matrix upon exposure to water or humidity, which releases the encapsulated ethylene blocking agent.
Claim limits	The ethylene blocking agent is limited to cyclopropene, 3,3-dimethylcyclopropene, methylenecyclopropane and derivatives; the host is limited to cyclodextrins or crown ethers; the water-soluble polymer is limited to methylcellulose, propylcellulose, ethylcellulose, or hydroxyethylcellulose.
Overlap	Directly overlaps with the humidity-responsive, controlled-release 1-MCP packaging concept by claiming a film where a 1-MCP/cyclodextrin complex is intermingled with water-soluble polymers to enable moisture-triggered release. It addresses the core delivery mechanism but does not claim any antimicrobial component.
Design-around	To avoid infringement, one could use a water-insoluble but moisture-permeable polymer matrix, employ a different host system outside cyclodextrins and crown ethers, or incorporate an essential-oil antimicrobial component not covered by this claim.
Confidence	high

11. CN101416658B — 1- MCP/modified stabilizer coating material, production method and use thereof

Publication	CN101416658B
Priority date	2008-11-07
Publication date	2009-04-29
Jurisdictions	['CN']
Family seen	CN101416658B
Packaging	coating
1-MCP carrier	modified stabilizer
Confidence	low

12. CN101578978A — Ethylene absorbent and preparation method thereof

Publication	CN101578978A
Priority date	2009-06-11
Publication date	2009-11-18
Assignee	China Agricultural University
Inventors	曲桂芹, 李小红, 曹建康, 朱毅, 罗云波
Jurisdictions	CN
Family seen	CN101578978A
Claim 1	An ethylene absorbent prepared from a carrier substance, a metal promoter (1-5% of carrier weight), and a palladium divalent compound reaction host (0.3-5% of carrier weight).

Packaging	Granular absorbent placed in air-permeable paper bags or non-woven bags
Claim limits	Metal promoter weight must be 1-5% of carrier mass; reaction host (palladium divalent compound) weight must be 0.3-5% of carrier mass.
Design-around	Use metal promoters other than divalent copper, manganese, or molybdenum; use reaction hosts other than palladium divalent compounds; or adjust weight ratios outside the claimed 1-5% and 0.3-5% ranges.
Confidence	low

13. CN101715760A — 1-methylcyclopropene liquid preparation

Publication	CN101715760A
Priority date	2009-12-01
Publication	2010-06-02
Assignee	Shandong Nutrient Source Food Technology Co Ltd
Inventors	王庆国, 牟文良, 陈庆敏
Jurisdictions	CN
Family seen	CN101715760A
Claim 1	A liquid preparation comprising a 1-methylcyclopropene stable solution as host, mixed with an organic solvent, emulsifier, and synergist in specific weight percentages.
1-MCP carrier	1-methyl cyclopropene stable solution (alcohol/aldehyde mixture)
Manufacturing	Mixing 1-methyl cyclopropene stable solution with organic solvent, then adding and stirring in emulsifier and synergist.
Claim limits	Weight percentages: 1-MCP stable solution 1-80%, organic solvent 10-70%, emulsifier 1-50%, synergist 0.1-5%.
Confidence	low

14. CN104304242B — Maturation or curing inhibitor release from polymer, fiber, film, thin slice or packaging

Publication	CN104304242B
Priority date	2009-12-14
Publication	2017-06-30
Assignee	Cellresin Technologies, Llc
Jurisdictions	['US', 'EP', 'CN', 'JP', 'KR', 'AU', 'BR', 'CA', 'ES', 'MX', 'RU', 'WO']
Family seen	CN104304242B
Packaging	polymer, fiber, film, thin slice or packaging
Confidence	low

15. MX2012006797A — Maturation or ripening inhibitor release from polymer, fiber, film, sheet or packaging.

Publication	MX2012006797A
Priority date	2009-12-14
Publication	2012-07-23
Assignee	Cellresin Tech Llc
Inventors	['Willard E Wood', 'Neil J Beaverson', 'William J Kuduk']
Jurisdictions	['US', 'EP', 'JP', 'KR', 'CN', 'AU', 'BR', 'CA', 'ES', 'MX', 'RU', 'WO']
Family seen	MX2012006797A
Packaging	film, fiber, sheet, laminate, polymer chips, woven or nonwoven fabrics, closures
Release	humidity_moisture
Confidence	low

16. US9642356B2 — Maturation or ripening inhibitor release from polymer, fiber, film, sheet or packaging

Publication	US9642356B2
Priority date	2009-12-14
Publication	2017-05-09
Assignee	Verdant Technologies LLC
Inventors	['Willard E. Wood', 'Neil J. Beaverson', 'William J. Kuduk']
Jurisdictions	['US', 'EP', 'JP', 'KR', 'CN', 'AU', 'BR', 'CA', 'ES', 'MX', 'RU', 'WO']
Family seen	US9642356B2, WO2011081877A1
Claim 1	A controlled release composition comprising a cyclodextrin modified thermoplastic polymer with a covalently bonded cyclodextrin moiety, and 1-methylcyclopropene complexed with the cyclodextrin moiety, formed by contacting the polymer with gaseous 1-MCP under specific pressure, temperature, and moisture conditions.
Packaging	film, fiber, woven or nonwoven fabric, sheet, rigid or semi-rigid packaging material, package insert
1-MCP carrier	cyclodextrin
Release	humidity_moisture
Release control	Humidity-responsive desorption from cyclodextrin inclusion complex within the polymer matrix; moisture contact increases to trigger release proportional to relative humidity in the enclosed package volume.
Claim limits	Cyclodextrin moiety covalently bonded to thermoplastic polymer; 1-MCP complexed with cyclodextrin; formed by contacting polymer (<800 ppm water) with gaseous 1-MCP at molar ratio of about 0.5 to 10 moles inhibitor per mole cyclodextrin, in enclosed space at 1-15 atm pressure and 0°C to 100°C temperature.

Confidence low

17. CN101990936A — 1-methyl-3-(2-methyl cyclopropane)-1-cyclopropylene stable inclusion compound and preparation method thereof

Publication CN101990936A
Priority date 2010-10-19
Publication 2011-03-30
Assignee Xian Jiaotong University
Inventors 延卫, 常乐, 冯江涛
Jurisdictions CN
Family seen CN101990936A
1-MCP carrier cyclodextrin copper complex
Confidence low

18. US8481127B2 — Cyclodextrin compositions, articles, and methods

Publication US8481127B2
Priority date 2011-03-27
Publication 2013-07-09
Assignee Verdant Technologies LLC
Inventors ['Willard E. Wood', 'William J. Kuduk', 'Joseph S. Keute']
Jurisdictions ['US', 'EP', 'CN', 'JP', 'KR', 'AU', 'BR', 'CA', 'CL', 'CO', 'DK', 'ES', 'GB', 'HU', 'IL', 'MX', 'NZ', 'PL', 'PT', 'RU', 'SI', 'TR', 'TW', 'UA', 'WO', 'ZA']
Family seen US8481127B2
Claim 1 A cyclodextrin composition comprising one or more radiation polymerizable monomers, a photoinitiator and a cyclodextrin inclusion complex containing an olefinic inhibitor of ethylene generation in produce.
Packaging film, sheet, bag, punnet, dish, cup, cover, label, paperboard carton, or treated package insert
1-MCP carrier cyclodextrin inclusion complex (specifically α -cyclodextrin)
Overlap Directly overlaps with humidity-responsive, controlled-release 1-MCP packaging using cyclodextrin carriers.
Confidence low

19. CN102440237B — 1-methylcyclopropene preparation and preparation method thereof

Publication CN102440237B
Priority date 2011-09-30
Publication 2014-06-18
Jurisdictions ['CN']
Family seen CN102440237B
Confidence low

20. US9381163B2 — Floating capsules encapsulating particles loaded with one or more drugs

Publication US9381163B2
Priority date 2012-08-16
Publication 2016-07-05
Assignee NANYANG TECHNOLOGICAL UNIVERSITY, SINGAPORE
Inventors LOO, SAY CHYE JOACHIM; LEE, WEI LI; WEE, PEIXIN
Jurisdictions ['US', 'SG', 'WO']
Family seen US20150164815A1, US9381163B2
Confidence low

21. CN102823644A — Fresh-keeping method for muskmelon

Publication CN102823644A
Priority date 2012-09-04
Publication 2012-12-19
Jurisdictions ['CN']
Family seen CN102823644A
Packaging film
Confidence low

22. US20150272148A1 — Encapsulation of gases within cyclodextrins

Publication	US20150272148A1
Priority date	2012-10-10
Publication	2015-10-01
Applicant	THE UNIVERSITY OF QUEENSLAND, AUSTRALIA
Inventors	BHANDARI, BHESH; HO, BINH THANH
Jurisdictions	US, EP, JP, AU, NZ, WO, ZA
Family seen	US20150272148A1
1-MCP carrier	α -cyclodextrin (α -CD)
Release	humidity moisture
Overlap	Directly overlaps by disclosing gas (ethylene/1-MCP) encapsulation in α -cyclodextrin with release rates modulated by relative humidity.
Confidence	low

23. AU2013387471A1 — Methods and compositions for polymer matrix synthesized by polycondensation

Publication	AU2013387471A1
Priority date	2013-04-26
Publication	2015-11-12
Jurisdictions	['US', 'EP', 'JP', 'KR', 'CN', 'AU', 'BR', 'CA', 'CR', 'MX', 'WO']
Family seen	AU2013387471A1
Manufacturing	polymer matrix synthesized by polycondensation
Overlap	The document discloses a polymer matrix synthesized by polycondensation, which could theoretically serve as a controlled-release packaging material, but the provided text lacks specific claims regarding 1-MCP, humidity responsiveness, or antimicrobial components.
Design-around	Utilizing alternative polymer synthesis methods or different active ingredient carriers would avoid the claimed polycondensation-based matrix.
Uncertainty	The actual claims are not included in the provided document excerpt, making it impossible to determine specific claim limitations, release mechanisms, or whether 1-MCP/antimicrobials are claimed.
Confidence	medium

24. CA2910267A1 — Gel formulations for extended release of volatile compounds

Publication	CA2910267A1
Priority date	2013-04-26
Publication	2014-10-30
Jurisdictions	['US', 'EP', 'JP', 'KR', 'CN', 'AU', 'BR', 'CA', 'CR', 'MX', 'WO']
Family seen	CA2910267A1
Overlap	The document covers gel formulations for extended release of volatile compounds, which conceptually aligns with controlled-release packaging but lacks specific mention of methylcyclopropene, humidity responsiveness, or antimicrobial agents in the provided metadata.
Design-around	To avoid this document scope, one could specify non-gel packaging formats, explicitly incorporate methylcyclopropene and essential oils, and define humidity-responsive release mechanisms rather than relying on general gel matrices.
Confidence	medium

25. US20170101250A1 — Active compound formulation package and its subsequent release for use on plant and plant parts

Publication	US20170101250A1
Priority date	2013-05-05
Publication	2017-04-13
Applicant	Mirtech Inc
Inventors	Nazir Mir
Jurisdictions	['US']
Family seen	US20170101250A1
Packaging	aerosol
1-MCP carrier	cyclodextrin or a cyclodextrin derivative
Claim limits	at least one partially encapsulated ethylene response manipulation agent; at least one polyol liquid medium, or a hydrogel medium, or a combination of polyol and hydrogel medium, wherein particles comprising the ethylene response manipulation agent are dispersed or mixed in said medium; and at least one propellant.
Confidence	low

26. PH12016500070A1 — Humidity activated formulation for volatile compounds

Publication	PH12016500070A1
Priority date	2013-07-11
Publication date	2016-04-18
Assignee	Agrofresh Inc
Inventors	Williamson Alexander, Maclean Daniel
Jurisdictions	US, EP, CN, JP, KR, AR, AU, BR, CA, CL, ES, HR, IL, MX, PE, PH, PL, PT, RS, RU, SG, TW, WO
Family seen	PH12016500070A1
Claim 1	Delivery systems in the form of coatings or films that provide controlled release of volatile components (including 1-methylcyclopropene) in vapor form triggered by high relative humidity.
Packaging	coatings or films
Release	humidity_moisture
Release	Humidity activation controls the release rate of volatile components in vapor form.
Antimicrobial	volatile antimicrobial liquids including low molecular weight alcohols and/or aldehydes, 1-methylcyclopropene, and/or other volatile fungicides
Confidence	low

27. US20160330952A1 — Encapsulation and controlled release of volatile organic compounds

Publication	US20160330952A1
Priority date	2014-01-15
Publication date	2016-11-17
Assignee	University of Guelph
Inventors	Loong-Tak Lim, Suramya D.F. Mihindukulasuriya, Gopinadhan Paliyath, Jayasankar Subramanian, J. Alan Sullivan
Jurisdictions	US, CA, WO
Family seen	US20160330952A1
Claim 1	A method for encapsulating a VOC comprising preparing a stabilized system of biopolymer, surfactant, and VOC; mixing an encapsulating polymer solution with the stabilized system to form a fiber forming solution; and producing ultrafine fibers from the fiber forming solution.
Packaging	film, membrane, coating, patch, mat, sheet or lining
Antimicrobial	hexanal, allyl-isothiocyanate, acetaldehyde, 2E-hexanal
Manufacturing	electrospinning
Claim limits	stabilized system of biopolymer, surfactant, and VOC; mixing with encapsulating polymer solution; producing ultrafine fibers
Overlap	Discloses electrospun ultrafine fiber compositions for controlled VOC release in packaging formats, utilizing humidity-induced structural collapse and temperature-sensitive binding agents to modulate release kinetics.
Confidence	low

28. EP3161558B1 — Electrostatic printing of cyclodextrin compositions

Publication	EP3161558B1
Priority date	2014-06-26
Publication date	2021-05-19
Assignee	VERDANT TECHNOLOGIES, LLC
Inventors	WOOD, WILLARD E., KEUTE, JOSEPH S.
Jurisdictions	['US', 'EP', 'JP', 'KR', 'CN', 'AR', 'AU', 'BR', 'CA', 'CL', 'CO', 'ES', 'IL', 'MA', 'MX', 'NZ', 'PL', 'RU', 'TW', 'UA', 'WO']
Family seen	EP3161558B1
1-MCP carrier	cyclodextrin
Manufacturing	electrostatic printing
Confidence	low

29. US9332751B2 — Food packaging material with antibacterial, ethylene scavenging and barrier properties

Publication	US9332751B2
Priority date	2014-07-09
Publication date	2016-05-10
Assignee	Sabancı Universitesi
Inventors	Hayriye Unal, Serkan Unal, Yusuf Ziya Menceloglu, Fevzi Cakmak Cebeci
Jurisdictions	['US', 'EP', 'ES']
Family seen	US9332751B2
Claim 1	Claim 1 requires a method for producing a polymeric film for food packaging by incorporating halloysite nanotubes (a combination of empty and loaded with antibacterial agent), coating them over the film, arranging them in close proximity on the food product side, and specifically charging the surface, applying chitosan, and coating loaded nanotubes over the chitosan layer.

Packaging	polymeric film
1-MCP carrier	halloysite nanotubes (HNTs)
Release	controlled loading/release into/from halloysite nanotubes
Antimicrobial	thyme oil, oregano oil, cinnamon oil, lemongrass oil, peppermint oil, lavender oil, rosemary oil, coriander oil, clove oil and turmeric oil
Manufacturing	melt mixing, solution mixing, or layer-by-layer thin film coating involving charging the polymeric film surface, applying chitosan, and coating loaded nanotubes over the chitosan layer
Claim limits	combination of empty nanotubes in the polymeric film and nanotubes loaded with an antibacterial agent the halloysite nanotubes being coated over the polymeric film, and wherein the nanotubes coated over the polymeric film are being arranged in close proximity on the food product side of the packaging film, and wherein the coating of the nanotubes loaded with an antibacterial agent comprises the steps of charging surface of the polymeric film; applying chitosan over the charged surface; and coating loaded nanotubes over chitosan layer.
Overlap	Uses halloysite nanotubes as a nanocontainer for 1-MCP and essential oils in a polymeric food packaging film, providing controlled release, antimicrobial activity, and ethylene scavenging properties.
Confidence	high

30. US10449156B2 — Controlled release system including a gas or volatile encapsulated in a polymeric support and a matrix system, a method of preparing the system, and their use

Publication	US10449156B2
Priority date	2014-07-23
Publication	2019-10-22
Assignee	UNIVERSIDAD ANDRES BELLO, CHILE
Inventors	VELASQUEZ, LUIS; CAMPOS, REINALDO; GONZALEZ, DANILO; AND OTHERS
Jurisdictions	['US', 'WO']
Family seen	US10449156B2
Overlap	The document claims a controlled release system for a gas or volatile encapsulated in a polymeric support and matrix system, applicable to agribusiness. It overlaps with the project's controlled-release packaging goal but lacks specific details on 1-MCP or humidity responsiveness in the provided text.
Confidence	medium

31. US10351490B2 — Complexes of 1-methylcyclopropene with metal coordination polymer networks

Publication	US10351490B2
Priority date	2014-11-10
Publication	2019-07-16
Assignee	AgroFresh Inc
Inventors	Nazir Mir
Jurisdictions	US, EP, AU, BR, CA, IL, NZ, WO, ZA
Family seen	US10351490B2
1-MCP carrier	metal coordination polymer network (MCPN)
Release	humidity_moisture
Regulate	adsorption to a surface of the MCPN, released upon contact with aqueous fluid
Control limits	metal node selected from Mg, Mn, Ca, Cu, Al, Zn, Fe, or a combination thereof, and one or more ligands coupled to the metal node
Confidence	low

32. US20190037839A1 — Compositions for controlled release of active ingredients and methods of making same

Publication	US20190037839A1
Priority date	2016-02-19
Publication	2019-02-07
Assignee	Hazel Technologies Inc
Inventors	['Adam Truett Preslar', 'Aidan R. Mouat']
Jurisdictions	['US', 'EP', 'CN', 'JP', 'KR', 'AU', 'BR', 'CA', 'CL', 'CR', 'ES', 'IL', 'MX', 'NZ', 'PE', 'SG', 'WO']
Family seen	US20190037839A1
Claim 1	A composition comprising a carbon or silicate delivery material and a volatile/gaseous active ingredient (≥ 0.01 wt%), configured to release the ingredient passively without external wetting/hydrating, releasing 20-50% of the active ingredient over two weeks in a non-equilibrium condition.
Packaging	sachet
1-MCP carrier	carbon materials or silicate materials
Release	passive_diffusion
Regulate	Passive release governed by the delivery material's physicochemical properties (surface area, porosity, functionalization) and active ingredient concentration, releasing 20-50% over two weeks in a non-equilibrium condition.
Antimicrobial	essential oils, terpenes, terpenoids, carvone, thymol, hexanal, carvacrol, oregano oil, and thyme oil
Manufacturing	Modifying carbon material properties (hydrophobicity, hydrophilicity, etc.) and associating/loading the active ingredient.

Overlap	The patent directly claims porous carbon/silicate carriers for controlled release of 1-MCP (a cyclopropene) and optionally includes essential oils as antimicrobials. However, it explicitly claims passive release 'without addition of external wetting or hydrating,' which contrasts with a humidity-responsive trigger mechanism.
Design-around	To avoid claim 1, one could engineer a carrier that explicitly requires humidity/moisture activation for release (contradicting 'without addition of external wetting'), use a non-carbon/silicate porous material (e.g., MOFs or specific polymers), or target a release profile outside the 20-50% over two weeks window.
Uncertainty	The document does not specify how ambient humidity within standard packaging might passively influence the release kinetics without 'external' wetting, nor does it detail the exact pore size distribution or surface functionalization density required to achieve the claimed 20-50% release window.
Confidence	high

33. US20170318804A1 — Preparation and dispersion of stable emulsion formulations

Publication	US20170318804A1
Priority date	2016-05-04
Publication date	2017-11-09
Assignee	AgroFresh Inc
Inventors	Tirthankar Ghosh
Jurisdictions	US, EP, AR, AU, BR, CA, CL, TW, WO
Family seen	US20170318804A1
Claim 1	An aqueous-in-oil emulsion composition comprising an oil phase (25-70% by weight) and an aqueous phase containing a cyclopropene molecular encapsulating agent complex and water (30-75% by weight), with water at 0.5-20% by weight.
Release trigger	humidity, moisture
Manufacturing	Combining an oil phase and aqueous phase, adding a cyclopropene molecular encapsulating agent complex, and mixing to form the emulsion.
Claim limits	Oil phase 25-70% by weight; aqueous phase 30-75% by weight containing cyclopropene complex and water; water 0.5-20% by weight.
Confidence	low

34. US20190166861A1 — Method and system for the treatment of agricultural products

Publication	US20190166861A1
Priority date	2016-12-05
Publication date	2019-06-06
Assignee	Pace International LLC
Inventors	['David Felicetti', 'Scott Christie', 'Robert Fassel']
Jurisdictions	['US', 'AR', 'UY', 'WO']
Family seen	US20190166861A1
Claim 1	Deploys a portable treatment system with a compound generator and sampling subsystem, operating them to deliver fluid to a confined atmosphere and automatically obtain a sample at a preselected time.
Release control	Automated operation via timing circuits at a preselected time.
Claim limits	Requires a portable treatment system with an active generator and sampling subsystem, operated to deliver fluid and automatically sample at a preselected time.
Confidence	low

35. CN112218529A — Compositions and articles comprising complexes of 1-methylcyclopropene with alpha-cyclodextrin

Publication	CN112218529A
Priority date	2018-04-27
Publication date	2021-01-12
Jurisdictions	['US', 'EP', 'JP', 'KR', 'CN', 'AU', 'BR', 'CA', 'CL', 'CO', 'CR', 'EC', 'ES', 'IL', 'MX', 'PE', 'PH', 'PL', 'WO', 'ZA']
Family seen	CN112218529A
1-MCP carrier	alpha-cyclodextrin
Confidence	low

36. KR20210030899A — Compositions and articles comprising complexes of 1-methylcyclopropene and alpha-cyclodextrin

Publication	KR20210030899A
Priority date	2018-04-27
Publication date	2021-03-18
Assignee	프레쉬 인셋 에스.에이.
Inventors	안드르제즈 볼란, 마리우스 보시아크, 마르신 파쿨스키, 루시나 차츠코브스카, 카타르지나 안나 구라노브스카, 로크사나 카타르지나 라돌로브스카
Jurisdictions	KR, US, EP, CN, JP, AU, BR, CA, CL, CO, CR, EC, ES, IL, MX, PE, PH, PL, WO, ZA
Family seen	KR20210030899A

1-MCP carrier	α -cyclodextrin complex
Release control	Delayed release profile controlled by polymeric binder absorbing moisture; no release for ≥ 1 hour after exposure to 85% RH, followed by continuous release for ≥ 5 hours.
Confidence	low

37. US20200296978A1 — Modified 1-methylcyclopropene (1-MCP) nanobubbles

Publication	US20200296978A1
Priority date	2019-03-15
Publication date	2020-09-24
Assignee	NANOTEC SA
Inventors	Marlene Arismendi-Macuer, Patricio Jarpa, Danilo González, Reinaldo Campos Vargas
Jurisdictions	US, BR, CL, WO
Family seen	US20200296978A1
Claim 1	A stable nanobubble comprising an ethylene inhibitor and at least one food grade sugar.
1-MCP carrier	food-grade sugar-stabilized nanobubbles in aqueous solution
Release control	Stabilization via hydrogen bridges formed by food-grade sugars coating the nanobubble surface.
Manufacturing	In situ synthesis of 1-MCP gas, dispersion in aqueous solutions with food-grade sugars (0.01-3 ppm), and high-speed homogenization/cavitation at 25,000 rpm.
Claim limits	Nanobubble size: 50 to 600 nm; sugar concentration: 0.01-3 ppm; nanobubble concentration in aqueous solution: 0.01 ug/L-3.0 ug/L.
Confidence	low

38. US11033870B1 — Generators for 1-methylcyclopropene release from carrier complex

Publication	US11033870B1
Priority date	2019-11-16
Publication date	2021-06-15
Assignee	Individual
Inventors	Nazir Mir
Jurisdictions	[US]
Family seen	US11033870B1
1-MCP carrier	α -cyclodextrin, metal organic framework, zeolites, activated carbons, cucurbit[6]uril, and other polymeric or porous materials such as gelatin and pectin
Confidence	low

39. US20210235692A1 — Stable 1-Methylcyclopropene Compositions and Uses Thereof

Publication	US20210235692A1
Priority date	2020-02-03
Publication date	2021-08-05
Assignee	Fresh Inset SA
Inventors	Andrzej Jan Wolan, Lucyna Czajkowska, Katarzyna Anna Guranowska, Marcin Rakowiecki, Joanna Andrusiak, Mariusz Jan Bosiak
Jurisdictions	US, EP, CN, AU, BR, CA, CL, MX, PE, WO
Family seen	US20210235692A1
Claim 1	A composition comprising 1-methylcyclopropene and a stabilizer, wherein the concentration of the 1-methylcyclopropene does not change substantially after at least 4 hours, with the proviso that the 1-methylcyclopropene is not complexed to another molecule.
Release control	stabilization via low-volatility compounds (e.g., cis-2-butene, lower alkanes/alkenes) to inhibit free radical oligomerization and oxidation
Manufacturing	reacting a base with 3-chloro-2-methylpropene, condensing at $< -40^{\circ}\text{C}$, and mixing with stabilizer/propellant/gas; or dispensing into aerosol containers
Claim limits	concentration of the 1-methylcyclopropene does not change substantially after at least 4 hours; with the proviso that the 1-methylcyclopropene is not complexed to another molecule. Stabilizer ranges from about 0.1% to about 10% by weight/volume. 1-MCP ranges from about 0.1% to about 50% by weight or 1-10% by volume in gas.
Confidence	low

40. US20210331990A1 — Compositions and Methods for Differential Release of 1-Methylcyclopropene

Publication	US20210331990A1
Priority date	2020-04-27
Publication date	2021-10-28
Assignee	Verdant Technologies LLC
Inventors	Willard E. Wood; Joseph Frank Sarageno, JR.; Joseph S. Keute; Amanda Lundgren
Jurisdictions	US, EP, AR, AU, BR, CA, CL, CO, CR, IL, MX, PE, TW, WO
Family seen	US20210331990A1

Claim 1	A coated substrate comprising a coating disposed on a substrate surface, the coating consisting essentially of a 1-methylcyclopropene clathrate of α -cyclodextrin with a mean particle size between 3 μm and 15 μm , and a polymer.
Release	Particle size modification (comminuting, classifying, or blending) to inversely control release rate
Control limits	Mean particle size between 3 μm and 15 μm ; coating consists essentially of the clathrate and a polymer; coating weight/thickness ranges specified
Overlap	Directly overlaps with humidity-responsive, controlled-release 1-MCP packaging. The patent claims using cyclodextrin clathrate particles modified by size to control release triggered by ambient humidity (e.g., from biological respiration), matching the core mechanism but differing in the specific control method and lacking antimicrobial components.
Confidence	low